Admissible, Not Repeatable: Isolating Approximate Similarity Reads in Shared Agent Memory (Full Version: Appendices)

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This technical report holds the appendices of the paper of the same title, submitted to PVLDB Volume 20. Part I (§A) contains the proofs and the precise statements behind §§4 and 5 of the paper. Part II (§§B to F) documents the experiments that the paper summarizes in one sentence each. Appendix G discusses related work, and Appendix H collects all symbols and terms. Section, theorem, figure, and table numbers without a letter are the paper's.

A Proofs and Precise Statements

A.1 Conventions

States are finite sets of items whose vectors lie in a metric space with distance d; $\delta_i(S, q)$ is the i-th smallest distance from q to S, and reads ask for $k \le |S|$ items, as in §2. For the exact statements below we assume that the distances from a query to the items of a state are pairwise distinct; with ties, the statements hold for any fixed tie-breaking rule, with "<" between tied distances read as "precedes." A transaction is a deterministic function from the answers it has received so far to its next read or to its write set; Algorithm 1 fixes the queries of a transaction before its first read, and the theorems below are stated for such transactions. A write set consists of a set of insertions, each of a new item, and a set of deletions of items in the transaction's held answers (a replacement contributes to both). It is applied atomically with respect to reads: a read observes either all of it or none of it. On stores without multi-statement transactions this holds for write sets with one operation; for larger write sets Ariadne relies on the commit lock and on reads that are validated under it. A read-only transaction with one read is not validated, so on such stores Theorem A.4 needs write sets of one operation, or validation of those transactions as well.

Lemma A.1 (Final state). *Let I and U be the items inserted and deleted by the transactions of an execution. Every order π that satisfies Definition 4.1 for some c has last state $(S_0 \cup I) \setminus U$.*

Proof. An item $y \in U$ deleted by the j-th transaction in π lies in a held answer of that transaction, hence in S^{π}_{j-1}, because admissible answers consist of items of the state. So every deletion removes an item that is present, and no item is deleted twice: after its deletion it is absent from every later state, and a later held answer containing it would not be admissible. Every insertion adds a new item, which no earlier transaction can delete. Hence each item of $S_0 \cup I$ is present at the end exactly if it is not in U. □

A.2 Operational reading of Definition 4.1

Call a store a *c-store* if its similarity read $\textsc{Read}(q, k)$ in state S may return any element of $\mathrm{Adm}_c(S, q, k)$.

Proposition A.2. *An execution is c-serializable if and only if there is a total order π of its transactions such that running the transactions one at a time in the order π on a c-store, with every read returning its held answer in the execution, is a legal run in which every transaction produces its write set in the execution.*

Proof. Suppose the execution is c-serializable with order π. We show by induction on j that the serial run reaches state S^{π}_j after its j-th transaction. The j-th transaction starts in S^{π}_{j-1}. Its writes are buffered, so all its reads are evaluated in S^{π}_{j-1}, and by Definition 4.1 each held answer is in $\mathrm{Adm}_c(S^{\pi}_{j-1}, q, k)$, which a c-store may return. Shown the same answers, the transaction issues the same reads and the same write set by determinism, and the run reaches S^{π}_j. Conversely, a legal serial run with the held answers shows $A \in \mathrm{Adm}_c(S^{\pi}_{j-1}, q, k)$ for every read directly. □

For $c = 1$ and distinct distances $\mathrm{Adm}_1(S, q, k)$ is a singleton, a 1-store is an exact store, and the proposition is view serializability restricted to transactions whose reads are similarity reads.

A.3 Separation (Proposition 4.2) is tight

Proposition A.3. *For every $\tau > 0$, $c > 1$ and $\varepsilon > 0$ there is, on the real line with $d(x, y) = |x - y|$, a c-serializable execution of one τ-deduplicating insertion on a (τ/c)-separated state whose final state contains two items at distance less than $\tau/c + \varepsilon$.*

Proof. Let $k = 1$, $0 < \eta < \varepsilon$, the query $q = 0$, and $S_0 = \{a, b\}$ with $a = \tau/c + \eta$ and $b = -(\tau + \eta)$. The two items are farther apart than τ/c. The exact answer of $\textsc{Read}(q, 1)$ is (a), and the list (b) is c-admissible because $d(q, b) = \tau + \eta \le \tau + c\,\eta = c\,\delta_1(S_0, q)$. A transaction that observes (b) sees $d(q, a_1) > \tau$ and inserts q; the final state contains a and q at distance $\tau/c + \eta < \tau/c + \varepsilon$. □

With a validation read of approximation ratio ρ, Theorem 4.3 gives $(c\rho)$-serializability and therefore, if S_0 is $(\tau/(c\rho))$-separated, $(\tau/(c\rho))$-separation; the certified variant ($\rho = 1$) gives τ/c. §D.2 reports the smallest stored pairwise distances that we measured in the certified runs against these bounds.

A.4 Safety (Theorem 4.3) for general transactions

We restate the theorem for transactions with several reads and with insertions, deletions, and replacements, and construct the serial order explicitly from the order of the commit lock.

Theorem A.4. *Consider an execution in which every transaction runs Algorithm 1 under one commit lock, with validation reads issued*

after the lock is granted, on a store whose reads observe every write set applied before they are issued and none in part, and suppose every answer that the store returns is ρ-admissible in the state it is evaluated in. Then the execution is $(c\rho)$-serializable. A transaction that aborts at the round limit writes nothing and is not part of the execution.

Proof. Number the transactions that take the commit lock as $T_1, T_2, \dots$ in the order in which they hold it in their last iteration, and let S_j be the state after T_j applied its write set, with S_0 the initial state, before any write set is applied.

The validation state. All write sets are applied while the commit lock is held, and the lock is held from the validation reads to the application of the writes. Hence between the moment T_j acquires the lock in its last iteration and the moment it applies its writes no other write set is applied, and all write sets applied before are those of $T_1, \dots, T_{j-1}$. The validation reads of T_j are therefore evaluated in S_{j-1}, and T_j's write set transforms S_{j-1} into S_j.

Admissibility. Let A be the answer of a read Read(q,k) that T_j used in its last call of Think, and F the corresponding fresh answer. The test $A \sqsubseteq_c F$ gives $A \subseteq S_{j-1}$, $|A| = |F| = k$, and $d(q,a_i) \le c\, d(q,f_i)$ for all i. Since $\rho_F = \rho(F \mid S_{j-1}, q) \le \rho$,

$$d(q,a_i) \;\le\; c\, d(q,f_i) \;\le\; c\,\rho_F\, \delta_i(S_{j-1},q),$$

so $A \in \mathrm{Adm}_{c\rho}(S_{j-1},q,k)$ and $\rho(A \mid S_{j-1},q) \le c\,\rho_F$. The write set of T_j is the one that Think computed from these answers. The bound of Theorem 4.3 is attained: with $k=1$, a nearest item at distance 1 from q, a fresh answer $F=(f_1)$ at distance ρ_F, and a held answer $A=(a_1)$ at distance $c\rho_F$, the held answer passes the test with equality at $i=1$, and its ratio is $c\,\rho_F$.

Read-only transactions with one read. Such a transaction T' observes an answer that the store evaluated in some state. Write sets are applied atomically with respect to reads (§A.1), so this state is S_j for some j, and the answer is ρ-admissible in it. Insert T' into the order directly after T_j. It changes no state, so the states that the order assigns to the other transactions are unchanged.

The resulting total order explains every read by a state in which its answer is $(c\rho)$-admissible, as Definition 4.1 requires. □

Remarks. (1) Earlier iterations of the loop, and the answers they used, do not appear in the execution: only the write set of the last iteration is applied. (2) The proof uses the commit lock twice: all writers must take it, and the validation reads must observe the state at the time the lock is held. A lock per group of agents on a shared store violates the first requirement. A validation transaction whose snapshot was taken before the lock was granted violates the second; this is what happens on PostgreSQL at repeatable read, where the statement that requests the advisory lock fixes the snapshot (§E). (3) The theorem bounds the ratio of the held answer by $c\,\rho_F$ for the *realized* ratio ρ_F of each validation read, which is what the auditor checks commit by commit. (4) A read-only transaction with several reads is not serialized by its reads, because writes may commit between them; it is validated like any other transaction.

A.5 Characterization (Theorem 5.1) and the conflict radius

Lemma A.5 (General form). *Let A be the exact answer of* Read(q,k) *in S, with distances $r_1 < \cdots < r_k$, and let $S' = (S \cup V) \setminus U$ for a set V of inserted items and a set U of deleted items. Then $A \in \mathrm{Adm}_c(S',q,k)$ if and only if $A \cap U = \emptyset$ and $r_i \le c\,\delta_i(S',q)$ for all $i \le k$. If $A \cap U = \emptyset$, then $A \in \mathrm{Adm}_c(S \cup V,q,k)$ implies $A \in \mathrm{Adm}_c(S',q,k)$; in particular, deletions alone never make A inadmissible unless they delete an item of A.*

Proof. The first claim is Definition 2.1. For the second, removing items from a state cannot decrease any δ_i, so $\delta_i(S',q) \ge \delta_i(S \cup V,q)$. For $V=\emptyset$ and $A \cap U = \emptyset$, the items of A are the k nearest items of S and belong to $S \setminus U$, so they are its k nearest items as well, and A is the exact answer in S'. □

Theorem 5.1 is the case $V=\{v\}$, $U=\emptyset$: with $t = d(q,v) < r_k$ and rank $p = 1 + |\{i : r_i < t\}|$, the sorted distances of $S \cup \{v\}$ are $r_1, \dots, r_{p-1}, t, r_p, \dots$, so $\delta_i = r_i$ for $i < p$, $\delta_p = t$, and $\delta_i = r_{i-1}$ for $p < i \le k$; Lemma A.5 then gives Theorem 5.1.

Proof of Corollary 5.2. Put $r_0 = 0$ and let $J = \{i \in \{1,\dots,k\} : r_i > c\, r_{i-1}\}$. The index 1 belongs to J, so $j = \max J$ exists and $R_c(q) = r_j/c$, because the r_i increase. Let a new item v be inserted at distance $t = d(q,v)$ from the query q.

If $t \ge r_k$, the top-k is unchanged, A stays admissible, and $t \ge r_k \ge r_j/c = R_c(q)$, because $j \le k$ and $c \ge 1$.

Otherwise let $p \le k$ be the rank of v, so that $r_{p-1} \le t \le r_p$. If $p < j$, the second condition of Theorem 5.1 fails at $i = j$, so A is inadmissible, and $t \le r_p \le r_{j-1} < r_j/c = R_c(q)$. If $p = j$, no index $i > p$ lies in J, so the second condition holds and A is inadmissible exactly if the first fails, that is, if $t < r_j/c = R_c(q)$. If $p > j$, again the second condition holds, and $p \notin J$ gives $r_p \le c\, r_{p-1} \le c\, t$, so the first holds too and A is admissible; moreover $t \ge r_{p-1} \ge r_j \ge R_c(q)$. In every case A is inadmissible if and only if $t < R_c(q)$. The bounds $r_1/c \le R_c(q) \le r_k/c$ follow from $1 \in J$ and $j \le k$. □

Properties. $R_c(q)$ is non-increasing in c, and $R_1(q) = r_k$ for distinct distances: under strict semantics the whole top-k ball is protected, which is the predicate that re-execution validation enforces on an exact index. If all consecutive ratios r_i/r_{i-1}, $i \ge 2$, are at most c, then $R_c(q) = r_1/c$: only an insertion that becomes the new nearest neighbor by a factor of more than c conflicts.

Several insertions. The radius form does not extend to $|V| > 1$. For $k=3$, $r=(1, 1.09, 1.18)$ and $c=1.1$ we have $R_c(q) = 1/1.1 \approx 0.909$. Two insertions at distances 0.95 and 0.96 both lie outside $R_c(q)$, yet the new second-smallest distance is 0.96 and $r_2 = 1.09 > 1.1 \cdot 0.96$. The general test of Lemma A.5 applies, and the conflict law treats several insertions by simulation (§A.6).

A.6 Conflict law (Theorem 5.3)

Model. Around the query q the stored items form a Poisson process whose expected number of points within distance s is λs^D on the range of distances that matters for the read (the model is used for the $k+1$ nearest items). In the *mass coordinate* $\omega = \lambda s^D$ the process is homogeneous with unit rate, and a distance ratio c corresponds to the mass ratio $c^D = 1/x$. Let $\mu_1 < \mu_2 < \dots$ be the masses of the stored neighbors and $\mu_0 = 0$. A concurrent in-ball insertion with dimension ratio $\alpha = D_s/D \in (0,1]$ has mass ω with density proportional to $\omega^{\alpha-1}$ on $(0,\mu_k)$, independently of the ratios B_i of consecutive neighbor masses defined next.

Lemma A.6. *The ratios $B_i = \mu_i/\mu_{i+1}$, $i \geq 1$, are independent, B_i has density $i\,b^{i-1}$ on $[0,1]$, and $(B_1, \ldots, B_{k-1})$ is independent of μ_k.*

Proof. Given μ_{i+1}, the points $\mu_1, \ldots, \mu_i$ are the order statistics of i independent uniform variables on $(0, \mu_{i+1})$, so μ_i/μ_{i+1} is the maximum of i uniform variables on $(0,1)$, with density $i\,b^{i-1}$, and $(\mu_1, \ldots, \mu_i)/\mu_{i+1}$ is independent of μ_{i+1} and of the process beyond it. Applying this for $i = k-1, k-2, \ldots, 1$ gives the claim. □

Proof of Theorem 5.3. Because the density of ω on $(0, \mu_k)$ is $\alpha\omega^{\alpha-1}/\mu_k^\alpha$, for an event E determined by ω and the μ_i we have $\Pr[E] = \mathbb{E}\big[\mu_k^{-\alpha} \int_E \alpha\omega^{\alpha-1}\, d\omega\big]$. In mass coordinates Theorem 5.1 reads: the insertion has rank p and the answer stays c-admissible if and only if

$$\max(\mu_{p-1},\, x\,\mu_p) \leq \omega < \mu_p \quad \text{and} \quad B_i \geq x \;\text{ for } p \leq i < k.$$

With $B_0 = 0$, integrating $\alpha\omega^{\alpha-1}$ over the interval for ω gives

$$\mu_p^\alpha\big(1 - \max(B_{p-1}, x)^\alpha\big), \qquad \mu_p = \mu_k \textstyle\prod_{i=p}^{k-1} B_i.$$

With the factor $\mu_k^{-\alpha}$ and the conditions $B_i \geq x$ for $p \leq i < k$, the expectation is

$$\varphi_\alpha(p) = \mathbb{E}\Big[\big(1 - \max(B_{p-1}, x)^\alpha\big) \prod_{i=p}^{k-1} B_i^\alpha\, \mathbf{1}\{B_i \geq x\}\Big]$$

$$= \mathbb{E}\big[1 - \max(B_{p-1}, x)^\alpha\big] \prod_{i=p}^{k-1} \mathbb{E}\big[B_i^\alpha \mathbf{1}\{B_i \geq x\}\big]$$

by Lemma A.6. For $i \geq 1$, $\mathbb{E}[B_i^\alpha \mathbf{1}\{B_i \geq x\}] = \int_x^1 i\,b^{i-1+\alpha}\, db = \frac{i}{i+\alpha}(1 - x^{i+\alpha})$. For $p \geq 2$,

$$\mathbb{E}\big[1 - \max(B_{p-1}, x)^\alpha\big] = 1 - x^\alpha\, x^{p-1} - \tfrac{p-1}{p-1+\alpha}\big(1 - x^{p-1+\alpha}\big)$$
$$= \tfrac{\alpha}{p-1+\alpha}\big(1 - x^{p-1+\alpha}\big),$$

and for $p = 1$ the factor is $1 - x^\alpha$, which is the same expression. This is the stated $\varphi_\alpha(p)$. Setting $x = 0$ removes the admissibility conditions and gives the rank distribution

$$\theta_\alpha(p) = \frac{\alpha}{p-1+\alpha} \prod_{i=p}^{k-1} \frac{i}{i+\alpha} = \frac{\alpha\,\Gamma(p-1+\alpha)\,\Gamma(k)}{\Gamma(p)\,\Gamma(k+\alpha)}.$$

The mixture statement is linearity in the kind of the insertion. □

Consequences. (1) For $\alpha = 1$, $\theta_1(p) = 1/k$ and $\varphi_1(p) = \frac{1}{k}\prod_{i=p}^{k}(1-x^i)$, so $P_1(c) = 1 - \frac{1}{k}\sum_{p=1}^{k}\prod_{i=p}^{k}(1-x^i) = \frac{1}{k}\sum_{i=1}^{k} i\,x^i + O(x^3) = \frac{x}{k} + O(x^2)$, because each product differs from $1 - \sum_{i\geq p} x^i$ by terms of order x^{2p+1} or smaller. (2) Conditional on rank one, the insertion violates the first condition of Theorem 5.1 with probability $\Pr[\omega < x\mu_1 \mid \omega < \mu_1] = x^\alpha = c^{-D_s}$. (3) As $c \to \infty$, $1 - \varphi_\alpha(p)/\theta_\alpha(p) = x^{p-1+\alpha} + \sum_{i=p}^{k-1} x^{i+\alpha} + O(x^{1+2\alpha})$, whose leading term over all ranks is x^α at $p = 1$; therefore $P_\alpha(c) = \theta_\alpha(1)\, c^{-D_s}\,(1 + o(1))$ and a mixture behaves as $\sigma\,\theta_\alpha(1)\, c^{-D_s} + (1-\sigma)\, c^{-D}/k$. (4) The formulas remain valid for $\alpha > 1$, which describes writes that avoid the query, since the density $\alpha\omega^{\alpha-1}/\mu_k^\alpha$ then grows with ω.

Background rate. We model a background write as exchangeable with the stored items, so its rate needs no fitted parameter.

Lemma A.7. *Let v be exchangeable with the N items of S. Then v is among the k nearest items of $S \cup \{v\}$ to a fixed query with probability $k/(N+1)$, and given that it is, its rank is uniform on $\{1, \ldots, k\}$.*

Proof. By exchangeability the rank of $d(q, v)$ among the $N+1$ distances is uniform on $\{1, \ldots, N+1\}$. □

Summing over the n background writes that overlap a read gives the expected number $n\,k/(N+1)$ of background in-ball insertions, with N the store size at the read; no parameter is fitted.

Semi-parametric form. For a read with neighbor distances $r_1, \ldots, r_k$, a sibling insertion has $(t/r_k)^{D_s}$ uniform on $[0,1]$ given that it is in the ball, so by Corollary 5.2 it conflicts with probability $\Pr[t < R_c(q)] = (R_c(q)/r_k)^{D_s}$. For a background insertion, exchangeability says that, given the configuration of the nearest items after the insertion, the inserted item is equally likely to be any of them. We estimate the resulting conflict probability by leave-one-out on the read's own neighbors, whose configuration has the same law up to one item (N stored items against $N+1$): designate the p-th of the read's $k+1$ nearest neighbors, at distances $r_1 < \cdots < r_{k+1}$, as the inserted one, for $p \leq k$ (an in-ball insertion). The held answer would then be $r_1, \ldots, r_{p-1}, r_{p+1}, \ldots, r_{k+1}$, which is inadmissible if and only if $r_{i+1} > c\,r_i$ for some $p \leq i \leq k$. Averaging this indicator over $p = 1, \ldots, k$, which Lemma A.7 makes equally likely, and over reads gives the background prediction. With sibling shares σ_j and sibling dimensions $D_s^{(j)}$, one per sibling scale j, the predicted conflict probability of one in-ball insertion at a read is

$$\textstyle\sum_j \sigma_j\,(R_c/r_k)^{D_s^{(j)}} + \big(1 - \sum_j \sigma_j\big) \cdot (\text{leave-one-out}).$$

For reads with several in-ball insertions we draw the positions of the observed number of insertions from the same mixture around the read's actual neighbor list and apply Lemma A.5.

A.7 Estimation

All quantities are computed exact-to-exact from the logs of instrumented runs: the auditor reconstructs, for the first read of every task, the exact neighbor distances in the state at the read and at the first validation, and the insertions committed in between.

Sibling dimensions. For every in-ball insertion we record $y = \ln(r_k/t)$. Under the sibling model y is exponential with rate D_s. The background share w_{bg} is fixed at the ratio of the first-principles count $\sum n\,k/(N+1)$ to the observed count, and the background density of y is the empirical law of $\ln(r_{k+1}/r_i)$, $i \leq k$, over all reads, smoothed by a three-component exponential mixture and then frozen. We maximize the likelihood of the mixture $w_{\text{bg}} f_{\text{bg}}(y) + \sum_j \sigma_j D_s^{(j)} e^{-D_s^{(j)} y}$ over one or two sibling scales. Intervals come from a bootstrap that resamples reads; goodness of fit is a Kolmogorov–Smirnov (KS) distance with a parametric bootstrap. Conflict outcomes and ranks are never used for fitting.

Lag kernel. For a lag bucket b, $\hat{h}_b$ is the number of in-ball insertions whose source item lies at a lag in b behind or ahead of the reading item in the same stream, divided by the number of

writes at such lags that overlapped the read, and $\hat{D}_b$ is the reciprocal of the mean of y over these insertions. The static estimator replaces the concurrent run by the stream itself: for item j it takes the exact top-k ball in the state that contains the initial memory and the items before j, and counts the items $j + \ell$, $\ell \in b$, that fall into it. A prediction for a run uses the run's overlap counts o_b and the kernel: $\nu = \sum_b o_b \hat{h}_b$ in-ball insertions per read, and the mixture of sibling laws with weights $o_b \hat{h}_b / \nu$ and dimensions $\hat{D}_b$ in the semi-parametric form, one scale per lag bucket.

B PostgreSQL: Details of §3

B.1 Background: serializable snapshot isolation in PostgreSQL

PostgreSQL implements its serializable level as serializable snapshot isolation (SSI) [18, 117]. Every transaction reads from a snapshot, and SSI tracks *rw-antidependencies*: $T_r \rightarrow T_w$ when T_r reads a version that T_w later replaces, or when T_w inserts a row that a predicate read of T_r would have returned (a phantom). Reads are recorded by predicate locks (SIREAD locks), which never block: a read takes them on the tuples it returns and on the index pages, or the whole index relation, that its scan visits, and every write checks them through `CheckForSerializableConflictIn`. Every cycle that snapshot isolation admits contains two consecutive rw-antidependencies between concurrent transactions, a *dangerous structure* [42]; SSI aborts a transaction of every dangerous structure it detects, so it may also abort executions that are serializable. An index access method declares through `ampredlocks` whether it takes page-level predicate locks itself. B-tree, hash, GiST, and GIN do; for a method that does not, a scan locks the whole index relation, and `index_insert` checks the whole relation before it calls the method's insert routine. Both properties matter in §3.2: the relation-level lock makes every similarity read conflict with every insertion (coarseness), and the order of the check and of the insertion opens the window described next (unsoundness).

B.2 Code path and the fix

If the handler of an index access method does not set `ampredlocks`, `index_insert` calls `CheckForSerializableConflictIn` on the whole relation and then the method's insert routine. For pgvector's HNSW this routine, `hnswinsert`, searches the graph for the neighbors of the new element and then links it. A scan by another serializable transaction takes its relation-level predicate lock when it starts. If it starts after the writer's check, the writer has not seen its lock; if it finishes before the element is linked, it has not seen the element. Figure A1 repeats the check after the element is reachable. It is the smallest change that closes the window; it does not make the lock finer than the whole index relation. We reported the window and this change to the pgvector maintainers [87].

B.3 Tests

Table A1 lists all tests: the predicate locks that one read takes per access method, the deterministic reproduction with a debugger pause, the stress test without a debugger, and its controls. Figure A2 shows the share against the mean duration of an insertion: every configuration with slower insertions than the default lets a tightly

```
--- a/src/hnswinsert.c
+++ b/src/hnswinsert.c
@@ -3,6 +3,7 @@
 #include "hnsw.h"
+#include "storage/predicate.h"
@@ -793,6 +794,13 @@ hnswinsert(Relation index, ...
         /* Insert tuple */
         HnswInsertTuple(index, values, isnull, heap_tid);

+        /* repeat the SSI conflict-in check after the
+         * element is linked into the graph */
+        CheckForSerializableConflictIn(index,
+            (ItemPointer) NULL, InvalidBlockNumber);
+
         /* Delete memory context */
```

Figure A1: The change to pgvector (abridged diff) that closes the window of §3.2 in `hnswinsert`.

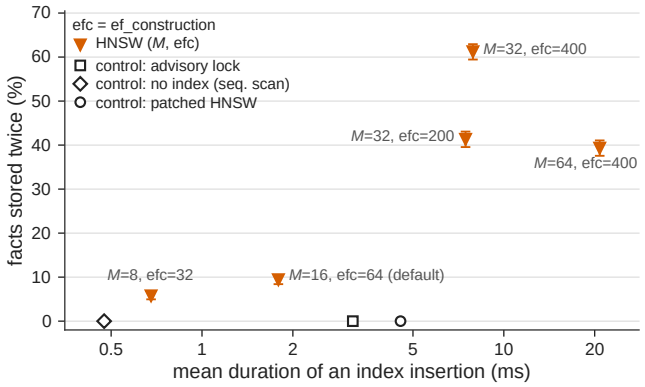


Figure A2: Share of tightly timed serializable pairs that store both rows, against the mean duration of an HNSW insertion (varied through M and *ef_construction*), with 95% Wilson intervals; hollow marks: the controls.

timed reader fall into the window more often, most often (61.2%) at a mean insertion time of 7.9 ms. The same pause on other access methods without predicate locking (SP-GiST, BRIN) produces a double commit; on B-tree, hash, and GIN, which take page-level predicate locks inside the insertion, one of the two transactions aborts; GiST, which does lock the pages a search visits, also commits both rows (§B.4), and the pause is never reached in bloom. Figure A3 complements Table 3 of the paper with the other concurrency levels and the control on a scalar key indexed by a B-tree.

B.4 GiST

Our reading of the observed behavior is the following. GiST takes predicate locks on the pages that a search visits and checks an insertion against the locks on the pages it modifies. A search for a point outside every bounding box of an internal page is pruned there and locks only that page. The first insertion of such a point enlarges a bounding box on the internal page, modifies it, and is checked against the lock. A second insertion of the same point now descends into the enlarged box and modifies only a leaf page, which no search locked. If two transactions search before either inserts, both find nothing, the first insertion conflicts with the second transaction's lock (one dependency), and the second insertion conflicts

Table A1: Soundness tests for serializable isolation over index access methods (generated from the test logs). In the ivfflat control, the reader's probes miss the committed row in some trials: an index miss, not a race.

access method	SIREAD locks of one read	both commit	trials
Predicate locks taken by a SERIALIZABLE index read (pg_locks)			
hnsw	relation: 1, tuple: 10		
ivfflat	relation: 1, tuple: 10		
btree	page: 1 (ampredlocks)		
gist	page: 1 (ampredlocks)		
spgist	relation: 1		
brin	relation: 2		
hash	page: 1 (ampredlocks)		
gin	page: 2 (ampredlocks)		
Deterministic reproduction (writer paused between its conflict check and the insertion)			
hnsw, window		8	8
hnsw, control (reader starts after the insertion)		0	8
ivfflat, window		8	8
ivfflat, control (reader starts after the insertion)		4	8
The same pause in other access methods			
spgist	relation	3	3
brin	relation	3	3
bloom	relation, tuple	breakpoint not reached	3
hash	page	0	3
gist	page	3	3
gin	page	0	3
btree	page	0	3
Two reads with almost antipodal queries, then two inserts (both commit / one aborts)			
hnsw	relation:mem_vec=1, tuple:mem=10	0 / 20	20
btree	page:kmem_key=1	20 / 0	20
Without a debugger: facts stored twice / facts (Wilson 95% interval)			
HNSW, m=16, ef_construction =64	insertion 1.79 ms	282 (9.4%, 8.4 to 10.5)	3000
HNSW, m=8, ef_construction =32	insertion 0.68 ms	172 (5.7%, 5.0 to 6.6)	3000
HNSW, m=32, ef_construction =200	insertion 7.47 ms	1239 (41.3%, 39.6 to 43.1)	3000
HNSW, m=32, ef_construction =400	insertion 7.90 ms	1836 (61.2%, 59.4 to 62.9)	3000
HNSW, m=64, ef_construction =400	insertion 20.8 ms	1179 (39.3%, 37.6 to 41.1)	3000
HNSW + advisory lock (control), m=16, ef_construction =64	insertion 3.16 ms	0 (0.0%, 0.0 to 0.1)	3000
no index, sequential scan (control)	insertion 0.47 ms	0 (0.0%, 0.0 to 0.1)	3000
IVFFlat, one probe (includes recall misses)	insertion 0.28 ms	1055 (35.2%, 33.5 to 36.9)	3000
HNSW, patched library (control), m=16, ef_construction =64	insertion 4.55 ms	0 (0.0%, 0.0 to 0.0)	30000

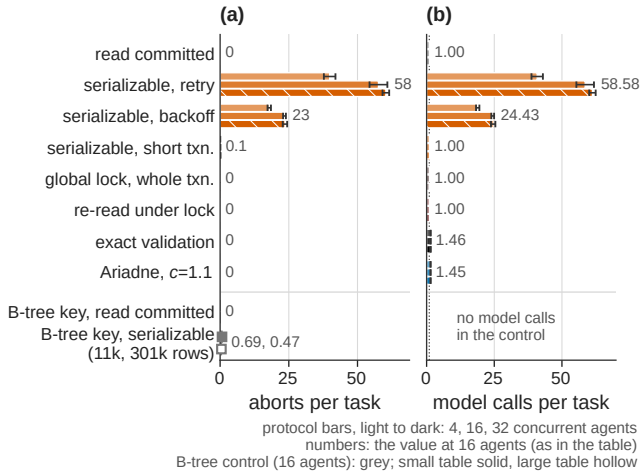


Figure A3: Aborts and model calls per task under serializable isolation and its alternatives, by concurrency level (light to dark: 4, 16, 32 agents; hatched: 32; "global lock, whole txn." is the lock across the thinking step); the printed numbers are the values at 16 agents, as in Table 3 of the paper.

with nothing, so no dangerous structure forms. The protocol of Kornacker et al. [73] attaches the search predicate to the node and propagates it to a child when the child's bounding predicate grows; the implementation moves predicate locks on page splits only. We observed double commits in 40 of 40 trials for equality and 40 for range searches outside all data, 2 of 40 for points inside the populated region, 0 for nearest-neighbor ordering, and 0 for the same test on a B-tree. We have not traced the mechanism in the source line by line, and we tested one major version.

C Conflict Law: Details of §§5 and 6.5

Besides the configurations of §6.1, the law runs include A256 (AG News, 256 agents, 12,000 writes) and A64-long (AG News, 64 agents, 30,000 writes), shown in Figure 6 of the paper and below.

Closed form. Figure A7 maps Theorem 5.3. For a pure sibling stream, the $c-1$ needed for a given conflict probability grows as the sibling dimension falls; background writes flatten this dependence where conflicts are likely. Figure A4 applies Theorem 5.1 to the neighbor list of one real read. Figure A8 compares the closed form with simulation of the model on 288 parameter settings; the largest standardized deviation is 2.9, and 3.6% of the tested combinations of setting and seed exceed two standard errors of the Monte Carlo estimate of the same parameter setting and seed.

Where insertions land. Figure A9 is the data-driven view of Corollary 5.2: for every read of a run we compute $R_c(q)/r_k$ and average the indicator that an insertion at normalized distance t/r_k conflicts. The marginal on top shows where real in-ball insertions land, split into the first-principles background and the excess over it. Background insertions land near the rim, where almost no tolerance is needed; the excess lies deeper. Figure A10 repeats Figure 5 of the paper for every configuration, Figure A11 shows the fitted position densities, and Figure A12 the rank distributions, unseen by the fit.

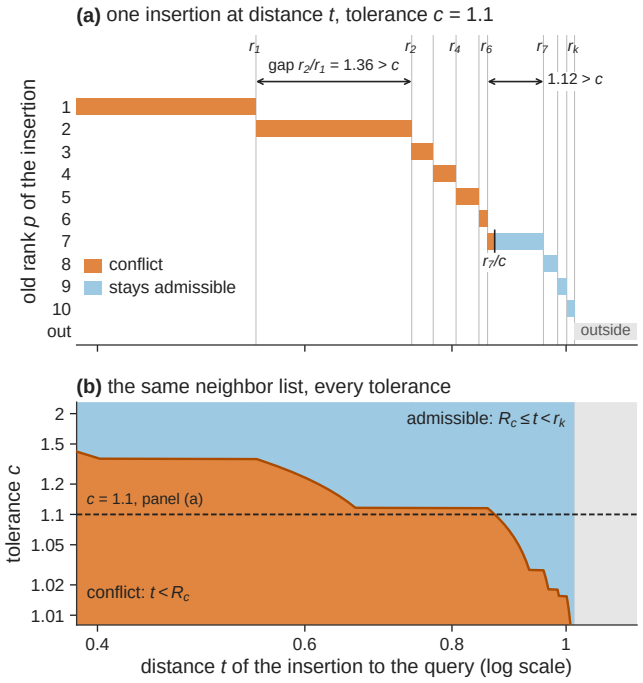


Figure A4: Theorem 5.1 on the neighbor list of one real read. (a) For each rank, the distances at which an insertion breaks admissibility; (b) the conflict region over all tolerances.

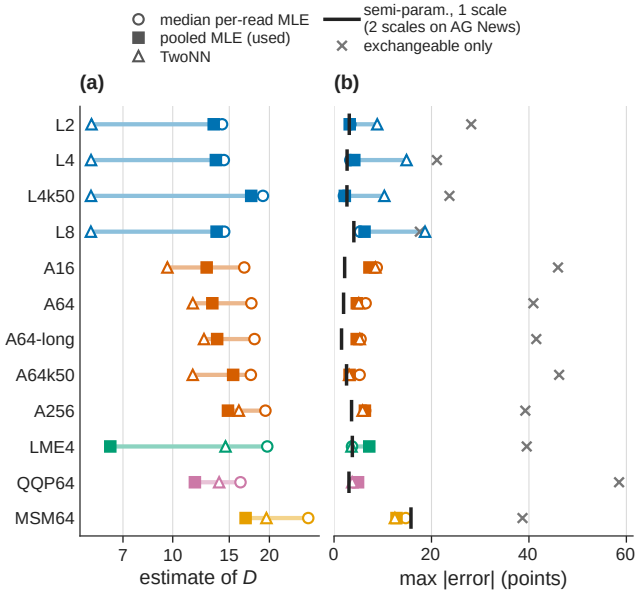


Figure A5: Sensitivity of the closed form to the estimator of the intrinsic dimension. Black ticks: the semi-parametric form (two sibling scales on AG News, one elsewhere).

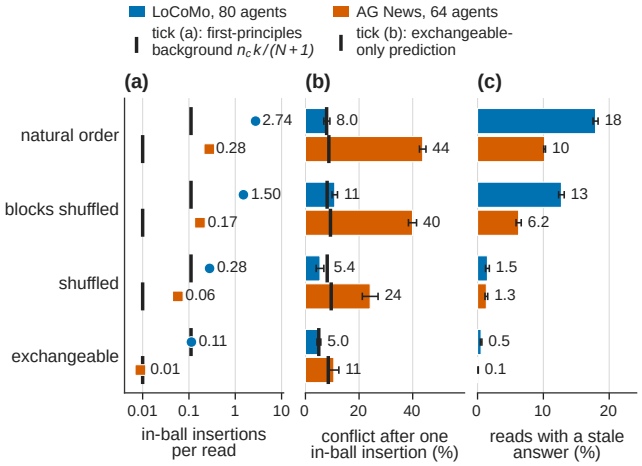


Figure A6: Controls on LoCoMo (one team of 80 agents, unlike L8) and A64 as the order of the stream is destroyed step by step (blocks of 64 items shuffled, all items shuffled, exchangeable split). (a) In-ball insertions per read (log scale); black ticks: first-principles background rate. (b) Conflict probability after one in-ball insertion at $c = 1.1$; black ticks: exchangeable-only prediction. (c) Share of all reads with a stale answer, for each order of the stream.

Ranks. The position fit never sees the ranks of the in-ball insertions. Its predicted share at rank one is within 3.9 points of the observed share on the six configurations of Table 6 of the paper, far closer than uniform ranks (Figure A12). A chi-square test of the whole rank distribution, pooled over seeds and corrected for the dependence of insertions within a burst, rejects the prediction on these configurations except L4, so the law describes the share at rank one, not the whole rank distribution.

All reads. For reads with any number of in-ball insertions, the simulation of §A.6 predicts the share whose held answer is inadmissible at its first validation; the largest error over configurations and tolerances is 1.8 points (two sibling scales on AG News), against 12.6 for background writes only. Stratified by the number of in-ball insertions of the read, the errors are 11.3, 18.0, and 24.2 points for one, two, and three or more (largest over the configurations with at least 30 such reads; this check uses one sibling scale per configuration). The larger errors are on AG News, which needs two scales (Table 6); on LoCoMo, which one scale describes, the stratum of reads with one in-ball insertion errs least.

Tails and dimension estimators. Figure A13 compares the log-log slope of the observed conflict probability with the fitted sibling dimensions, and Figure A5 shows how the error of the closed form depends on the estimator of the intrinsic dimension (the semi-parametric form needs none). Stratifying the reads by their local intrinsic dimension raises the error of the closed form with one pooled D to 8.7 points on average and that of the semi-parametric form to 7.1; both rise by similar amounts. On a synthetic store that mixes two manifolds of different dimension, the closed form errs more than the semi-parametric form (Figure A14).

Planted law and controls. On synthetic streams the background items are exchangeable with the stored ones and siblings are planted at chord distance $R\,\xi^{1/D_s}$ from a parent a few positions earlier, with ξ uniform on $(0, 1)$ and a radius R above the $(k{+}1)$-th neighbor distance of almost every read. Figure A15 compares planted and recovered sibling dimensions over 630 configurations and seeds (the figure draws means over seeds); the planted value lies inside the 95% interval of its run in 592 of them. On these synthetic streams (not drawn), the pooled in-ball rate of background insertions is 1.00 times the first-principles rate, and at $k = 50$ their rank-one share is 2.0% (95% interval 1.9 to 2.2; uniform ranks give 2.0%). Figure A6 shows the controls of §6.5 on real streams. In the exchangeable split the rank-one share of in-ball insertions is 10.4% (95% interval 9.0 to 11.9, against 10.0% for uniform ranks), and a chi-square test of the whole rank distribution does not reject uniform ranks ($p = 0.43$).

Transfers. Figure A16 shows every prediction that was made without fitting on its target: across concurrency levels with pooled parameters and with the lag kernel, across k, across time, from the static stream, and across collections. Figure A17 shows the fitted parameters for other values of k and other embedding models.

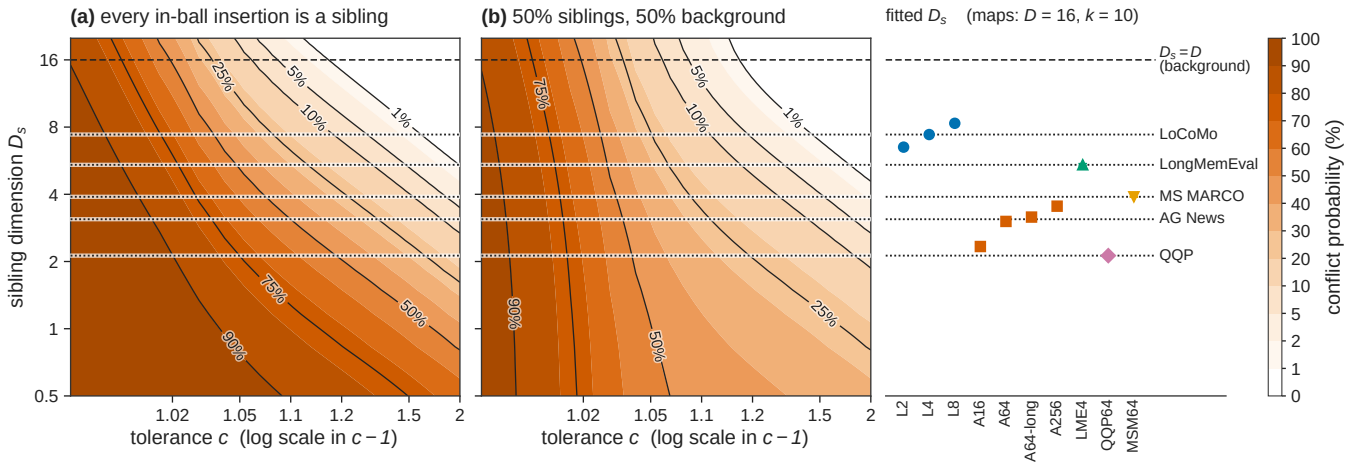


Figure A7: Conflict probability of one in-ball insertion by Theorem 5.3 over the tolerance and the sibling dimension, for two sibling shares. The strip on the right marks the fitted sibling dimension D_s of each workload.

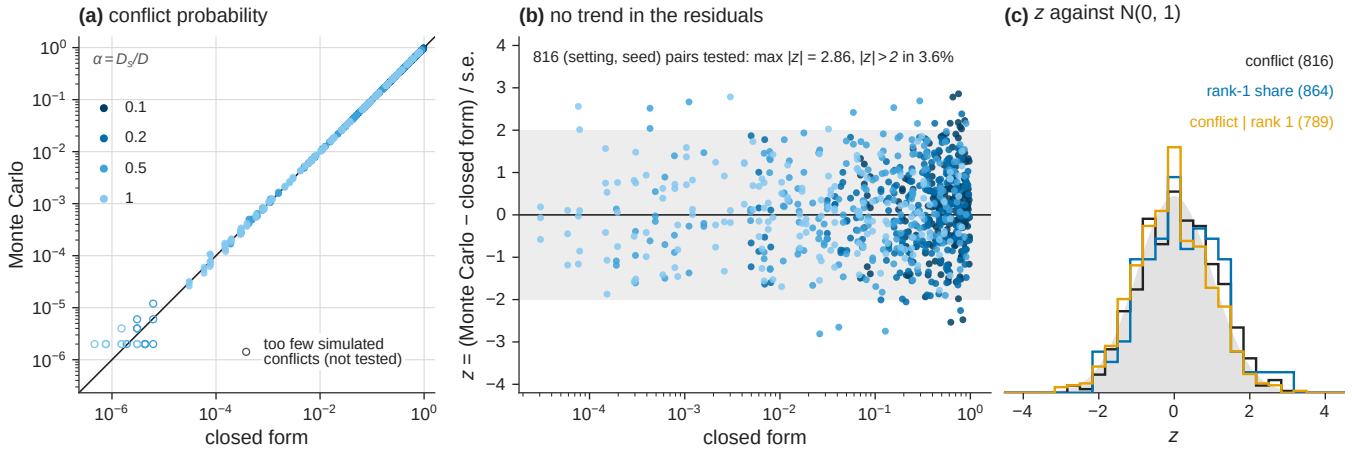


Figure A8: Closed form of Theorem 5.3 against Monte Carlo simulation of the model, per setting and seed; hollow marks: too few simulated conflicts for a standardized deviation. (c) Standardized deviations of the conflict probability, the rank-one share, and the conflict probability given rank one; the maximum and the share above two in the text refer to the conflict probability.

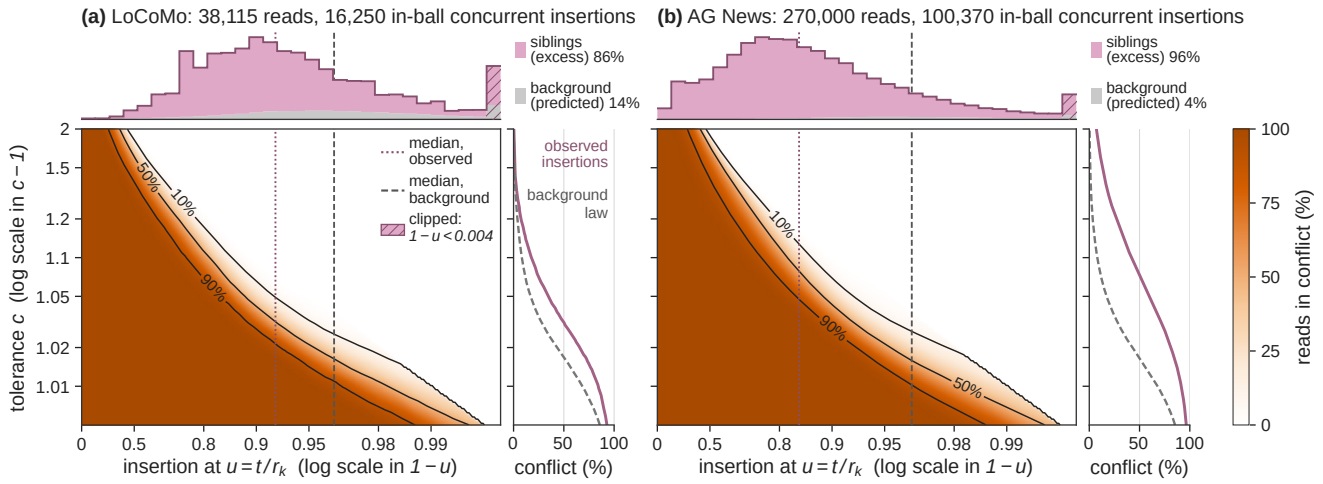


Figure A9: Probability that an insertion at normalized distance t/r_k breaks c-admissibility, averaged over the real neighbor lists of the runs of a data set, and where concurrent insertions land. Dotted lines: median position of the observed insertions and of the background law; the hatched last bin also holds the insertions with $1 - u < 0.004$. Right: conflict probability by tolerance for the observed insertions (solid) and the background law (dashed).

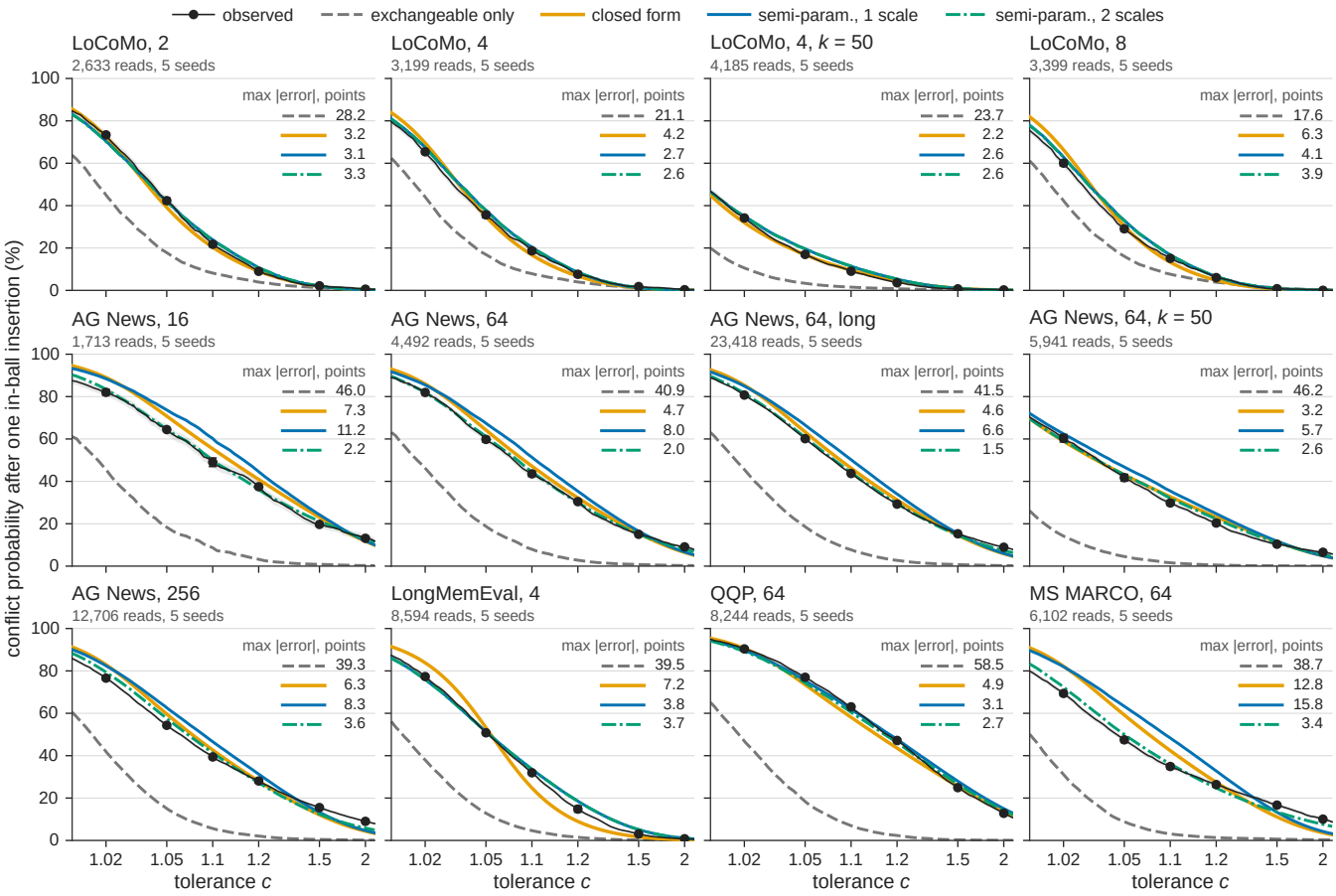


Figure A10: Conflict probability of reads with one in-ball insertion, all configurations (the paper shows two of them).

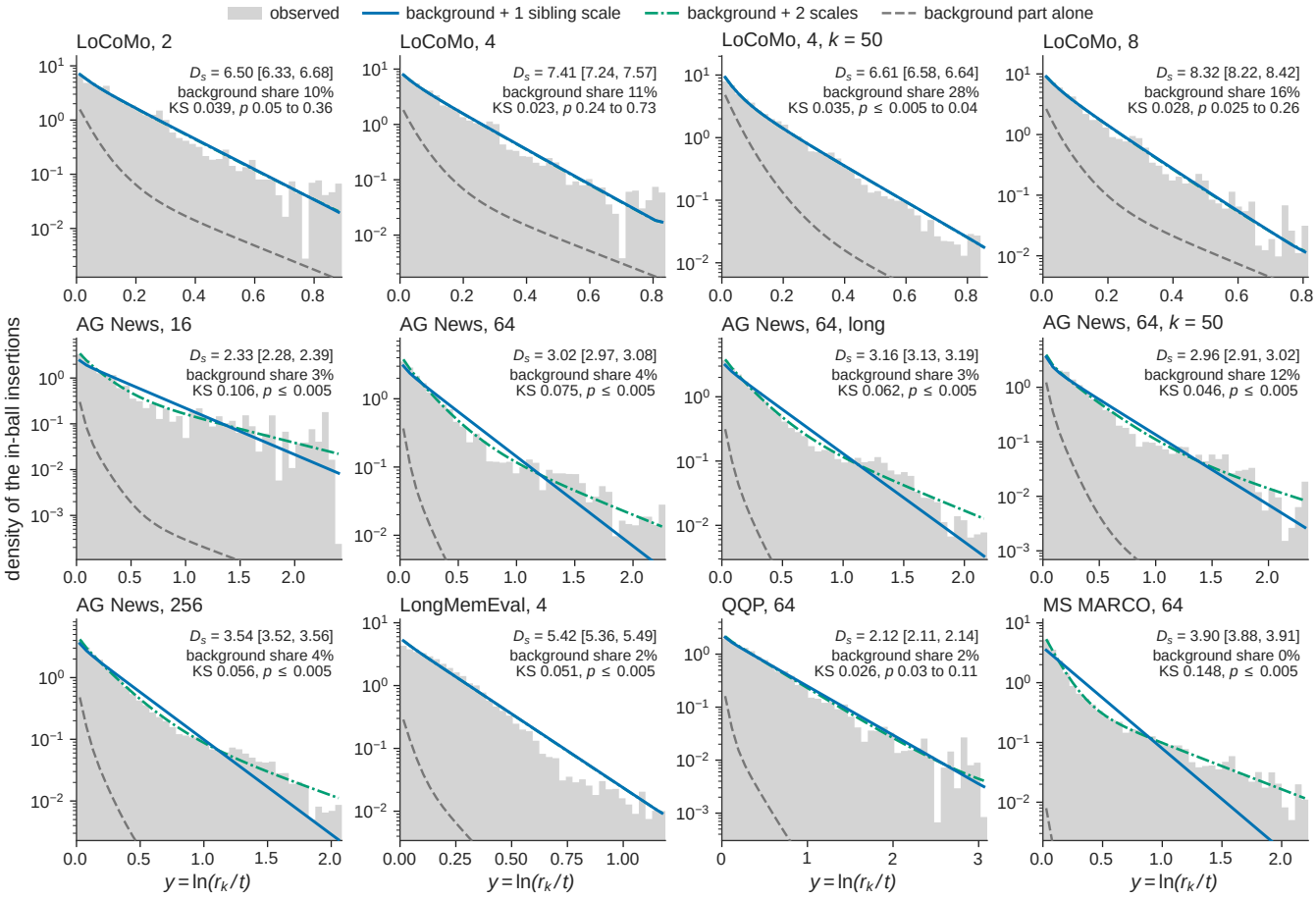


Figure A11: Positions $y = \ln(r_k/t)$ of in-ball insertions with the fitted mixtures and the background shape. Printed: KS distance (mean over seeds; the paper quotes the largest single run) with its parametric bootstrap p (range over seeds; $p \le 0.005$ is the smallest value that 200 bootstrap draws can give).

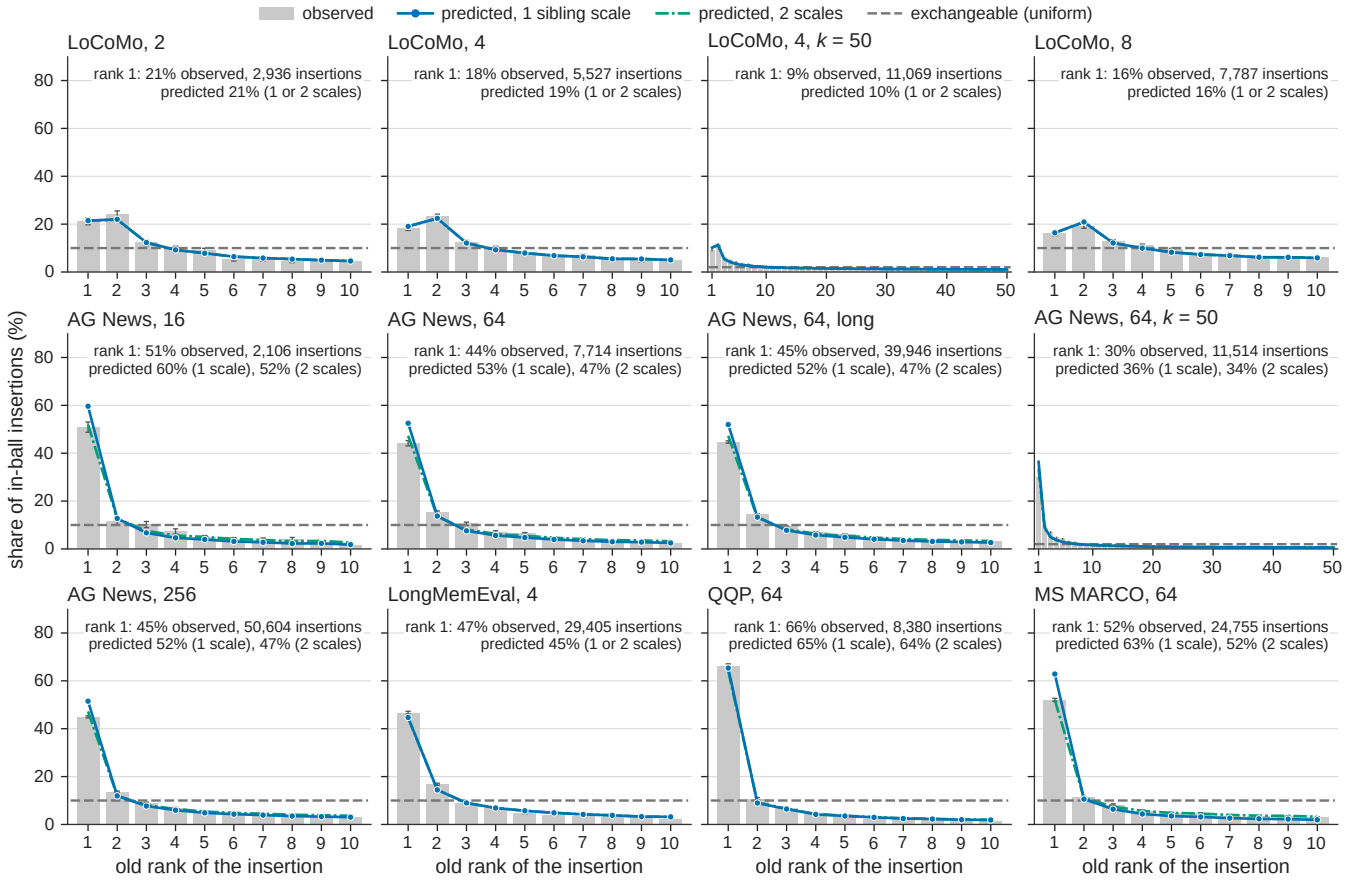


Figure A12: Old rank of in-ball insertions: observed, predicted from the position fit with one and with two sibling scales (never fitted on ranks; the text uses one scale on LoCoMo and two on AG News), and the uniform distribution of exchangeable writes.

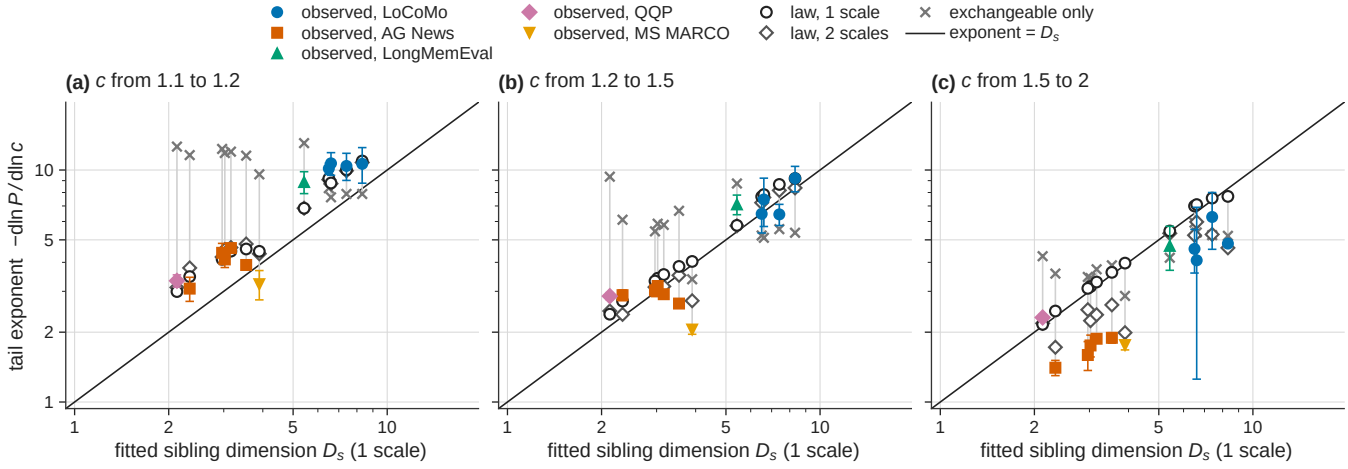


Figure A13: Tail exponents of the conflict probability against the fitted one-scale sibling dimensions.

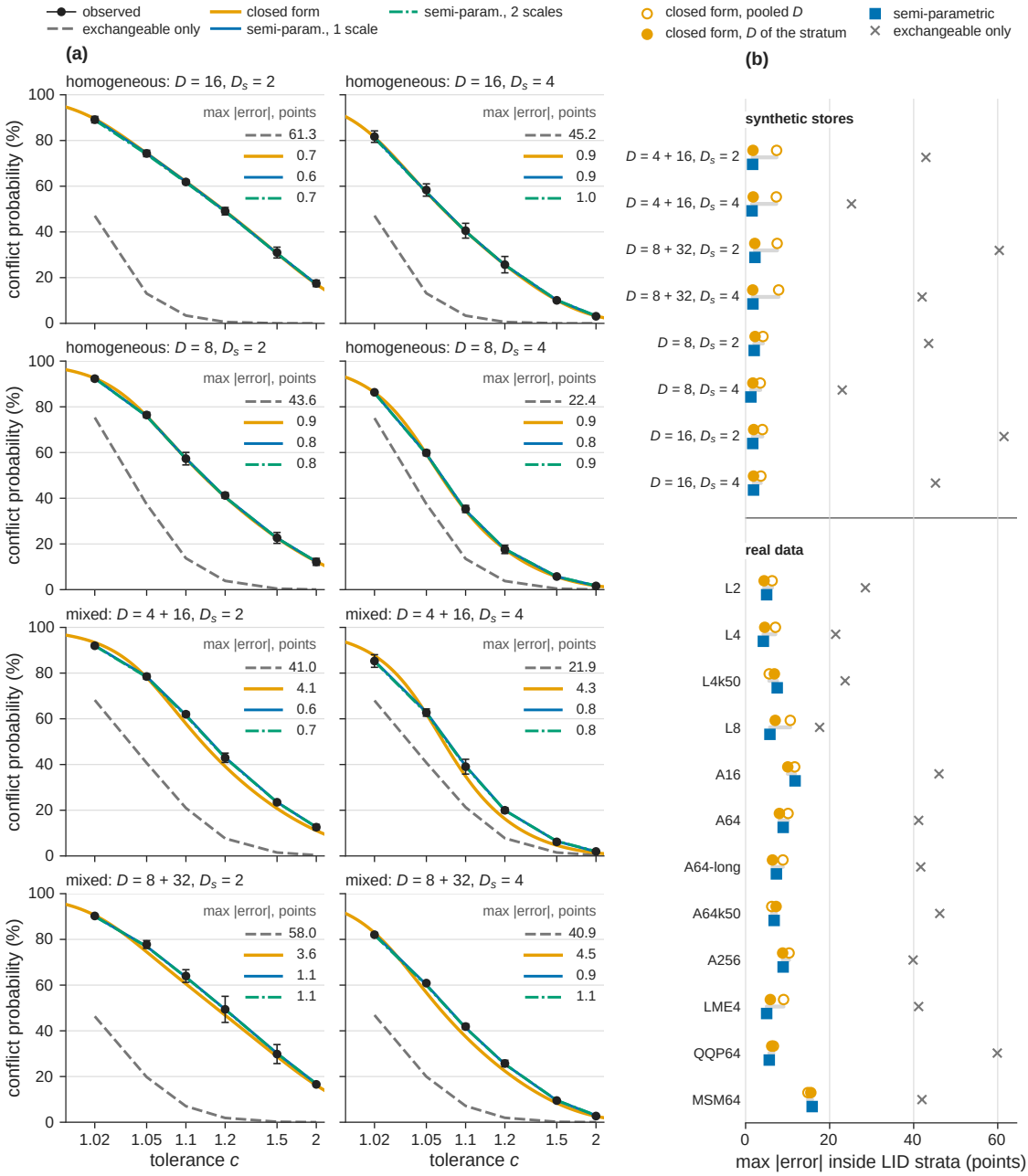


Figure A14: Heterogeneous intrinsic dimension: (a) closed form with one pooled D against the semi-parametric form on synthetic stores; (b) errors inside strata of local dimension, synthetic and real data.

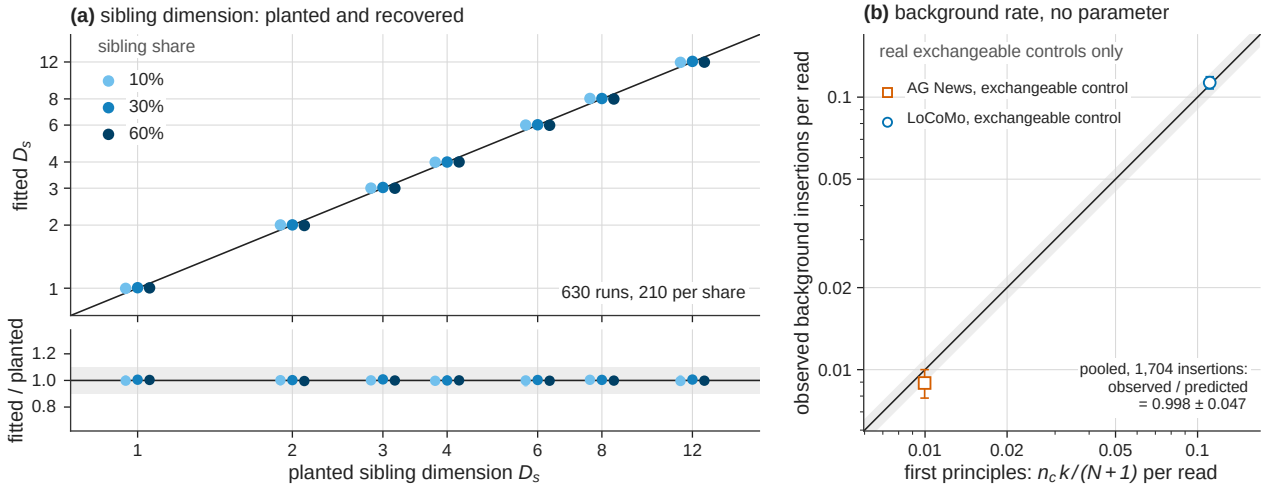


Figure A15: (a) Recovered against planted sibling dimension on synthetic streams; (b) observed against first-principles background rate on the exchangeable controls of real data.

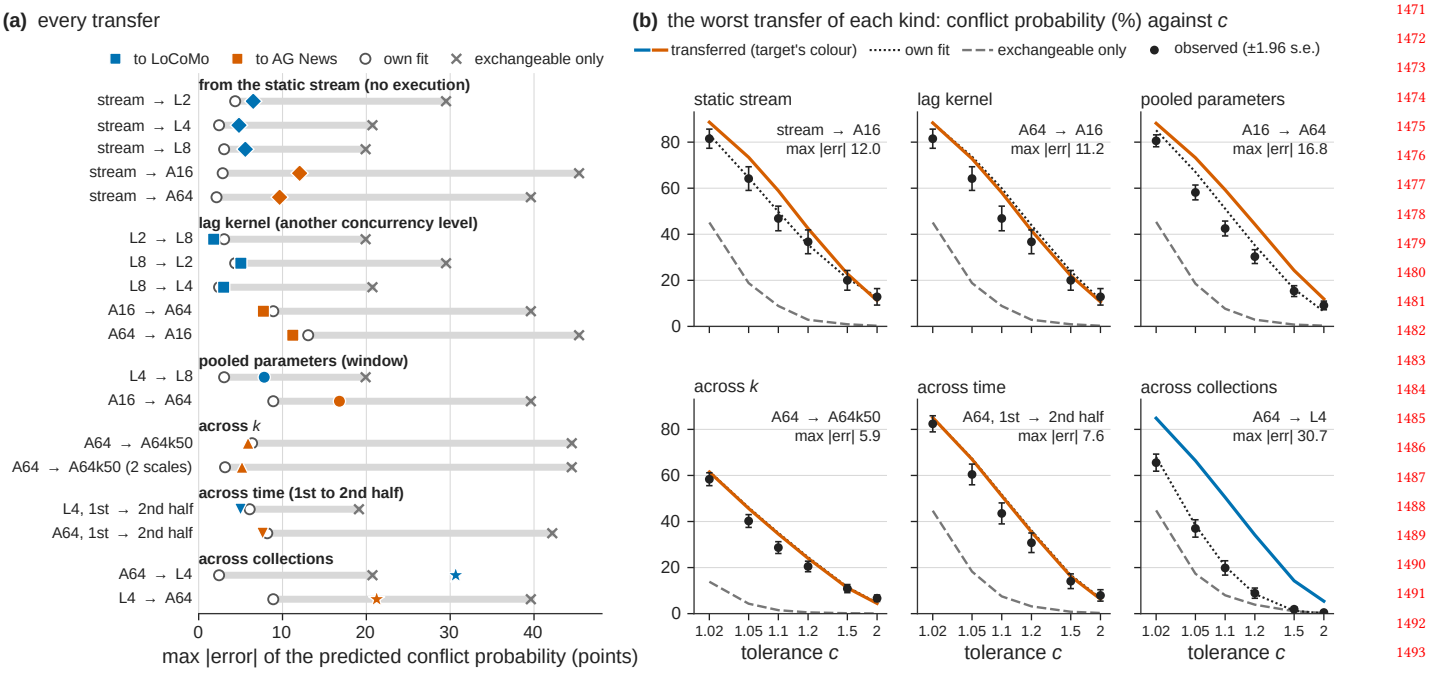


Figure A16: Predictions without fitting on the target, by transfer mode (marker shape: kind of transfer). The target's own fit uses the scales of its row: two on AG News for the static stream, those of the transfer otherwise, so one target can show two own-fit errors, one per row in which it appears.

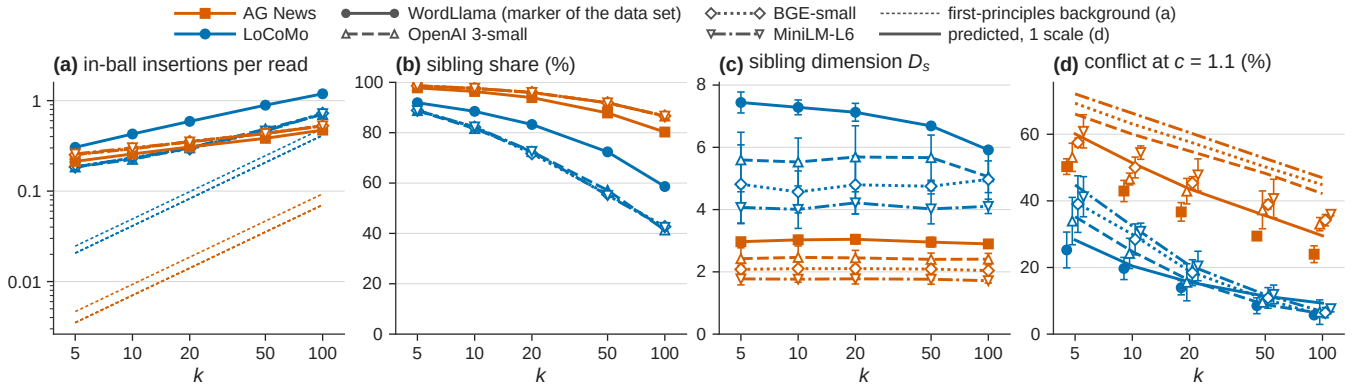


Figure A17: Other retrieval depths and embedding models. (a) In-ball insertions per read with the first-principles background; (b, c) the fitted sibling share and dimension; (d) observed (marks) and predicted (lines) conflict probability at $c = 1.1$, observed marks of the models offset slightly along k. Color: data set; line style and marker: embedding model (WordLlama: the data set's filled marker; the others: hollow markers of their own).

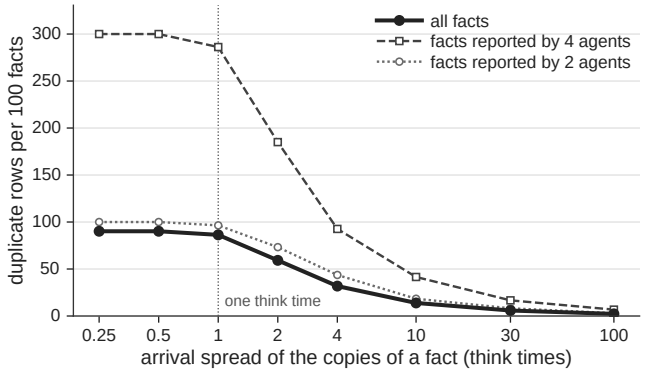


Figure A18: Unprotected check-then-insert on pgvector: duplicates against how tightly co-observations arrive.

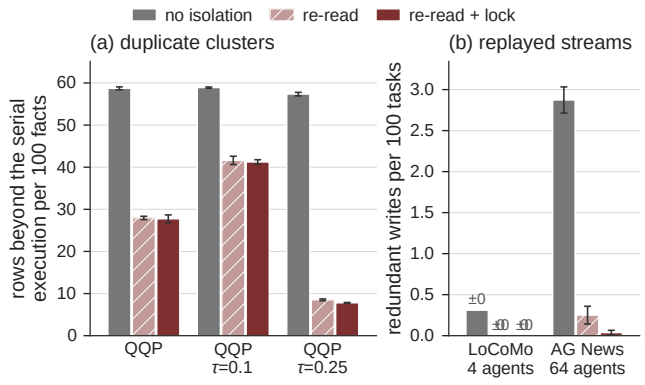


Figure A19: Real paraphrases and replayed streams.

D Additional Experiments

D.1 The race (§3.1)

Figure A20 shows the race per store at every concurrency level, Figure A18 its dependence on the arrival spread of the copies, and Figure A21 the remaining factors one at a time: thinking time, fan-out, threshold, retrieval depth, topic locality, store size, and concurrency. Of the timing and load parameters the arrival spread relative to the thinking time matters most; concurrency and the others shift the rate far less; the fan-out sets how many copies there are to collide, and the threshold does not matter while it exceeds the distance between copies. Figure A19 repeats the measurement with real duplicate questions and with replayed streams, where the auditor counts insertions made although an item within the threshold existed. Figure A22 breaks the anomalies down by workload and protocol. Through Mem0 (its default pipeline with Chroma as the store, one shared user, a target of eight calls in flight), about 379 concurrent add calls per run for 200 facts (three runs) stored 78.8 duplicates per 100 facts (serial: 0.0; with a lock around the call: 0.0). Writing first and verifying afterward is no remedy either: on the workload of Table 2 of the paper, deleting the own copy when a near duplicate with a smaller identifier is found leaves 15.9 duplicates per 100 facts, and deleting it whenever one is found leaves 0.0 and loses 48.1 facts per 100, whose copies all deleted themselves.

D.2 Correctness checks (§4)

Table A2 lists, per experiment, the number of commits on which the inequality of Theorem 4.3 was checked and the number of violations. Runs of the protocol as specified are separated from runs of variants that we weakened on purpose (no existence check although items are deleted, no commit lock or a lock narrower than the read scope, a snapshot taken before the lock is granted). Violations in the second group come from the runs without the existence check; they show that the check is not vacuous. The variants of the lock and of the snapshot pass it, since the inequality holds between the held answer and the validation read they take; when the workload exposes them, they fail through duplicate writes and fresh answers above the

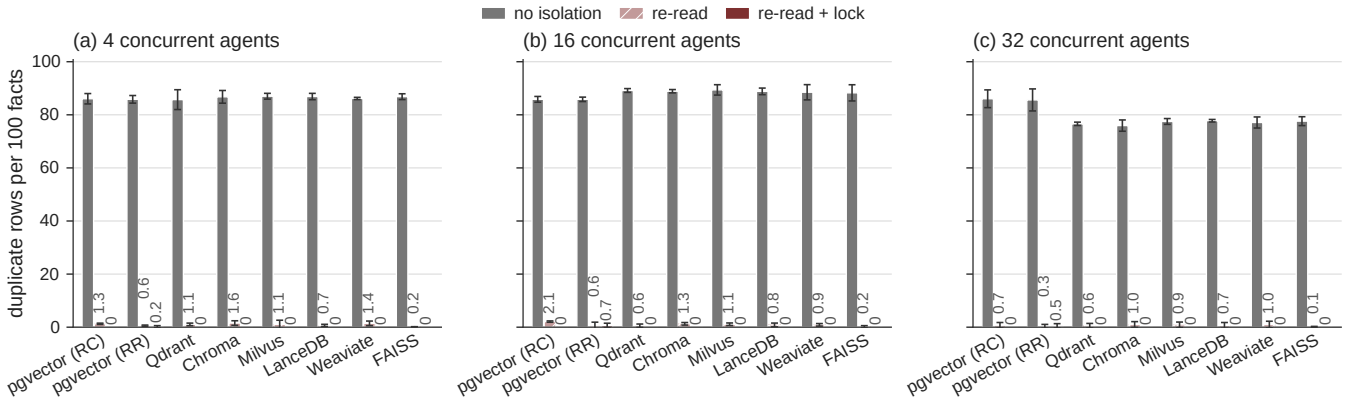


Figure A20: Duplicates per 100 facts by store and protocol.

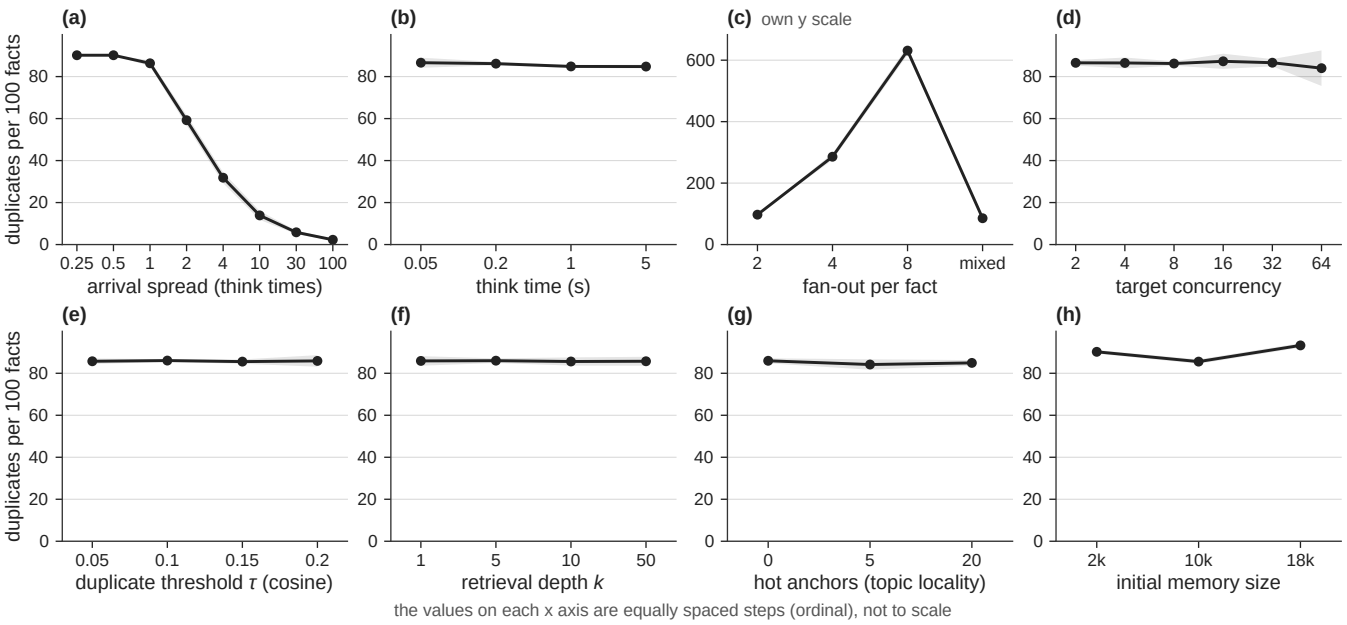


Figure A21: The race against each workload factor, one at a time (unprotected check-then-insert on pgvector; steps equally spaced; the fan-out panel has its own y scale).

Table A2: Audited commits and violations of the safety bound, per experiment, for the protocol as specified and for variants weakened on purpose (generated).

	as specified			weakened on purpose		
experiment	runs	commits	violations	runs	commits	violations
A4_anomaly_taxonomy	160	784 920	0	160	784 920	6089
B1_safety_theorem	864	4 304 232	0	–	–	–
B2_separation	96	712 368	0	–	–	–
B3_multi_read	270	1 153 035	0	–	–	–
B4_deletes_updates	36	151 110	0	36	151 110	1251
B5a_broken_variants	9	29 046	0	27	87 138	0
C10_vs_pessimistic	90	309 645	0	–	–	–
C11_starvation	48	230 901	0	–	–	–
C12_k_tau_sensitivity	216	922 428	0	–	–	–
C13_read_mix	30	66 666	0	–	–	–
C1_calls_vs_policy	321	1 400 232	0	–	–	–
C3_llm_decisions	90	332 460	0	–	–	–
C6_index_operating_points	432	1 844 856	0	–	–	–
C7_choosing_c	84	358 722	0	–	–	–
C8_think_and_arrival	120	512 460	0	–	–	–
C8b_open_loop	21	126 000	0	–	–	–
C9_scalability	24	318 000	0	–	–	–
E1_predicate_components	27	87 138	0	–	–	–
E2_lock_scope	50	256 230	0	100	512 460	0
E3_retry_strategy	72	294 232	0	–	–	–
E4_validation_budget	30	128 115	0	–	–	–
E5_isolation_of_phases	12	42 504	0	12	42 504	0
E6_per_decision_tolerance	30	106 710	0	–	–	–
F1_stores	30	90 000	0	–	–	–
F2_embedders_and_datasets	45	238 869	0	–	–	–
F3_seeds_and_variance	60	256 230	0	–	–	–
F4_client_processes	18	108 000	0	–	–	–
F5_long_run	2	150 000	0	–	–	–
F6_failure_injection	24	144 000	0	–	–	–
F7_metric_l2	8	24 000	0	–	–	–
total	3319	15 483 109	0	335	1 578 132	7340

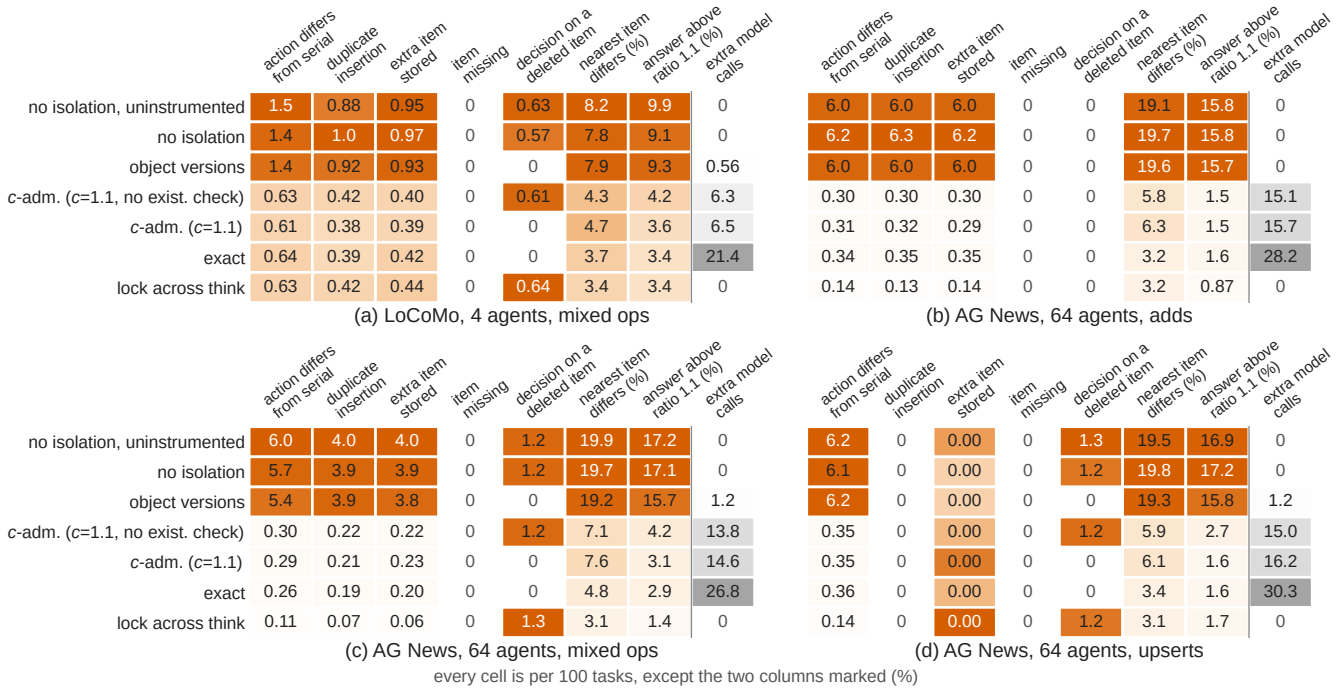


Figure A22: Anomalies by workload (adds, upserts, mixed operations) and protocol; counts per 100 tasks unless marked %, shading relative to the column maximum within a panel (0.00: a mean below 0.005 that is not zero; gray: a cost, not an anomaly). The c-admissible rows validate with an approximate read, so their answers are bounded by $c\rho$ rather than c; in the other rows, answers above the ratio come from the index itself.

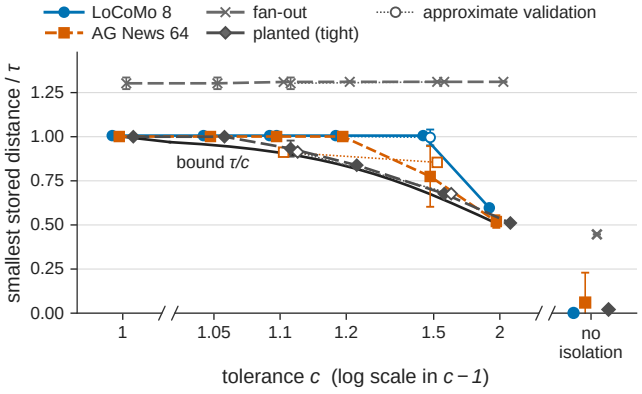


Figure A23: Smallest stored pairwise distance against the tolerance, with the bound τ/c of Proposition 4.2 for certified runs; with approximate validation reads the bound is $\tau/(c\rho)$ (§A.3). Caps: 95% Student-t intervals of the mean over three seeds, cut at 0; series are offset slightly along c so that coinciding marks of different series stay visible.

ratio (§E). Figure A23 shows the smallest stored pairwise distance against the bound τ/c of Proposition 4.2, including a synthetic stream whose siblings are planted around the threshold so that the bound is approached. The auditor computes the ratio of a held answer against the state at the start of its task. With several reads per task or with deletions, a later read can return an item inserted after that state or miss one deleted after it, so on A64multi and A64mix the share of held answers with a ratio above c is not zero even in certified runs. The safety inequality, which compares the held answer with the fresh answer in the commit state, has no violation in these runs. On LongMemEval and the question stream, which contain exact copies, a few certified answers hold an item at distance zero; the store returns that distance as zero and the auditor recomputes it from the vectors as the float32 resolution of the chord distance, so these exact answers show a large ratio. Every certified answer above c outside A64multi and A64mix is of this kind.

D.3 Cost and savings (§§6.2 to 6.4)

Table A3 extends Table 4 of the paper to all predicates and configurations, and Figure A26 extends the frontier to all workloads. Figure A24 shows the distribution of the approximation ratio of held answers per predicate against that of fresh answers; Figure A25 sweeps c on a fine grid (over the grid, extra calls fall and differing decisions do not fall as c grows) with the quantiles of the index's own ratio as reference; Figure A27 varies the index and its search

Table A3: Cost of every validation predicate on every configuration, all on pgvector (generated; eight seeds on L2, L4, and A64, three on the others). On MS MARCO, whose base of one million passages makes each run slow, only the predicates none and exact and the tolerances $c \in \{1.05, 1.1, 1.2\}$ were run.

	extra calls	dup. writes	decision differs from exact (%)					ρ	latency
predicate	/ 100 tasks	/ 100 tasks	dup	tag	link	top-3	set	p99	p99 (s)
LoCoMo, 2 agents									
no isolation	0	0.23 ± 0.03	0.23 ± 0.03	6.08 ± 0.28	5.27 ± 0.25	14.7 ± 0.5	27.4 ± 0.5	1.317 ± 0.027	0.38 ± 0.01
exact	19.3 ± 0.5	0	0	1.15 ± 0.05	0.89 ± 0.05	2.11 ± 0.06	6.97 ± 0.08	1.048 ± 0.003	0.75 ± 0.01
set-exact	19.4 ± 0.5	0	0	1.20 ± 0.06	0.90 ± 0.06	2.18 ± 0.05	7.02 ± 0.39	1.048 ± 0.004	0.75 ± 0.02
top-1	4.63 ± 0.14	0	0	4.48 ± 0.17	0.88 ± 0.04	10.1 ± 0.3	22.6 ± 0.6	1.166 ± 0.011	0.50 ± 0.01
overlap 80%	0.070 ± 0.030	0.24 ± 0.03	0.24 ± 0.03	6.02 ± 0.19	5.55 ± 0.30	14.8 ± 0.3	27.7 ± 0.5	1.330 ± 0.013	0.39 ± 0.01
overlap 90%	1.22 ± 0.08	0.23 ± 0.03	0.23 ± 0.03	5.69 ± 0.24	5.19 ± 0.18	13.9 ± 0.4	26.4 ± 0.4	1.311 ± 0.019	0.43 ± 0.01
top-3	12.1 ± 0.4	0	0	2.92 ± 0.17	0.88 ± 0.04	2.10 ± 0.07	15.0 ± 0.2	1.068 ± 0.003	0.60 ± 0.01
radius (c=1.1)	0	0.24 ± 0.03	0.24 ± 0.03	6.19 ± 0.15	5.34 ± 0.21	14.8 ± 0.3	27.5 ± 0.4	1.311 ± 0.009	0.38 ± 0.01
rank-1 (c=1.1)	2.36 ± 0.19	0	0	5.17 ± 0.16	3.06 ± 0.15	12.2 ± 0.3	24.6 ± 0.5	1.182 ± 0.014	0.44 ± 0.01
sum-dist. (c=1.1)	0.040	0.18 ± 0.06	0.18 ± 0.06	6.45 ± 0.34	5.63 ± 0.29	14.7 ± 0.3	27.2 ± 0.5	1.307 ± 0.021	0.39 ± 0.01
radius (c=1.2)	0	0.23 ± 0.03	0.23 ± 0.03	6.40 ± 0.32	5.60 ± 0.44	14.8 ± 0.2	27.6 ± 0.3	1.317 ± 0.011	0.39 ± 0.01
rank-1 (c=1.2)	1.42 ± 0.12	0	0	5.58 ± 0.26	4.07 ± 0.14	13.3 ± 0.2	26.0 ± 0.5	1.198 ± 0.006	0.43 ± 0.01
sum-dist. (c=1.2)	0	0.23 ± 0.05	0.23 ± 0.05	6.13 ± 0.19	5.54 ± 0.21	14.7 ± 0.3	27.4 ± 0.4	1.316 ± 0.012	0.38 ± 0.01
c-adm. (c=1.02)	14.5 ± 0.4	0	0	2.28 ± 0.12	0.95 ± 0.05	2.71 ± 0.13	11.8 ± 0.3	1.046 ± 0.002	0.66 ± 0.01
c-adm. (c=1.05)	9.18 ± 0.21	0	0	3.45 ± 0.18	1.38 ± 0.11	5.78 ± 0.24	18.1 ± 0.5	1.050 ± 0.002	0.58 ± 0.01
c-adm. (c=1.1)	4.66 ± 0.28	0	0	4.71 ± 0.25	2.45 ± 0.11	9.65 ± 0.33	22.2 ± 0.4	1.093 ± 0.002	0.51 ± 0.01
c-adm. (c=1.2)	2.21 ± 0.14	0	0	5.17 ± 0.12	3.48 ± 0.22	12.3 ± 0.3	24.7 ± 0.3	1.161 ± 0.007	0.45 ± 0.01
c-adm. (c=1.5)	0.57 ± 0.07	0	0	5.80 ± 0.39	5.00 ± 0.23	14.0 ± 0.4	26.6 ± 0.3	1.260 ± 0.013	0.40
certified (c=1.05)	10.2 ± 0.2	0	0	3.09 ± 0.11	0.72 ± 0.09	4.66 ± 0.30	16.4 ± 0.4	1.043	0.59 ± 0.01
certified (c=1.1)	5.38 ± 0.36	0	0	4.33 ± 0.17	1.92 ± 0.16	8.83 ± 0.25	21.5 ± 0.3	1.085 ± 0.001	0.53 ± 0.01
certified (c=1.2)	2.18 ± 0.15	0	0	5.34 ± 0.17	3.53 ± 0.13	12.2 ± 0.2	25.1 ± 0.3	1.153 ± 0.005	0.45 ± 0.01
oracle (duplicate)	0.22 ± 0.05	0	0	6.28 ± 0.23	5.21 ± 0.22	14.5 ± 0.3	27.2 ± 0.3	1.294 ± 0.020	0.39 ± 0.01
oracle (link)	4.74 ± 0.26	0	0	4.47 ± 0.19	0.89 ± 0.04	9.88 ± 0.41	22.7 ± 0.4	1.161 ± 0.012	0.49 ± 0.01
oracle (set)	19.3 ± 0.6	0	0	1.18 ± 0.06	0.90 ± 0.05	2.10 ± 0.06	7.11 ± 0.29	1.048 ± 0.003	0.74 ± 0.02
oracle (tag)	4.77 ± 0.48	0.17 ± 0.02	0.17 ± 0.02	1.21 ± 0.06	3.74 ± 0.13	11.6 ± 0.3	22.7 ± 0.4	1.257 ± 0.011	0.50 ± 0.01
oracle (top-3)	11.9 ± 0.2	0	0	2.87 ± 0.14	0.86 ± 0.04	2.04 ± 0.08	15.1 ± 0.4	1.070 ± 0.002	0.60 ± 0.01
object versions	0	0.25 ± 0.04	0.25 ± 0.04	6.19 ± 0.29	5.64 ± 0.24	15.2 ± 0.3	27.7 ± 0.4	1.319 ± 0.024	0.38 ± 0.01
LoCoMo, 4 agents									
no isolation	0	0.30 ± 0.01	0.30 ± 0.01	10.5 ± 0.4	7.72 ± 0.21	21.0 ± 0.3	38.1 ± 0.4	1.390 ± 0.008	0.40 ± 0.03
exact	35.7 ± 0.9	0	0	1.28 ± 0.08	0.93 ± 0.06	2.19 ± 0.09	7.03 ± 0.11	1.052 ± 0.005	0.86 ± 0.02
set-exact	36.7 ± 0.8	0	0	1.27 ± 0.07	0.91 ± 0.06	2.15 ± 0.08	7.10 ± 0.28	1.050 ± 0.004	0.87 ± 0.02
top-1	6.92 ± 0.21	0	0	7.93 ± 0.34	0.88 ± 0.05	14.4 ± 0.3	32.1 ± 0.6	1.191 ± 0.008	0.53 ± 0.01
overlap 80%	1.29 ± 0.11	0.29 ± 0.01	0.29 ± 0.01	9.96 ± 0.34	7.45 ± 0.21	20.0 ± 0.3	37.0 ± 0.3	1.377 ± 0.008	0.44 ± 0.02
overlap 90%	7.07 ± 0.33	0.25 ± 0.03	0.25 ± 0.03	7.91 ± 0.25	5.73 ± 0.19	15.7 ± 0.4	32.6 ± 0.4	1.321 ± 0.005	0.55 ± 0.01
top-3	19.7 ± 0.6	0	0	4.70 ± 0.12	0.87 ± 0.06	2.21 ± 0.11	20.9 ± 0.8	1.075 ± 0.004	0.70 ± 0.01
radius (c=1.1)	0	0.30 ± 0.01	0.30 ± 0.01	10.6 ± 0.4	7.91 ± 0.25	21.3 ± 0.5	37.8 ± 0.4	1.383 ± 0.004	0.39 ± 0.01
rank-1 (c=1.1)	3.66 ± 0.20	0	0	8.91 ± 0.32	4.19 ± 0.18	17.6 ± 0.4	35.4 ± 0.3	1.213 ± 0.007	0.48 ± 0.02
sum-dist. (c=1.1)	0.12 ± 0.03	0.27 ± 0.02	0.27 ± 0.02	10.5 ± 0.4	7.81 ± 0.11	21.1 ± 0.3	37.9 ± 0.4	1.380 ± 0.008	0.39 ± 0.01
radius (c=1.2)	0	0.31 ± 0.01	0.31 ± 0.01	10.5 ± 0.3	7.91 ± 0.25	20.7 ± 0.6	37.9 ± 0.4	1.397 ± 0.009	0.38 ± 0.01
rank-1 (c=1.2)	1.79 ± 0.12	0	0	9.65 ± 0.32	5.87 ± 0.30	19.0 ± 0.7	36.6 ± 0.6	1.226 ± 0.009	0.48 ± 0.06
sum-dist. (c=1.2)	0	0.30 ± 0.01	0.30 ± 0.01	10.6 ± 0.3	8.01 ± 0.15	21.2 ± 0.3	38.3 ± 0.3	1.388 ± 0.007	0.38 ± 0.01
c-adm. (c=1.02)	25.8 ± 0.4	0	0	3.29 ± 0.14	0.98 ± 0.04	3.07 ± 0.14	15.9 ± 0.3	1.048 ± 0.003	0.78 ± 0.02
c-adm. (c=1.05)	14.7 ± 0.3	0	0	6.04 ± 0.37	1.53 ± 0.09	8.21 ± 0.24	26.4 ± 0.7	1.053 ± 0.003	0.65 ± 0.02
c-adm. (c=1.1)	7.88 ± 0.32	0	0	7.88 ± 0.37	3.15 ± 0.11	13.9 ± 0.3	31.6 ± 0.6	1.094 ± 0.003	0.56 ± 0.01
c-adm. (c=1.2)	3.28 ± 0.20	0	0	9.31 ± 0.43	5.28 ± 0.17	17.9 ± 0.3	35.0 ± 0.5	1.169 ± 0.003	0.48 ± 0.01
c-adm. (c=1.5)	0.57 ± 0.10	0	0	10.3 ± 0.2	7.41 ± 0.19	20.5 ± 0.3	37.5 ± 0.5	1.313 ± 0.008	0.40 ± 0.01
certified (c=1.05)	15.9 ± 0.4	0	0	5.55 ± 0.39	0.87 ± 0.09	7.46 ± 0.28	25.3 ± 0.4	1.044	0.66 ± 0.01
certified (c=1.1)	8.56 ± 0.34	0	0	7.78 ± 0.33	2.77 ± 0.23	13.5 ± 0.4	31.3 ± 0.6	1.090 ± 0.001	0.56 ± 0.01
certified (c=1.2)	3.49 ± 0.15	0	0	9.16 ± 0.50	5.13 ± 0.19	18.3 ± 0.4	35.7 ± 0.4	1.171 ± 0.002	0.48 ± 0.01
oracle (duplicate)	0.34 ± 0.08	0	0	10.3 ± 0.3	7.54 ± 0.41	21.0 ± 0.5	37.9 ± 0.5	1.354 ± 0.013	0.40 ± 0.01
oracle (link)	7.41 ± 0.25	0	0	7.48 ± 0.43	0.90 ± 0.06	14.4 ± 0.3	32.1 ± 0.3	1.192 ± 0.004	0.55 ± 0.01
oracle (set)	36.0 ± 0.8	0	0	1.23 ± 0.10	0.88 ± 0.06	2.15 ± 0.08	6.95 ± 0.17	1.050 ± 0.003	0.86 ± 0.01
oracle (tag)	8.99 ± 0.27	0.22 ± 0.03	0.22 ± 0.03	1.22 ± 0.07	5.57 ± 0.27	15.9 ± 0.2	30.6 ± 0.5	1.342 ± 0.019	0.54 ± 0.01
oracle (top-3)	20.3 ± 0.4	0	0	4.81 ± 0.15	0.85 ± 0.06	2.18 ± 0.10	21.3 ± 0.3	1.076 ± 0.006	0.68 ± 0.01
object versions	0	0.31	0.31	10.6 ± 0.3	8.07 ± 0.22	21.1 ± 0.2	38.7 ± 1.0	1.390 ± 0.007	0.38

Table A3 (continued)

	extra calls	dup. writes	decision differs from exact (%)					ρ	latency
predicate	/ 100 tasks	/ 100 tasks	dup	tag	link	top-3	set	p99	p99 (s)
LoCoMo, 8 agents									
no isolation	0	0.31	0.31	13.2 ± 1.4	9.08 ± 0.21	25.0 ± 0.7	46.3 ± 0.8	1.405 ± 0.009	0.34 ± 0.02
exact	51.4 ± 1.3	0	0	1.31 ± 0.30	0.88 ± 0.14	2.17 ± 0.44	7.07 ± 0.44	1.049 ± 0.009	0.98 ± 0.01
set-exact	51.0 ± 1.2	0	0	1.22 ± 0.26	0.98 ± 0.10	2.24 ± 0.10	7.00 ± 0.25	1.056 ± 0.008	1.01 ± 0.01
top-1	8.42 ± 0.68	0	0	10.7 ± 0.6	0.89 ± 0.19	17.1 ± 0.8	39.9 ± 0.7	1.197 ± 0.008	0.56 ± 0.06
overlap 80%	2.93 ± 0.62	0.29 ± 0.04	0.29 ± 0.04	12.0 ± 0.7	7.78 ± 0.30	23.2 ± 0.8	44.4 ± 0.9	1.380 ± 0.015	0.48 ± 0.02
overlap 90%	12.7 ± 0.9	0.16	0.16	9.22 ± 0.60	5.81 ± 0.36	16.8 ± 0.9	37.6 ± 1.0	1.311 ± 0.028	0.60 ± 0.08
top-3	25.1 ± 1.8	0	0	6.26 ± 0.97	0.88 ± 0.06	2.15 ± 0.19	27.7 ± 1.4	1.088 ± 0.021	0.74 ± 0.08
radius (c=1.1)	0.013 ± 0.057	0.31	0.31	13.6 ± 1.3	9.22 ± 0.40	25.2 ± 0.3	46.3 ± 1.4	1.416 ± 0.026	0.34 ± 0.01
rank-1 (c=1.1)	4.38 ± 0.41	0	0	11.6 ± 1.1	4.59 ± 0.40	21.5 ± 2.0	43.1 ± 2.1	1.223 ± 0.011	0.48
sum-dist. (c=1.1)	0.12	0.28	0.28	13.6 ± 0.6	9.14 ± 0.98	25.0 ± 0.2	46.0 ± 0.2	1.395 ± 0.016	0.35 ± 0.01
radius (c=1.2)	0	0.31	0.31	13.7 ± 0.4	9.21 ± 0.36	25.0 ± 1.0	46.7 ± 2.6	1.392 ± 0.013	0.35 ± 0.03
rank-1 (c=1.2)	2.13 ± 0.09	0	0	12.5 ± 0.5	6.95 ± 0.39	23.0 ± 0.4	44.4 ± 1.3	1.239 ± 0.024	0.47 ± 0.10
sum-dist. (c=1.2)	0	0.31	0.31	13.5 ± 0.7	8.98 ± 0.15	24.9 ± 0.9	45.4 ± 3.7	1.408 ± 0.001	0.35 ± 0.01
c-adm. (c=1.02)	35.1 ± 0.5	0	0	4.33 ± 0.60	0.98 ± 0.19	3.75 ± 1.13	20.8 ± 2.1	1.047 ± 0.006	0.89 ± 0.05
c-adm. (c=1.05)	18.1 ± 0.2	0	0	8.26 ± 0.78	1.85 ± 0.36	10.2 ± 0.9	33.2 ± 2.3	1.054 ± 0.003	0.72 ± 0.01
c-adm. (c=1.1)	9.10 ± 0.55	0	0	10.5 ± 1.2	3.75 ± 0.23	16.7 ± 1.0	39.4 ± 1.1	1.097 ± 0.001	0.56 ± 0.01
c-adm. (c=1.2)	3.83 ± 0.46	0	0	12.2 ± 0.8	6.02 ± 0.61	20.9 ± 0.7	43.3 ± 1.0	1.179 ± 0.004	0.47 ± 0.04
c-adm. (c=1.5)	0.76 ± 0.15	0	0	13.0 ± 2.6	8.62 ± 1.19	24.4 ± 0.6	45.9 ± 1.5	1.317 ± 0.033	0.36 ± 0.01
certified (c=1.05)	19.5 ± 1.0	0	0	7.78 ± 0.89	1.02 ± 0.20	9.22 ± 0.69	32.6 ± 0.7	1.046 ± 0.003	0.69 ± 0.04
certified (c=1.1)	10.1 ± 1.2	0	0	10.6 ± 0.6	3.44 ± 0.34	16.4 ± 0.5	39.5 ± 1.7	1.092 ± 0.003	0.57 ± 0.04
certified (c=1.2)	3.86 ± 0.10	0	0	12.2 ± 1.0	6.30 ± 0.29	22.0 ± 0.8	43.4 ± 0.8	1.173 ± 0.003	0.46 ± 0.04
oracle (duplicate)	0.24 ± 0.10	0	0	13.1 ± 1.5	8.90 ± 0.61	24.8 ± 0.5	45.5 ± 0.7	1.372 ± 0.039	0.41 ± 0.15
oracle (link)	8.74 ± 0.84	0	0	9.92 ± 0.26	0.87 ± 0.19	17.3 ± 1.1	39.7 ± 1.1	1.190 ± 0.007	0.56 ± 0.04
oracle (set)	51.7 ± 1.7	0	0	1.23 ± 0.40	0.91 ± 0.25	2.15 ± 0.14	6.99 ± 0.29	1.049 ± 0.004	1.02 ± 0.08
oracle (tag)	11.7 ± 0.9	0.24 ± 0.17	0.24 ± 0.17	1.21 ± 0.32	6.34 ± 0.44	18.6 ± 1.9	37.3 ± 0.7	1.360 ± 0.018	0.55 ± 0.04
oracle (top-3)	25.2 ± 0.7	0	0	6.07 ± 0.83	0.86 ± 0.15	2.23 ± 0.48	28.3 ± 1.8	1.082 ± 0.010	0.74 ± 0.06
object versions	0	0.31	0.31	13.8 ± 1.7	9.16 ± 0.21	24.5 ± 0.6	45.7 ± 0.3	1.403 ± 0.025	0.34 ± 0.07
AG News, 16 agents									
no isolation	0	1.25 ± 0.07	1.25 ± 0.07	2.00 ± 0.16	6.38 ± 0.33	10.6 ± 0.3	19.2 ± 0.4	2.003 ± 0.069	0.39 ± 0.02
exact	6.61 ± 0.29	0.057 ± 0.029	0.057 ± 0.029	1.88 ± 0.11	3.25 ± 0.17	6.18 ± 0.06	13.4 ± 0.3	1.162 ± 0.024	0.55 ± 0.04
set-exact	6.83 ± 0.43	0.050	0.050	1.90 ± 0.11	3.35 ± 0.19	6.33 ± 0.20	13.6 ± 0.1	1.172 ± 0.026	0.55 ± 0.02
top-1	3.23 ± 0.27	0.050	0.050	1.95 ± 0.05	3.33 ± 0.22	7.66 ± 0.40	16.4 ± 0.3	1.197 ± 0.017	0.47 ± 0.02
overlap 80%	0.15 ± 0.05	1.20 ± 0.10	1.20 ± 0.10	1.98 ± 0.09	6.32 ± 0.39	10.5 ± 0.6	19.0 ± 0.8	1.945 ± 0.097	0.39 ± 0.02
overlap 90%	0.79 ± 0.23	0.96 ± 0.08	0.96 ± 0.08	2.00 ± 0.07	5.87 ± 0.47	9.89 ± 0.62	18.5 ± 0.9	1.711 ± 0.027	0.41 ± 0.03
top-3	4.63 ± 0.29	0.043 ± 0.029	0.043 ± 0.029	1.92 ± 0.21	3.31 ± 0.15	6.18 ± 0.35	15.3 ± 0.2	1.175 ± 0.009	0.50 ± 0.03
radius (c=1.1)	0.007 ± 0.029	1.24 ± 0.18	1.24 ± 0.18	2.01 ± 0.10	6.42 ± 0.33	10.4 ± 0.8	19.3 ± 0.3	2.017 ± 0.146	0.38 ± 0.01
rank-1 (c=1.1)	2.42 ± 0.20	0.057 ± 0.029	0.057 ± 0.029	1.89 ± 0.13	3.98 ± 0.27	8.25 ± 0.34	17.1 ± 0.1	1.208 ± 0.010	0.45 ± 0.02
sum-dist. (c=1.1)	0.30 ± 0.11	0.96 ± 0.03	0.96 ± 0.03	1.93 ± 0.09	6.06 ± 0.30	10.2 ± 0.5	18.7 ± 0.4	1.681 ± 0.072	0.39 ± 0.02
radius (c=1.2)	0	1.22 ± 0.07	1.22 ± 0.07	2.02 ± 0.21	6.42 ± 0.23	10.6 ± 0.3	19.2 ± 0.5	1.957 ± 0.082	0.39 ± 0.01
rank-1 (c=1.2)	2.13 ± 0.34	0.070 ± 0.043	0.070 ± 0.043	1.92 ± 0.06	4.42 ± 0.20	8.82 ± 0.22	17.5 ± 0.3	1.226 ± 0.052	0.45 ± 0.01
sum-dist. (c=1.2)	0.11 ± 0.03	1.16 ± 0.13	1.16 ± 0.13	1.96 ± 0.25	6.27 ± 0.31	10.5 ± 0.3	19.0 ± 0.9	1.866 ± 0.246	0.39 ± 0.02
c-adm. (c=1.02)	5.95 ± 0.34	0.043 ± 0.029	0.043 ± 0.029	1.84 ± 0.03	3.30 ± 0.10	6.42 ± 0.18	14.3 ± 0.1	1.158 ± 0.029	0.53 ± 0.03
c-adm. (c=1.05)	4.10 ± 0.49	0.050	0.050	1.93 ± 0.11	3.61 ± 0.24	7.13 ± 0.17	15.5 ± 0.4	1.154 ± 0.014	0.49 ± 0.01
c-adm. (c=1.1)	3.51 ± 0.25	0.043 ± 0.029	0.043 ± 0.029	1.90 ± 0.11	3.80 ± 0.23	7.68 ± 0.26	16.5 ± 0.4	1.166 ± 0.027	0.48 ± 0.02
c-adm. (c=1.2)	2.38 ± 0.25	0.057 ± 0.029	0.057 ± 0.029	1.97 ± 0.14	4.26 ± 0.13	8.67 ± 1.00	17.0 ± 0.2	1.184 ± 0.018	0.46 ± 0.02
c-adm. (c=1.5)	1.39 ± 0.23	0.100 ± 0.050	0.100 ± 0.050	1.95 ± 0.23	5.08 ± 0.41	9.45 ± 0.59	18.2 ± 0.2	1.298 ± 0.037	0.43 ± 0.04
certified (c=1.05)	7.08 ± 0.13	0	0	0.75 ± 0.05	1.58 ± 0.19	4.70 ± 0.14	12.9 ± 0.2	1.037 ± 0.002	0.65 ± 0.02
certified (c=1.1)	4.89 ± 0.13	0	0	1.13 ± 0.04	2.48 ± 0.17	6.24 ± 0.18	14.8 ± 0.5	1.074	0.63 ± 0.01
certified (c=1.2)	3.15 ± 0.28	0	0	1.53 ± 0.04	3.54 ± 0.18	7.55 ± 0.28	16.1 ± 0.3	1.127 ± 0.004	0.58 ± 0.04
oracle (duplicate)	1.16 ± 0.09	0.050	0.050	1.96 ± 0.13	5.14 ± 0.30	9.57 ± 0.14	18.1 ± 0.3	1.320 ± 0.018	0.42 ± 0.01
oracle (link)	3.29 ± 0.40	0.033 ± 0.038	0.033 ± 0.038	1.92 ± 0.28	3.36 ± 0.27	7.83 ± 0.23	16.6 ± 0.3	1.205 ± 0.021	0.48 ± 0.04
oracle (set)	7.17 ± 0.28	0.057 ± 0.029	0.057 ± 0.029	1.84 ± 0.31	3.28 ± 0.23	6.18 ± 0.11	13.4 ± 0.3	1.160 ± 0.044	0.55 ± 0.03
oracle (tag)	0.12 ± 0.06	1.21 ± 0.09	1.21 ± 0.09	1.91 ± 0.08	6.30 ± 0.16	10.6 ± 0.1	19.2 ± 0.3	1.970 ± 0.166	0.39
oracle (top-3)	4.77 ± 0.37	0.050	0.050	1.87 ± 0.12	3.30 ± 0.06	6.27 ± 0.12	15.2 ± 0.3	1.160 ± 0.023	0.52 ± 0.02
object versions	0	1.24 ± 0.10	1.24 ± 0.10	1.99 ± 0.10	6.38 ± 0.12	10.6 ± 0.4	19.2 ± 0.4	1.996 ± 0.082	0.39 ± 0.02

Table A3 (continued)

	extra calls	dup. writes	decision differs from exact (%)					ρ	latency
predicate	/ 100 tasks	/ 100 tasks	dup	tag	link	top-3	set	p99	p99 (s)
AG News, 64 agents									
no isolation	0	2.91 ± 0.05	2.91 ± 0.05	2.09 ± 0.04	12.6 ± 0.1	19.9 ± 0.2	31.0 ± 0.5	3.201 ± 0.052	0.40 ± 0.12
exact	25.1 ± 0.2	0.045 ± 0.008	0.045 ± 0.008	1.93 ± 0.03	3.34 ± 0.04	6.29 ± 0.07	13.5 ± 0.1	1.170 ± 0.013	0.88 ± 0.02
set-exact	25.0 ± 0.4	0.043 ± 0.009	0.043 ± 0.009	1.90 ± 0.04	3.33 ± 0.04	6.33 ± 0.25	13.5 ± 0.3	1.165 ± 0.008	0.87 ± 0.02
top-1	9.64 ± 0.16	0.045 ± 0.008	0.045 ± 0.008	2.02 ± 0.04	3.31 ± 0.06	11.6 ± 0.2	23.9 ± 0.2	1.301 ± 0.010	0.56 ± 0.01
overlap 80%	1.22 ± 0.08	2.61 ± 0.07	2.61 ± 0.07	2.10 ± 0.06	11.7 ± 0.1	19.1 ± 0.1	30.6 ± 0.1	2.943 ± 0.141	0.40 ± 0.01
overlap 90%	4.62 ± 0.07	2.07 ± 0.06	2.07 ± 0.06	2.01 ± 0.08	10.1 ± 0.2	16.9 ± 0.2	28.7 ± 0.4	2.509 ± 0.101	0.49 ± 0.01
top-3	16.4 ± 0.2	0.048 ± 0.006	0.048 ± 0.006	2.01 ± 0.05	3.32 ± 0.09	6.31 ± 0.13	19.6 ± 0.3	1.196 ± 0.004	0.69 ± 0.02
radius (c=1.1)	0.073 ± 0.009	2.87 ± 0.03	2.87 ± 0.03	2.08 ± 0.06	12.6 ± 0.1	19.9 ± 0.1	31.1 ± 0.1	3.154 ± 0.045	0.35 ± 0.02
rank-1 (c=1.1)	6.80 ± 0.15	0.074 ± 0.004	0.074 ± 0.004	2.04 ± 0.05	5.77 ± 0.17	14.1 ± 0.2	26.1 ± 0.2	1.325 ± 0.012	0.51 ± 0.01
sum-dist. (c=1.1)	1.00 ± 0.06	2.24 ± 0.06	2.24 ± 0.06	2.06 ± 0.06	11.7 ± 0.2	19.1 ± 0.2	30.7 ± 0.2	2.467 ± 0.092	0.37 ± 0.01
radius (c=1.2)	0	2.90 ± 0.04	2.90 ± 0.04	2.10 ± 0.05	12.6 ± 0.2	19.8 ± 0.2	31.1 ± 0.3	3.188 ± 0.054	0.37 ± 0.02
rank-1 (c=1.2)	5.45 ± 0.08	0.090 ± 0.015	0.090 ± 0.015	2.06 ± 0.05	7.19 ± 0.09	15.2 ± 0.1	27.1 ± 0.2	1.333 ± 0.010	0.49 ± 0.01
sum-dist. (c=1.2)	0.14 ± 0.02	2.76 ± 0.03	2.76 ± 0.03	2.09 ± 0.03	12.5 ± 0.2	19.7 ± 0.2	31.0 ± 0.1	3.018 ± 0.100	0.35 ± 0.01
c-adm. (c=1.02)	20.9 ± 0.3	0.048 ± 0.006	0.048 ± 0.006	1.91 ± 0.06	3.38 ± 0.06	6.85 ± 0.10	16.6 ± 0.2	1.178 ± 0.011	0.79 ± 0.02
c-adm. (c=1.05)	15.3 ± 0.3	0.045 ± 0.008	0.045 ± 0.008	1.98 ± 0.05	4.00 ± 0.07	8.78 ± 0.16	20.3 ± 0.4	1.172 ± 0.010	0.70 ± 0.01
c-adm. (c=1.1)	10.6 ± 0.4	0.045 ± 0.008	0.045 ± 0.008	2.00 ± 0.04	5.17 ± 0.10	11.4 ± 0.3	23.4 ± 0.4	1.178 ± 0.008	0.62 ± 0.02
c-adm. (c=1.2)	7.24 ± 0.13	0.048 ± 0.006	0.048 ± 0.006	2.05 ± 0.03	6.69 ± 0.10	14.0 ± 0.2	26.0 ± 0.2	1.194 ± 0.002	0.54 ± 0.01
c-adm. (c=1.5)	3.63 ± 0.13	0.19 ± 0.03	0.19 ± 0.03	2.04 ± 0.04	8.96 ± 0.13	16.4 ± 0.3	28.3 ± 0.2	1.389 ± 0.007	0.46 ± 0.01
certified (c=1.05)	18.1 ± 0.3	0	0	0.79 ± 0.02	1.97 ± 0.03	6.18 ± 0.13	17.4 ± 0.3	1.043	3.62 ± 0.06
certified (c=1.1)	12.8 ± 0.2	0	0	1.20 ± 0.02	3.76 ± 0.06	9.90 ± 0.11	21.8 ± 0.1	1.087 ± 0.001	3.53 ± 0.05
certified (c=1.2)	8.07 ± 0.17	0	0	1.57 ± 0.03	6.16 ± 0.16	13.3 ± 0.3	25.3 ± 0.1	1.173 ± 0.001	2.62 ± 0.09
oracle (duplicate)	2.81 ± 0.10	0.042 ± 0.009	0.042 ± 0.009	2.06 ± 0.05	9.43 ± 0.15	17.1 ± 0.4	28.8 ± 0.2	1.496 ± 0.011	0.44 ± 0.01
oracle (link)	9.95 ± 0.33	0.048 ± 0.006	0.048 ± 0.006	2.03 ± 0.02	3.31 ± 0.05	11.8 ± 0.1	24.1 ± 0.2	1.298 ± 0.004	0.55 ± 0.01
oracle (set)	25.1 ± 0.1	0.040 ± 0.009	0.040 ± 0.009	1.92 ± 0.04	3.30 ± 0.06	6.26 ± 0.07	13.5 ± 0.1	1.166 ± 0.010	0.87 ± 0.02
oracle (tag)	0.28 ± 0.04	2.85 ± 0.04	2.85 ± 0.04	1.88 ± 0.06	12.5 ± 0.1	19.7 ± 0.2	30.8 ± 0.3	3.135 ± 0.076	0.36 ± 0.01
oracle (top-3)	16.4 ± 0.2	0.048 ± 0.011	0.048 ± 0.011	1.93 ± 0.06	3.32 ± 0.05	6.31 ± 0.08	19.7 ± 0.2	1.198 ± 0.004	0.70 ± 0.02
object versions	0	2.92 ± 0.05	2.92 ± 0.05	2.08 ± 0.10	12.6 ± 0.2	20.0 ± 0.3	31.0 ± 0.5	3.200 ± 0.053	0.35 ± 0.01
AG News, 64 agents, k=50									
no isolation	0	2.89 ± 0.14	2.89 ± 0.14	0.93 ± 0.14	10.9 ± 0.5	17.5 ± 0.4	53.5 ± 0.4	3.104 ± 0.054	0.32 ± 0.01
exact	38.9 ± 0.9	0.020	0.020	0.89 ± 0.08	1.57 ± 0.07	3.68 ± 0.19	34.2 ± 0.4	1.078 ± 0.010	1.10 ± 0.06
set-exact	39.3 ± 0.4	0.020	0.020	0.95 ± 0.16	1.54 ± 0.20	3.66 ± 0.25	34.1 ± 0.2	1.076 ± 0.009	1.12 ± 0.03
top-1	9.91 ± 0.20	0.020	0.020	0.92 ± 0.04	1.53 ± 0.14	9.30 ± 0.23	48.3 ± 0.6	1.243 ± 0.025	0.55 ± 0.01
overlap 80%	0.013 ± 0.029	2.86 ± 0.19	2.86 ± 0.19	0.94 ± 0.03	10.9 ± 0.2	17.5 ± 0.2	53.6 ± 0.5	3.124 ± 0.091	0.33 ± 0.05
overlap 90%	0.28 ± 0.09	2.81 ± 0.32	2.81 ± 0.32	0.87 ± 0.13	10.9 ± 0.5	17.3 ± 0.3	53.6 ± 0.3	3.012 ± 0.376	0.33 ± 0.01
top-3	16.7 ± 0.6	0.020	0.020	0.93 ± 0.04	1.50 ± 0.19	3.62 ± 0.23	44.9 ± 1.0	1.121 ± 0.029	0.69
radius (c=1.1)	0	2.87 ± 0.16	2.87 ± 0.16	0.93 ± 0.04	10.9 ± 0.2	17.4 ± 0.1	53.5 ± 0.5	3.113 ± 0.063	0.33 ± 0.03
rank-1 (c=1.1)	7.05 ± 0.17	0.030	0.030	0.91 ± 0.03	4.20 ± 0.06	11.8 ± 0.7	49.7 ± 0.8	1.265 ± 0.028	0.48
sum-dist. (c=1.1)	0	2.88 ± 0.12	2.88 ± 0.12	0.90 ± 0.11	11.0 ± 0.3	17.5 ± 0.1	53.6 ± 0.2	3.118 ± 0.021	0.32 ± 0.02
radius (c=1.2)	0	2.87 ± 0.23	2.87 ± 0.23	0.92 ± 0.16	10.9 ± 0.4	17.4 ± 0.4	53.5 ± 0.4	3.130 ± 0.205	0.34 ± 0.05
rank-1 (c=1.2)	5.11 ± 0.55	0.077 ± 0.014	0.077 ± 0.014	0.89 ± 0.07	5.42 ± 0.36	12.8 ± 0.9	50.7 ± 0.6	1.283 ± 0.021	0.47 ± 0.02
sum-dist. (c=1.2)	0	2.89 ± 0.13	2.89 ± 0.13	0.91 ± 0.09	11.0 ± 0.4	17.6 ± 0.2	53.7 ± 0.4	3.146 ± 0.040	0.33 ± 0.04
c-adm. (c=1.02)	24.5 ± 0.7	0.020	0.020	0.89 ± 0.06	1.58 ± 0.16	4.12 ± 0.20	42.1 ± 0.7	1.076 ± 0.001	0.87 ± 0.05
c-adm. (c=1.05)	16.3 ± 0.8	0.020	0.020	0.92 ± 0.09	2.10 ± 0.16	6.13 ± 0.34	45.8 ± 0.7	1.075 ± 0.006	0.73 ± 0.06
c-adm. (c=1.1)	11.5 ± 0.3	0.020	0.020	0.90 ± 0.07	3.34 ± 0.44	8.92 ± 0.09	48.0 ± 1.5	1.096 ± 0.002	0.62 ± 0.07
c-adm. (c=1.2)	7.52 ± 0.44	0.020	0.020	0.90 ± 0.05	5.04 ± 0.20	11.5 ± 0.2	49.7 ± 0.6	1.175 ± 0.002	0.53 ± 0.04
c-adm. (c=1.5)	3.24 ± 0.30	0.16 ± 0.04	0.16 ± 0.04	0.89 ± 0.07	7.19 ± 0.21	14.1 ± 0.3	51.2 ± 0.5	1.359 ± 0.027	0.44 ± 0.02
certified (c=1.05)	18.0 ± 0.8	0	0	0.56 ± 0.03	1.39 ± 0.10	4.99 ± 0.10	42.1 ± 0.3	1.043 ± 0.001	3.85 ± 0.19
certified (c=1.1)	12.1 ± 0.3	0	0	0.65 ± 0.11	2.89 ± 0.19	8.31 ± 0.13	44.5 ± 2.0	1.087 ± 0.001	3.40 ± 0.12
certified (c=1.2)	7.13 ± 0.72	0	0	0.77 ± 0.04	4.88 ± 0.07	11.4 ± 0.2	48.4 ± 0.2	1.170 ± 0.003	2.50 ± 0.05
oracle (duplicate)	2.77 ± 0.29	0.013 ± 0.029	0.013 ± 0.029	0.89 ± 0.07	7.80 ± 0.07	14.7 ± 0.3	51.5 ± 2.1	1.455 ± 0.036	0.44 ± 0.02
oracle (link)	9.92 ± 0.23	0.020	0.020	0.92 ± 0.06	1.53 ± 0.09	9.22 ± 0.65	48.4 ± 0.4	1.245 ± 0.026	0.54 ± 0.05
oracle (set)	38.3 ± 0.7	0.020	0.020	0.89 ± 0.15	1.50 ± 0.06	3.53 ± 0.14	34.1 ± 0.4	1.070 ± 0.014	1.09 ± 0.10
oracle (tag)	0.11 ± 0.07	2.84 ± 0.13	2.84 ± 0.13	0.90 ± 0.05	10.9 ± 0.1	17.5 ± 0.1	53.4 ± 0.3	3.102 ± 0.082	0.32 ± 0.04
oracle (top-3)	16.6 ± 1.1	0.013 ± 0.029	0.013 ± 0.029	0.93 ± 0.19	1.50 ± 0.06	3.57 ± 0.04	45.2 ± 0.7	1.117 ± 0.007	0.69 ± 0.02
object versions	0	2.88 ± 0.16	2.88 ± 0.16	0.92 ± 0.04	11.0 ± 0.3	17.5 ± 0.3	53.5 ± 0.3	3.132 ± 0.053	0.31 ± 0.01

Table A3 (continued)

	extra calls	dup. writes	decision differs from exact (%)					ρ	latency
predicate	/ 100 tasks	/ 100 tasks	dup	tag	link	top-3	set	p99	p99 (s)
LongMemEval, 4 agents									
no isolation	0	1.20 ± 0.07	1.20 ± 0.07	24.4 ± 0.6	34.2 ± 0.8	48.5 ± 0.4	63.2 ± 0.4	3.350 ± 0.348	0.38 ± 0.01
exact	70.7 ± 1.3	0.013 ± 0.029	0.013 ± 0.029	3.32 ± 0.43	2.81 ± 0.52	6.10 ± 0.27	15.0 ± 0.5	1.300 ± 0.112	1.31 ± 0.01
set-exact	70.0 ± 1.4	0.020	0.020	3.36 ± 0.46	2.84 ± 0.09	6.01 ± 0.31	15.0 ± 0.6	1.305 ± 0.076	1.30 ± 0.01
top-1	31.7 ± 0.4	0.013 ± 0.029	0.013 ± 0.029	10.7 ± 0.5	2.78 ± 0.36	23.9 ± 1.2	44.8 ± 1.3	1.429 ± 0.059	0.75 ± 0.03
overlap 80%	7.96 ± 0.80	0.84 ± 0.15	0.84 ± 0.15	18.5 ± 1.4	27.6 ± 0.5	42.6 ± 0.1	59.3 ± 0.6	2.755 ± 0.139	0.58 ± 0.03
overlap 90%	22.7 ± 0.5	0.51 ± 0.15	0.51 ± 0.15	11.9 ± 0.5	19.5 ± 1.5	33.3 ± 1.1	51.1 ± 1.0	2.131 ± 0.135	0.76 ± 0.04
top-3	50.7 ± 0.2	0.020	0.020	6.45 ± 0.79	2.77 ± 0.12	6.15 ± 0.11	31.6 ± 0.6	1.318 ± 0.042	1.00 ± 0.08
radius (c=1.1)	0.100 ± 0.050	1.16 ± 0.10	1.16 ± 0.10	24.3 ± 0.5	34.4 ± 0.3	48.3 ± 1.0	62.9 ± 0.5	3.245 ± 0.243	0.38 ± 0.02
rank-1 (c=1.1)	17.6 ± 0.2	0.043 ± 0.029	0.043 ± 0.029	15.3 ± 0.9	15.1 ± 0.4	35.6 ± 0.6	53.6 ± 0.6	1.459 ± 0.149	0.64 ± 0.04
sum-dist. (c=1.1)	1.44 ± 0.18	0.65 ± 0.16	0.65 ± 0.16	22.9 ± 0.6	32.4 ± 1.0	47.2 ± 1.4	63.1 ± 3.3	2.237 ± 0.259	0.44 ± 0.03
radius (c=1.2)	0.013 ± 0.029	1.14 ± 0.03	1.14 ± 0.03	24.1 ± 0.6	34.6 ± 1.1	48.9 ± 0.5	64.0 ± 2.2	3.335 ± 0.319	0.38 ± 0.01
rank-1 (c=1.2)	10.7 ± 0.3	0.073 ± 0.014	0.073 ± 0.014	19.0 ± 2.2	22.6 ± 1.5	40.9 ± 0.9	57.2 ± 0.4	1.494 ± 0.075	0.57 ± 0.04
sum-dist. (c=1.2)	0.11 ± 0.07	1.07 ± 0.04	1.07 ± 0.04	24.4 ± 0.7	34.3 ± 0.4	48.5 ± 0.4	63.2 ± 0.7	3.060 ± 0.431	0.39 ± 0.01
c-adm. (c=1.02)	58.5 ± 0.8	0.013 ± 0.029	0.013 ± 0.029	5.33 ± 0.77	3.08 ± 0.31	7.48 ± 0.34	24.0 ± 0.6	1.312 ± 0.006	1.21 ± 0.05
c-adm. (c=1.05)	42.5 ± 1.3	0.007 ± 0.029	0.007 ± 0.029	8.49 ± 0.35	5.45 ± 0.43	15.3 ± 0.5	36.3 ± 1.2	1.310 ± 0.011	1.04 ± 0.06
c-adm. (c=1.1)	28.8 ± 1.4	0.013 ± 0.029	0.013 ± 0.029	11.9 ± 0.9	10.8 ± 1.4	25.5 ± 0.6	45.8 ± 1.3	1.312 ± 0.008	0.87 ± 0.02
c-adm. (c=1.2)	14.1 ± 0.7	0.027 ± 0.014	0.027 ± 0.014	17.2 ± 0.2	20.4 ± 1.2	37.5 ± 1.1	55.2 ± 0.7	1.323 ± 0.022	0.67 ± 0.02
c-adm. (c=1.5)	2.80 ± 0.29	0.10 ± 0.06	0.10 ± 0.06	22.5 ± 0.9	30.9 ± 0.4	46.1 ± 1.2	61.2 ± 0.9	1.486 ± 0.006	0.47 ± 0.01
certified (c=1.05)	48.6 ± 0.3	0	0	7.56 ± 0.20	4.35 ± 0.58	14.4 ± 0.9	35.9 ± 0.7	1.050	2.33 ± 0.32
certified (c=1.1)	31.0 ± 0.1	0	0	11.7 ± 0.4	11.1 ± 0.8	26.3 ± 0.9	47.0 ± 0.2	1.099 ± 0.001	2.07 ± 0.17
certified (c=1.2)	14.7 ± 0.9	0.020	0.020	17.6 ± 0.1	21.0 ± 0.4	38.9 ± 0.4	57.6 ± 1.2	1.198	1.50 ± 0.14
oracle (duplicate)	1.03 ± 0.23	0.020	0.020	23.4 ± 1.2	33.3 ± 0.9	47.4 ± 0.9	62.3 ± 0.9	1.859 ± 0.026	0.41 ± 0.02
oracle (link)	30.9 ± 1.7	0.020	0.020	10.9 ± 0.5	2.80 ± 0.29	24.2 ± 0.4	45.0 ± 0.7	1.453 ± 0.069	0.75 ± 0.02
oracle (set)	70.4 ± 1.2	0.013 ± 0.029	0.013 ± 0.029	3.58 ± 0.34	2.91 ± 0.20	6.13 ± 0.24	15.0 ± 0.2	1.311 ± 0.006	1.29 ± 0.06
oracle (tag)	18.6 ± 0.3	0.67 ± 0.16	0.67 ± 0.16	3.34 ± 0.53	21.4 ± 0.5	36.3 ± 1.0	52.5 ± 0.9	2.371 ± 0.378	0.61 ± 0.02
oracle (top-3)	49.8 ± 1.3	0.020	0.020	6.30 ± 0.91	2.80 ± 0.16	5.97 ± 0.17	31.1 ± 0.9	1.311 ± 0.036	1.01 ± 0.01
object versions	0	1.17 ± 0.20	1.17 ± 0.20	24.6 ± 0.7	34.9 ± 0.6	48.7 ± 0.8	63.3 ± 0.9	3.387 ± 0.328	0.39 ± 0.05
QQP, 64 agents									
no isolation	0	5.33 ± 0.04	5.33 ± 0.04	3.60 ± 0.17	25.6 ± 0.4	34.2 ± 0.4	47.0 ± 0.2	4.050 ± 0.110	0.38 ± 0.10
exact	26.1 ± 1.3	0.13 ± 0.01	0.13 ± 0.01	1.93 ± 0.12	8.29 ± 0.19	14.2 ± 0.2	26.8 ± 0.2	1.405 ± 0.049	0.58 ± 0.06
set-exact	25.9 ± 0.3	0.13 ± 0.04	0.13 ± 0.04	1.96 ± 0.13	8.29 ± 0.09	14.2 ± 0.5	26.9 ± 0.5	1.411 ± 0.035	0.59 ± 0.02
top-1	17.3 ± 0.4	0.14 ± 0.03	0.14 ± 0.03	2.48 ± 0.11	8.27 ± 0.37	18.1 ± 0.5	33.7 ± 0.4	1.454 ± 0.039	0.55 ± 0.03
overlap 80%	1.10 ± 0.11	5.04 ± 0.07	5.04 ± 0.07	3.47 ± 0.27	24.5 ± 0.4	33.1 ± 0.4	46.4 ± 0.3	3.608 ± 0.059	0.41 ± 0.01
overlap 90%	1.85 ± 0.23	4.87 ± 0.21	4.87 ± 0.21	3.48 ± 0.14	23.9 ± 0.3	32.6 ± 0.3	46.0 ± 0.3	3.558 ± 0.065	0.43 ± 0.02
top-3	21.2 ± 0.9	0.14 ± 0.10	0.14 ± 0.10	2.16 ± 0.04	8.24 ± 0.69	14.3 ± 0.5	30.2 ± 0.6	1.438 ± 0.013	0.56 ± 0.04
radius (c=1.1)	0.093 ± 0.080	5.51 ± 0.84	5.51 ± 0.84	3.68 ± 0.23	25.3 ± 0.4	33.9 ± 0.2	46.7 ± 0.3	4.044 ± 0.072	0.38 ± 0.03
rank-1 (c=1.1)	13.8 ± 0.2	0.14 ± 0.07	0.14 ± 0.07	2.70 ± 0.14	11.6 ± 0.4	21.4 ± 0.7	36.5 ± 0.4	1.456 ± 0.086	0.53 ± 0.02
sum-dist. (c=1.1)	1.01 ± 0.07	4.49 ± 0.06	4.49 ± 0.06	3.44 ± 0.07	24.4 ± 0.3	33.1 ± 0.5	46.7 ± 2.0	3.126 ± 0.031	0.39 ± 0.03
radius (c=1.2)	0.067 ± 0.087	5.28 ± 0.18	5.28 ± 0.18	3.70 ± 0.22	25.6 ± 0.3	34.1 ± 0.4	47.1 ± 0.3	4.051 ± 0.102	0.36 ± 0.06
rank-1 (c=1.2)	11.4 ± 0.4	0.22 ± 0.09	0.22 ± 0.09	2.86 ± 0.06	14.0 ± 0.8	23.5 ± 0.2	38.3 ± 0.5	1.469 ± 0.047	0.53 ± 0.01
sum-dist. (c=1.2)	0.057 ± 0.080	5.18 ± 0.50	5.18 ± 0.50	3.69 ± 0.04	25.4 ± 0.5	33.9 ± 0.4	46.9 ± 0.5	3.903 ± 0.386	0.38 ± 0.05
c-adm. (c=1.02)	23.2 ± 0.5	0.14 ± 0.07	0.14 ± 0.07	2.03 ± 0.18	8.33 ± 0.12	14.6 ± 0.3	28.8 ± 0.3	1.423 ± 0.010	0.57 ± 0.01
c-adm. (c=1.05)	20.3 ± 0.5	0.12 ± 0.09	0.12 ± 0.09	2.19 ± 0.25	9.08 ± 0.17	16.2 ± 0.4	31.2 ± 0.2	1.426 ± 0.003	0.58 ± 0.03
c-adm. (c=1.1)	16.6 ± 0.3	0.12 ± 0.06	0.12 ± 0.06	2.45 ± 0.28	10.5 ± 0.3	19.1 ± 0.4	34.1 ± 0.3	1.411 ± 0.035	0.57 ± 0.06
c-adm. (c=1.2)	12.8 ± 0.1	0.14 ± 0.03	0.14 ± 0.03	2.79 ± 0.09	13.4 ± 0.3	22.5 ± 0.5	37.1 ± 0.1	1.404 ± 0.046	0.55 ± 0.02
c-adm. (c=1.5)	7.03 ± 0.27	0.51 ± 0.10	0.51 ± 0.10	3.09 ± 0.13	18.7 ± 0.3	27.9 ± 0.2	41.9 ± 0.5	1.477 ± 0.014	0.50 ± 0.02
certified (c=1.05)	29.2 ± 0.6	0	0	1.10 ± 0.10	3.00 ± 0.16	7.84 ± 0.20	20.0 ± 0.2	1.046 ± 0.001	4.95 ± 0.23
certified (c=1.1)	21.7 ± 0.4	0	0	1.72 ± 0.11	6.96 ± 0.13	14.0 ± 0.2	27.0 ± 0.7	1.091 ± 0.001	4.45 ± 1.25
certified (c=1.2)	15.3 ± 0.4	0.023 ± 0.014	0.023 ± 0.014	2.19 ± 0.16	11.1 ± 0.3	19.3 ± 0.1	32.2 ± 0.3	1.173 ± 0.002	3.44 ± 0.13
oracle (duplicate)	5.22 ± 0.23	0.13 ± 0.06	0.13 ± 0.06	3.21 ± 0.14	20.2 ± 0.8	29.2 ± 0.2	42.9 ± 0.3	1.780 ± 0.026	0.49 ± 0.01
oracle (link)	17.0 ± 0.6	0.13	0.13	2.50 ± 0.12	8.32 ± 0.44	18.2 ± 0.5	33.8 ± 0.3	1.450 ± 0.008	0.56 ± 0.02
oracle (set)	25.9 ± 1.0	0.13 ± 0.04	0.13 ± 0.04	1.88 ± 0.01	8.19 ± 0.10	14.0 ± 0.2	26.8 ± 0.5	1.424 ± 0.027	0.59 ± 0.04
oracle (tag)	1.65 ± 0.14	4.95 ± 0.05	4.95 ± 0.05	1.91 ± 0.20	24.1 ± 0.1	32.7 ± 1.0	45.4 ± 0.4	3.650 ± 0.119	0.44 ± 0.10
oracle (top-3)	21.7 ± 0.1	0.13 ± 0.09	0.13 ± 0.09	2.20 ± 0.28	8.41 ± 0.10	14.2 ± 0.5	30.0 ± 0.7	1.414 ± 0.013	0.56 ± 0.01
object versions	0	5.58 ± 1.02	5.58 ± 1.02	3.63 ± 0.22	25.6 ± 0.1	34.2 ± 0.4	46.9 ± 0.1	4.028	0.40 ± 0.05
MS MARCO, 64 agents									
no isolation	0	7.03 ± 0.09	7.03 ± 0.09	n/a	32.3 ± 0.4	42.6 ± 1.3	58.9 ± 0.9	5.684 ± 0.164	0.51 ± 0.05
exact	51.5 ± 0.9	0.36 ± 0.08	0.36 ± 0.08	n/a	11.0 ± 0.6	18.9 ± 1.1	33.9 ± 0.2	1.453 ± 0.022	1.10 ± 0.04
c-adm. (c=1.05)	28.7 ± 0.6	0.38 ± 0.10	0.38 ± 0.10	n/a	13.3 ± 0.6	24.9 ± 0.4	46.0 ± 1.2	1.438 ± 0.049	0.87 ± 0.05
c-adm. (c=1.1)	21.2 ± 0.1	0.35 ± 0.05	0.35 ± 0.05	n/a	16.1 ± 0.4	29.1 ± 0.3	49.5 ± 0.7	1.446 ± 0.023	0.77 ± 0.01
c-adm. (c=1.2)	15.2 ± 0.1	0.39 ± 0.06	0.39 ± 0.06	n/a	19.6 ± 0.3	33.1 ± 1.0	51.7 ± 0.4	1.448 ± 0.048	0.68 ± 0.02

Table A4: Two language models, GPT-4o mini (gpt-4o-mini-2024-07-18) and Qwen2.5-7B-Instruct, decide on L4, L8, and LongMemEval, five seeds per cell; changed decisions from a replay of 2,000 tasks per run, F1 and judge accuracy of the downstream questions on the final memory of every seed (generated; the questions exist for L4 and LongMemEval, so L8 has none).

predicate	extra calls / 100	prompt tok. / task	compl. tok. / task	p50 (s)	decision changed (%)	index floor (%)	F1	judge acc.	ΔF1 vs. exact
GPT-4o mini, LoCoMo, 4 agents									
no isolation	0	921 ± 2	23.1 ± 0.1	0.72 ± 0.02	3.88 ± 1.92	1.50 ± 0.38	0.352 ± 0.002	0.613 ± 0.011	-1.58 [-1.71, -1.45]
exact	23.9 ± 0.3	1140 ± 4	28.6 ± 0.2	0.86 ± 0.02	1.47 ± 0.42	1.47 ± 0.42	0.368 ± 0.001	0.652 ± 0.014	ref.
overlap 90%	2.41 ± 0.52	943 ± 1	23.6 ± 0.1	0.74 ± 0.01	3.76 ± 0.30	2.02 ± 0.77	0.365 ± 0.001	0.645 ± 0.019	-0.30 [-0.36, -0.24]
c-adm. (c=1.05)	12.0 ± 1.0	1029 ± 9	25.7 ± 0.3	0.78 ± 0.04	1.58 ± 0.17	1.22 ± 0.40	0.367 ± 0.002	0.647 ± 0.015	-0.10 [-0.13, -0.07]
c-adm. (c=1.1)	7.24 ± 0.38	990 ± 7	24.7 ± 0.2	0.77 ± 0.02	1.50 ± 0.40	1.42 ± 0.96	0.368 ± 0.002	0.648 ± 0.013	-0.05 [-0.08, -0.02]
c-adm. (c=1.2)	4.71 ± 0.27	964 ± 3	24.0 ± 0.1	0.76 ± 0.01	2.21 ± 0.43	1.34 ± 0.38	0.365 ± 0.002	0.644 ± 0.015	-0.31 [-0.37, -0.26]
GPT-4o mini, LoCoMo, 8 agents									
no isolation	0	919 ± 3	23.0 ± 0.2	0.73 ± 0.02	4.48 ± 0.41	1.59 ± 0.21	n/a	n/a	n/a
exact	39.7 ± 1.3	1282 ± 12	32.1 ± 0.5	0.98 ± 0.03	1.87 ± 0.45	1.87 ± 0.45	n/a	n/a	n/a
overlap 90%	5.08 ± 0.65	964 ± 7	24.1 ± 0.1	0.74 ± 0.01	4.19 ± 1.68	1.03 ± 1.10	n/a	n/a	n/a
c-adm. (c=1.05)	18.6 ± 1.2	1091 ± 15	27.4 ± 0.1	0.85 ± 0.02	1.72 ± 0.47	1.55 ± 0.44	n/a	n/a	n/a
c-adm. (c=1.1)	11.6 ± 0.8	1028 ± 6	25.7 ± 0.2	0.78 ± 0.01	2.10 ± 0.49	1.19 ± 0.18	n/a	n/a	n/a
c-adm. (c=1.2)	7.11 ± 0.47	984 ± 6	24.5 ± 0.3	0.78 ± 0.03	2.54 ± 0.79	1.27 ± 0.44	n/a	n/a	n/a
GPT-4o mini, LongMemEval, 4 agents									
no isolation	0	922 ± 3	23.1 ± 0.2	0.74 ± 0.01	4.28 ± 1.20	1.42 ± 0.40	0.334 ± 0.054	0.692 ± 0.135	+0.20 [-0.62, +1.14]
exact	18.6 ± 0.4	1092 ± 6	27.3 ± 0.1	0.83 ± 0.01	1.44 ± 0.57	1.44 ± 0.57	0.332 ± 0.047	0.738 ± 0.145	ref.
overlap 90%	1.93 ± 0.10	939 ± 2	23.3 ± 0.3	0.73 ± 0.02	2.80 ± 0.54	1.18 ± 0.14	0.331 ± 0.048	0.754 ± 0.125	-0.11 [-0.34, +0.00]
c-adm. (c=1.05)	11.1 ± 0.5	1022 ± 5	25.6 ± 0.2	0.78 ± 0.01	1.12 ± 0.39	1.18 ± 0.30	0.332 ± 0.047	0.738 ± 0.145	+0.00 [+0.00, +0.00]
c-adm. (c=1.1)	7.37 ± 0.26	986 ± 3	24.7 ± 0.1	0.77 ± 0.01	1.62 ± 0.60	1.30 ± 0.33	0.329 ± 0.050	0.738 ± 0.145	-0.30 [-0.89, +0.00]
c-adm. (c=1.2)	5.31 ± 0.41	969 ± 3	24.3 ± 0.1	0.76 ± 0.01	1.48 ± 0.90	0.86 ± 0.59	0.327 ± 0.053	0.723 ± 0.145	-0.47 [-1.27, +0.00]
Qwen2.5-7B-Instruct, LoCoMo, 4 agents									
no isolation	0	1028 ± 6	25.8 ± 0.2	0.39 ± 0.01	5.05 ± 1.75	2.44 ± 0.34	0.343 ± 0.007	0.598 ± 0.010	-1.57 [-1.70, -1.45]
exact	25.7 ± 0.5	1292 ± 6	32.4 ± 0.3	0.48 ± 0.02	2.42 ± 0.29	2.42 ± 0.29	0.358 ± 0.005	0.641 ± 0.015	ref.
overlap 90%	2.75 ± 0.30	1057 ± 3	26.4 ± 0.1	0.39 ± 0.02	4.81 ± 0.35	2.96 ± 0.68	0.356 ± 0.005	0.632 ± 0.015	-0.28 [-0.34, -0.22]
c-adm. (c=1.05)	12.7 ± 0.6	1158 ± 6	29.0 ± 0.2	0.42 ± 0.02	2.45 ± 0.25	2.16 ± 0.30	0.357 ± 0.006	0.638 ± 0.014	-0.14 [-0.19, -0.10]
c-adm. (c=1.1)	7.76 ± 0.47	1107 ± 6	27.7 ± 0.2	0.40 ± 0.02	2.56 ± 0.41	2.36 ± 0.96	0.358 ± 0.005	0.639 ± 0.017	-0.03 [-0.06, -0.00]
c-adm. (c=1.2)	4.82 ± 0.42	1077 ± 4	26.8 ± 0.2	0.40 ± 0.02	3.13 ± 0.59	2.28 ± 0.28	0.355 ± 0.006	0.635 ± 0.015	-0.28 [-0.34, -0.23]
Qwen2.5-7B-Instruct, LoCoMo, 8 agents									
no isolation	0	1026 ± 3	25.6	0.39 ± 0.02	6.34 ± 0.64	2.32 ± 0.50	n/a	n/a	n/a
exact	42.4 ± 0.9	1460 ± 11	36.6 ± 0.8	0.51 ± 0.03	2.57 ± 0.49	2.57 ± 0.49	n/a	n/a	n/a
overlap 90%	5.34 ± 0.33	1081 ± 6	27.0 ± 0.2	0.41 ± 0.02	3.42 ± 0.78	1.76 ± 1.19	n/a	n/a	n/a
c-adm. (c=1.05)	19.5 ± 1.7	1226 ± 18	30.9 ± 0.4	0.44 ± 0.03	3.04 ± 1.08	2.28 ± 0.75	n/a	n/a	n/a
c-adm. (c=1.1)	12.8 ± 0.7	1155 ± 8	28.8 ± 0.2	0.42 ± 0.02	2.87 ± 0.49	1.92 ± 0.29	n/a	n/a	n/a
c-adm. (c=1.2)	7.37 ± 0.34	1100 ± 4	27.5 ± 0.1	0.42 ± 0.02	3.34 ± 0.76	2.00 ± 0.22	n/a	n/a	n/a
Qwen2.5-7B-Instruct, LongMemEval, 4 agents									
no isolation	0	1025 ± 5	25.8 ± 0.1	0.38 ± 0.01	6.46 ± 1.13	2.32 ± 0.57	0.306 ± 0.069	0.385 ± 0.214	-1.38 [-2.71, -0.12]
exact	20.2 ± 0.4	1234 ± 7	30.9 ± 0.1	0.45 ± 0.02	2.31 ± 0.35	2.31 ± 0.35	0.320 ± 0.069	0.477 ± 0.247	ref.
overlap 90%	2.07 ± 0.28	1047 ± 4	26.1 ± 0.4	0.40 ± 0.01	4.23 ± 1.42	2.08 ± 0.23	0.310 ± 0.077	0.477 ± 0.218	-0.97 [-2.30, +0.00]
c-adm. (c=1.05)	12.4 ± 0.5	1150 ± 2	28.8 ± 0.3	0.43 ± 0.02	2.19 ± 0.32	2.08 ± 0.13	0.318 ± 0.068	0.477 ± 0.247	-0.17 [-0.51, +0.00]
c-adm. (c=1.1)	8.10 ± 0.28	1108 ± 5	27.8 ± 0.2	0.41 ± 0.01	2.65 ± 0.32	2.20 ± 0.46	0.322 ± 0.073	0.462 ± 0.224	+0.20 [+0.00, +0.59]
c-adm. (c=1.2)	5.72 ± 0.29	1084 ± 5	27.1 ± 0.3	0.40 ± 0.01	2.64 ± 1.58	1.76 ± 0.73	0.319 ± 0.067	0.508 ± 0.240	-0.13 [-0.74, +0.46]

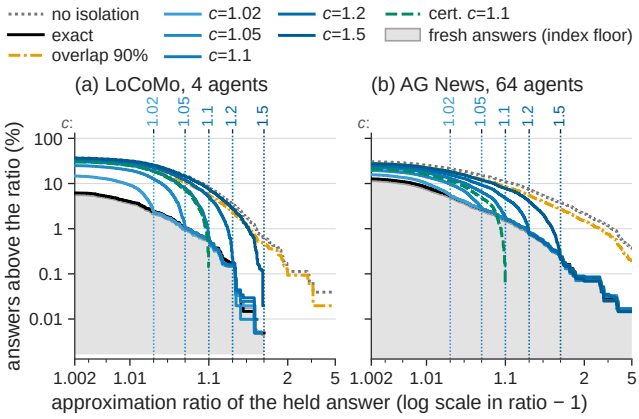


Figure A24: Approximation ratio of held answers per predicate (complementary cumulative distribution function, CCDF), and of fresh answers for comparison.

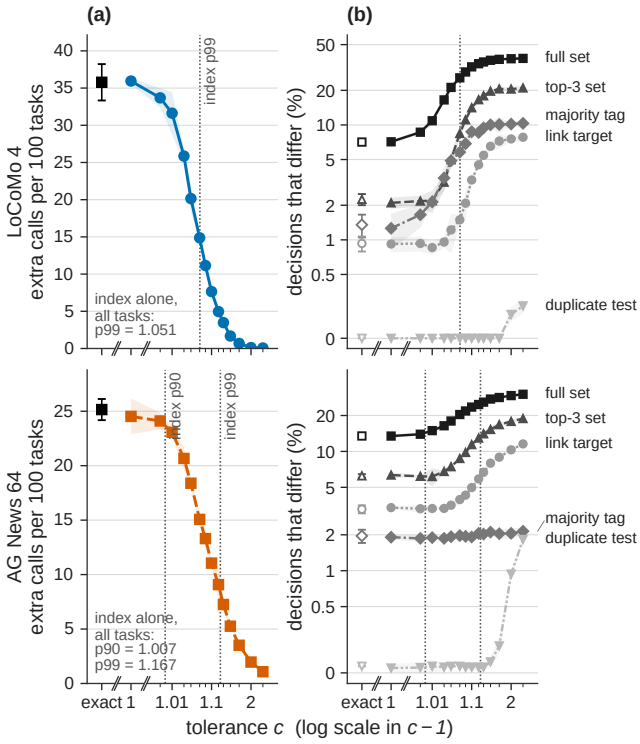


Figure A25: Choosing c: extra calls and differing decisions on a fine grid of tolerances. Vertical lines: 90th and 99th percentiles (where above 1; on LoCoMo the 90th is 1) over all tasks of the index's own approximation ratio (worst validation read per task), averaged over the runs.

budget. Table A4 is the generated version of Table 5 of the paper for both models and all three workloads (L4, L8, and LongMemEval).

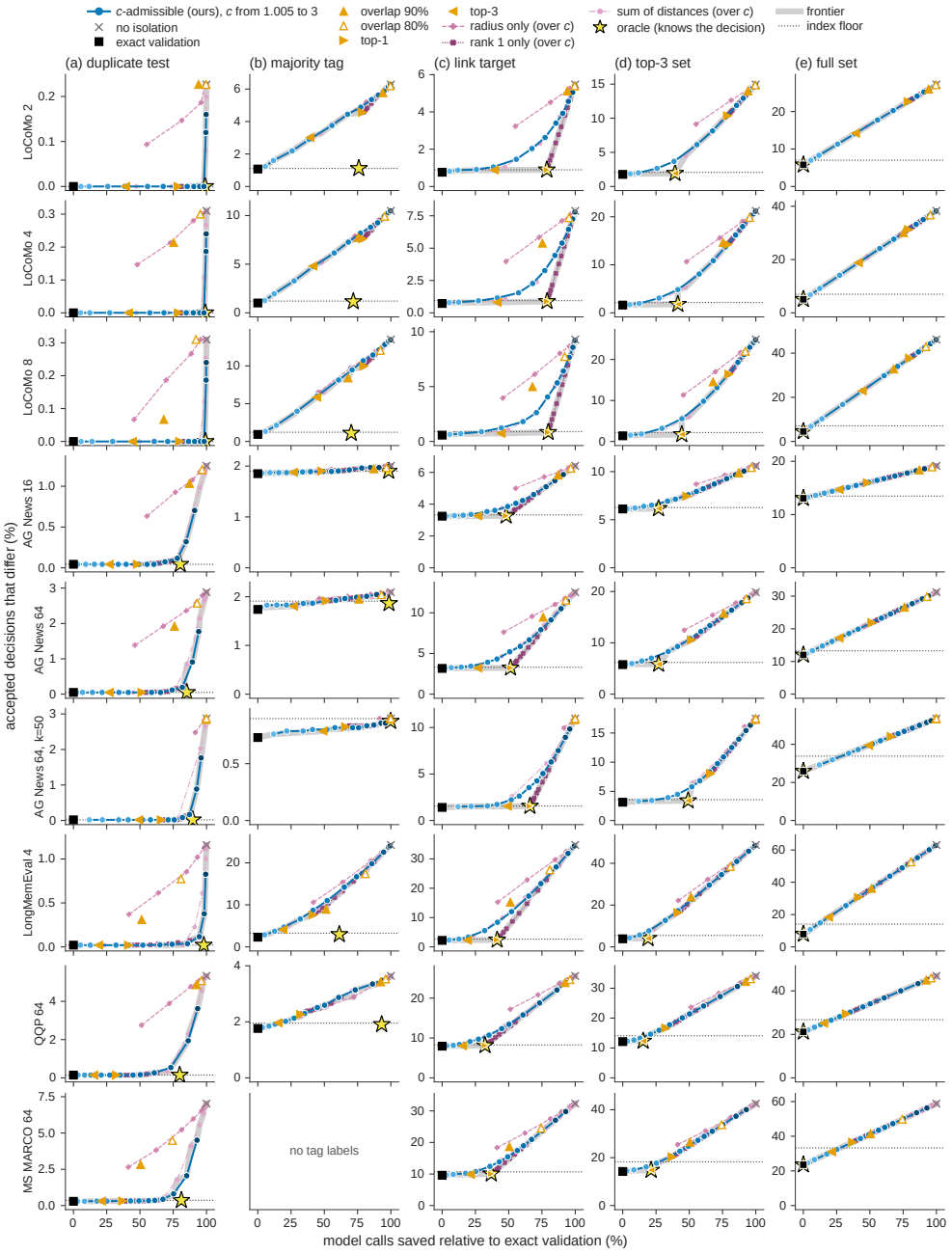


Figure A26: Calls saved against differing decisions, all workloads (rows: data set and number of agents). The index floor counts all transactions, the points only accepted decisions, so a point can lie below it. MS MARCO passages have no tag labels, so the majority-tag panel shows no points for MS MARCO.

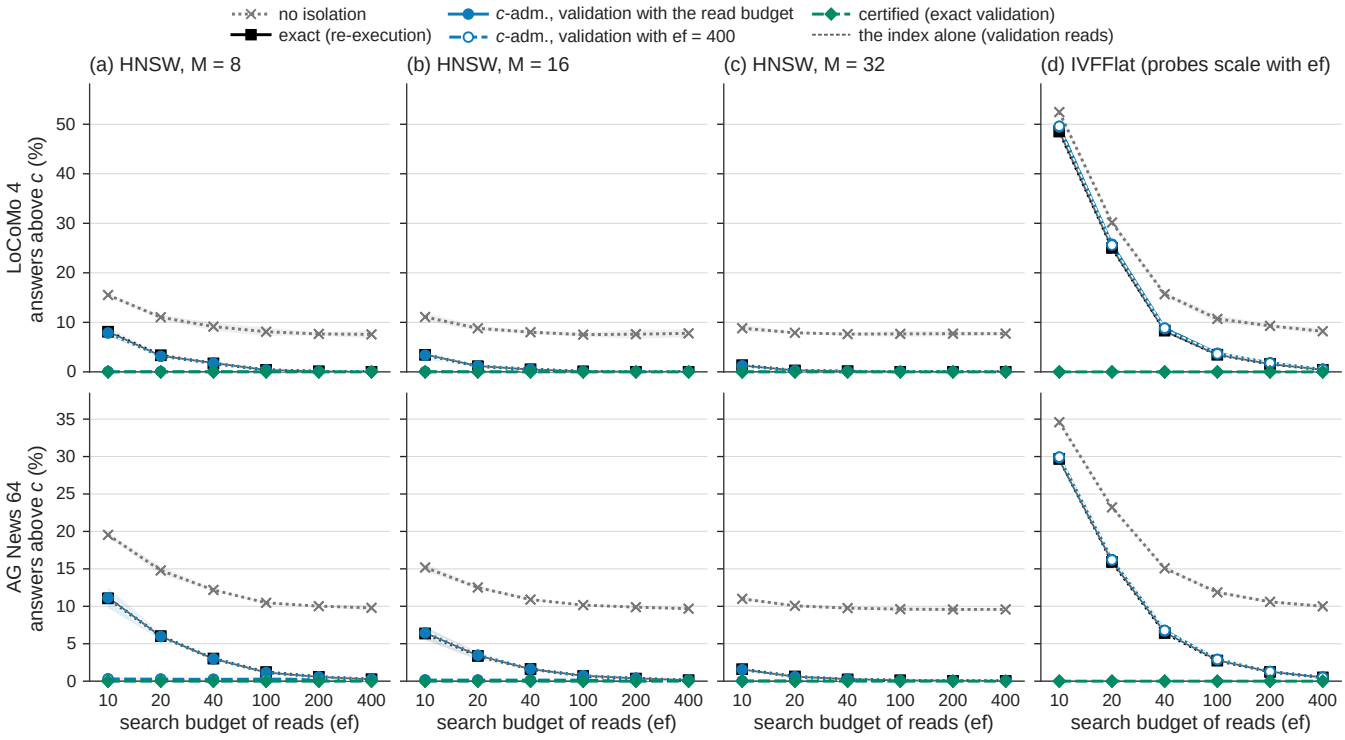


Figure A27: Answers above the tolerance ($c = 1.1$) against the search budget of the index.

D.4 Performance (§6.6)

Figure A32 and Table A5 compare validation with ball predicate locks on every workload at every tolerance; on the LoCoMo and AG News streams, tolerance reduces the extra calls of both designs, and on the fan-out workload, where the tasks of a team read the same neighborhood, the extra calls of both stay level. Figure A33 gives the distribution of rounds per task in the worst cases. Figure A28 shows throughput and tail latency against the number of agents at both thinking times: from 16 agents on, $c = 1.1$ and re-execution keep about 87% and 77% of the throughput without isolation (87% and 76% at 256 agents, where $c = 1.1$ holds the commit lock 77% of the time, 8% with thinking times of 2 s); certified validation, whose exact scan holds the lock 20.3 ms per task, saturates it from 16 agents on at 0.2 s (96%) and stays near 49 tasks per second, and at 2 s it saturates only from 128 agents on (54% at 64, 99% at 256); ball locks fall further behind as their deadlocks grow with the agents; Figure A29 breaks the critical section down, Figure A30 varies the distribution of the thinking time, Figure A31 replaces the closed loop by open-loop arrivals, and Figure A34 varies the size of the memory and of the run. The long run keeps only its summary rows, so its time series is not shown. Throughput times mean latency equals the number of clients within a factor of 0.8 to 1.1 in every closed-loop run, so no client waits outside the measured latency; with one process per agent the smallest ratio is 0.98.

Table A5: Optimistic against pessimistic: all six workloads at every tolerance, three seeds; "validation" is certified validation (exact validation read), and both designs run in the client library on the in-memory store (generated).

protocol	extra calls / 100	lock wait (ms)	waits / 100	deadlocks / 100	stale reads / 100	p99 (s)	tasks / s
LoCoMo, 2 agents							
no isolation	0	0	n/a	n/a	n/a	0.39 ± 0.01	97.5 ± 2.7
validation, c=1	81.3 ± 2.8	0.15 ± 0.11	n/a	n/a	n/a	0.96 ± 0.08	54.4 ± 1.8
validation, c=1.05	10.0 ± 0.2	0.051 ± 0.029	n/a	n/a	n/a	0.59 ± 0.04	87.5 ± 4.7
validation, c=1.1	5.43 ± 0.43	0.035 ± 0.007	n/a	n/a	n/a	0.53 ± 0.03	92.3 ± 2.1
validation, c=1.2	2.09 ± 0.19	0.056 ± 0.036	n/a	n/a	n/a	0.44 ± 0.02	95.4 ± 3.8
validation, c=1.5	0.55	0.055 ± 0.029	n/a	n/a	n/a	0.44 ± 0.01	97.2 ± 1.7
lock across think	0	3836 ± 90	n/a	n/a	n/a	4.41 ± 0.11	4.93 ± 0.15
ball lock, c=1	331 ± 93	4.85 ± 1.33	0.079	332 ± 94	0.079	35.7 ± 5.1	19.9 ± 5.1
ball lock, c=1.05	132 ± 57	2.60 ± 0.32	0.079	133 ± 57	0.079	20.0 ± 16.3	38.2 ± 7.5
ball lock, c=1.1	64.9 ± 19.1	2.54 ± 1.01	0.079	65.1 ± 19.2	0.079	1.25 ± 0.17	53.3 ± 6.9
ball lock, c=1.2	21.8 ± 6.8	2.21 ± 0.87	0.079	21.9 ± 6.8	0.052 ± 0.056	1.04 ± 0.10	73.5 ± 6.8
ball lock, c=1.5	3.70 ± 1.19	0.67 ± 0.54	0.039	3.71 ± 1.22	0.039	0.82 ± 0.06	92.9 ± 6.5
LoCoMo, 4 agents							
no isolation	0	0	n/a	n/a	n/a	0.38 ± 0.01	195 ± 9
validation, c=1	148 ± 3	0.28 ± 0.21	n/a	n/a	n/a	2.10 ± 0.08	78.9 ± 3.1
validation, c=1.05	16.0 ± 1.1	0.11 ± 0.06	n/a	n/a	n/a	0.67 ± 0.01	169 ± 2
validation, c=1.1	9.28 ± 0.38	0.094 ± 0.012	n/a	n/a	n/a	0.62 ± 0.03	178 ± 9
validation, c=1.2	3.36 ± 0.14	0.12 ± 0.05	n/a	n/a	n/a	0.56 ± 0.01	189 ± 5
validation, c=1.5	0.66 ± 0.06	0.098 ± 0.021	n/a	n/a	n/a	0.54 ± 0.01	195 ± 2
lock across think	0	7881 ± 180	n/a	n/a	n/a	8.87 ± 0.14	4.88 ± 0.06
ball lock, c=1	602 ± 168	13.6 ± 7.1	0.25 ± 0.06	605 ± 168	0.24 ± 0.10	36.7 ± 5.3	24.1 ± 6.6
ball lock, c=1.05	209 ± 85	7.67 ± 1.61	0.22 ± 0.06	210 ± 86	0.21 ± 0.06	33.0 ± 1.7	56.0 ± 10.7
ball lock, c=1.1	87.8 ± 4.8	7.81 ± 1.96	0.22 ± 0.06	88.0 ± 4.9	0.20	14.2 ± 0.6	90.8 ± 3.5
ball lock, c=1.2	33.2 ± 8.4	6.31 ± 3.25	0.18 ± 0.06	33.3 ± 8.5	0.14 ± 0.06	1.18 ± 0.10	129 ± 11
ball lock, c=1.5	4.45 ± 1.10	2.18 ± 2.23	0.13 ± 0.06	4.49 ± 1.11	0.12	0.84 ± 0.03	182 ± 7
LoCoMo, 8 agents							
no isolation	0	0	n/a	n/a	n/a	0.34 ± 0.02	387 ± 6
validation, c=1	207 ± 5	0.45 ± 0.25	n/a	n/a	n/a	1.76 ± 0.11	127 ± 3
validation, c=1.05	19.6 ± 0.5	0.22 ± 0.05	n/a	n/a	n/a	0.69 ± 0.05	326 ± 4
validation, c=1.1	10.1 ± 0.6	0.18 ± 0.11	n/a	n/a	n/a	0.58 ± 0.04	349 ± 5
validation, c=1.2	3.54 ± 0.26	0.27 ± 0.06	n/a	n/a	n/a	0.46 ± 0.04	376 ± 9
validation, c=1.5	0.82 ± 0.06	0.21 ± 0.08	n/a	n/a	n/a	0.45 ± 0.04	385 ± 10
lock across think	0	15 589 ± 527	n/a	n/a	n/a	17.3 ± 0.7	4.98 ± 0.17
ball lock, c=1	862 ± 263	33.3 ± 7.6	0.60 ± 0.06	866 ± 265	0.56 ± 0.06	41.4 ± 4.6	34.3 ± 11.5
ball lock, c=1.05	257 ± 96	17.7 ± 1.0	0.52 ± 0.06	258 ± 96	0.47	34.3 ± 2.7	95.5 ± 19.7
ball lock, c=1.1	119 ± 34	17.5 ± 5.6	0.49 ± 0.06	119 ± 34	0.43	17.3 ± 3.6	149 ± 32
ball lock, c=1.2	35.6 ± 11.3	15.3 ± 7.0	0.46 ± 0.11	35.7 ± 11.3	0.37 ± 0.06	1.25 ± 0.15	241 ± 29
ball lock, c=1.5	5.65 ± 1.34	5.30 ± 5.79	0.33 ± 0.06	5.69 ± 1.35	0.29 ± 0.06	0.86 ± 0.06	351 ± 24
AG News, 16 agents							
no isolation	0	0	n/a	n/a	n/a	0.39 ± 0.02	78.3 ± 0.8
validation, c=1	27.5 ± 1.5	0.51 ± 0.32	n/a	n/a	n/a	0.55 ± 0.01	60.8 ± 1.5
validation, c=1.05	6.94 ± 0.59	0.25 ± 0.11	n/a	n/a	n/a	0.46 ± 0.02	72.7 ± 2.0
validation, c=1.1	4.84 ± 0.20	0.15 ± 0.07	n/a	n/a	n/a	0.45 ± 0.01	72.8 ± 2.1
validation, c=1.2	3.00 ± 0.23	0.24 ± 0.09	n/a	n/a	n/a	0.43 ± 0.01	74.7 ± 0.9
validation, c=1.5	1.43 ± 0.04	0.23 ± 0.10	n/a	n/a	n/a	0.43 ± 0.01	75.6 ± 1.3
lock across think	0	3042 ± 20	n/a	n/a	n/a	3.57 ± 0.10	4.87 ± 0.06
ball lock, c=1	223 ± 54	67.3 ± 31.0	1.22 ± 0.25	224 ± 54	1.16 ± 0.36	36.1 ± 1.9	19.8 ± 4.3
ball lock, c=1.05	176 ± 67	36.3 ± 7.2	1.07 ± 0.25	177 ± 67	1.00 ± 0.21	30.6 ± 1.8	24.2 ± 4.5
ball lock, c=1.1	110 ± 27	38.1 ± 17.2	1.06 ± 0.13	110 ± 27	0.92 ± 0.10	17.1 ± 3.1	30.4 ± 5.6
ball lock, c=1.2	56.5 ± 16.2	30.1 ± 14.7	0.91 ± 0.25	56.6 ± 16.2	0.71 ± 0.19	1.50 ± 0.49	39.8 ± 6.4
ball lock, c=1.5	18.6 ± 5.8	11.2 ± 10.1	0.72 ± 0.02	18.6 ± 5.8	0.61 ± 0.06	1.04 ± 0.14	58.9 ± 8.2

Table A5 (continued)

protocol	extra calls / 100	lock wait (ms)	waits / 100	deadlocks / 100	stale reads / 100	p99 (s)	tasks / s
AG News, 64 agents							
no isolation	0	0	n/a	n/a	n/a	0.39 ± 0.01	311 ± 3
validation, c=1	102 ± 1	0.92 ± 0.21	n/a	n/a	n/a	1.54 ± 0.03	155 ± 2
validation, c=1.05	18.2 ± 0.6	1.07 ± 0.42	n/a	n/a	n/a	0.69 ± 0.01	264 ± 3
validation, c=1.1	12.7 ± 0.3	0.90 ± 0.23	n/a	n/a	n/a	0.64	277 ± 2
validation, c=1.2	7.66 ± 0.20	0.98 ± 0.36	n/a	n/a	n/a	0.60	290 ± 2
validation, c=1.5	3.70 ± 0.05	1.14 ± 0.23	n/a	n/a	n/a	0.57	301 ± 2
lock across think	0	12 619 ± 81	n/a	n/a	n/a	14.0 ± 0.2	4.97 ± 0.04
ball lock, c=1	811 ± 209	295 ± 90	5.37 ± 0.40	815 ± 210	5.07 ± 0.75	44.3 ± 0.7	26.0 ± 7.0
ball lock, c=1.05	462 ± 166	164 ± 22	4.82 ± 0.53	463 ± 167	4.52 ± 0.52	43.6 ± 0.9	43.1 ± 9.0
ball lock, c=1.1	291 ± 78	160 ± 67	4.47 ± 0.64	292 ± 78	3.88 ± 0.25	43.8 ± 1.7	57.0 ± 14.6
ball lock, c=1.2	146 ± 41	137 ± 60	4.13 ± 1.15	146 ± 41	3.26 ± 0.95	32.7 ± 23.3	82.5 ± 20.7
ball lock, c=1.5	48.8 ± 13.6	45.9 ± 43.2	2.94 ± 0.28	48.9 ± 13.6	2.47 ± 0.10	1.55 ± 0.71	162 ± 46
fan-out							
no isolation	0	0	n/a	n/a	n/a	0.45 ± 0.04	149 ± 2
validation, c=1	102 ± 12	0.21 ± 0.08	n/a	n/a	n/a	0.88 ± 0.10	75.9 ± 2.6
validation, c=1.05	97.8 ± 0.5	0.061 ± 0.013	n/a	n/a	n/a	0.86 ± 0.07	77.8 ± 1.5
validation, c=1.1	104 ± 1	0.045 ± 0.022	n/a	n/a	n/a	0.88 ± 0.01	75.7 ± 2.9
validation, c=1.2	102 ± 8	0.062 ± 0.014	n/a	n/a	n/a	0.87 ± 0.05	75.8 ± 6.6
validation, c=1.5	104 ± 8	0.066 ± 0.021	n/a	n/a	n/a	0.89 ± 0.08	74.7 ± 4.6
lock across think	0	6443 ± 134	n/a	n/a	n/a	7.27 ± 0.09	4.73 ± 0.07
ball lock, c=1	819 ± 206	144 ± 36	2.61 ± 0.37	823 ± 208	2.45 ± 0.42	42.0 ± 5.2	13.6 ± 3.5
ball lock, c=1.05	822 ± 170	77.6 ± 22.5	2.29 ± 0.37	825 ± 171	2.12 ± 0.51	40.9 ± 3.0	13.9 ± 2.8
ball lock, c=1.1	780 ± 185	78.5 ± 47.7	2.19 ± 0.70	782 ± 186	1.86 ± 0.49	41.9 ± 4.0	13.7 ± 3.8
ball lock, c=1.2	832 ± 197	65.9 ± 33.9	1.96 ± 0.42	834 ± 197	1.57 ± 0.42	49.2 ± 4.1	11.5 ± 2.1
ball lock, c=1.5	816 ± 165	23.4 ± 23.2	1.47 ± 0.24	819 ± 167	1.24 ± 0.14	49.2 ± 11.2	11.6 ± 1.9

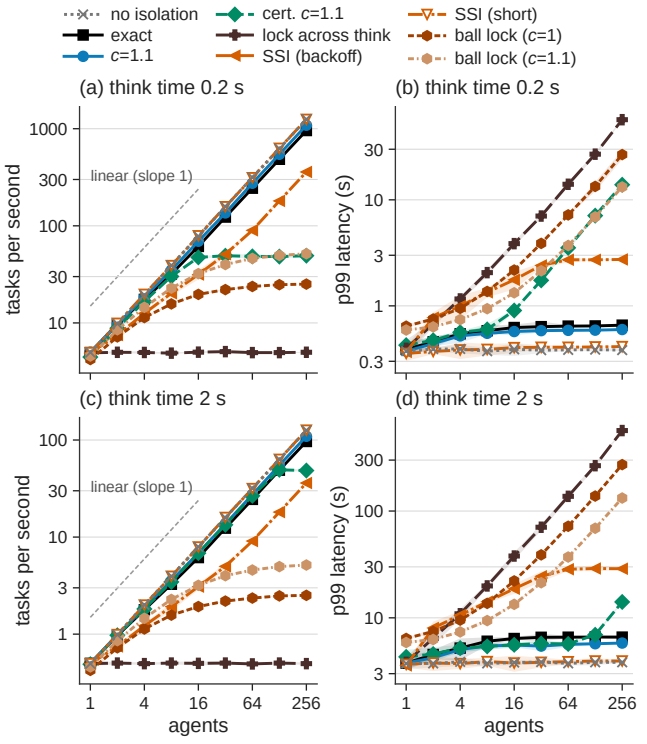


Figure A28: Throughput and tail latency against the number of agents at every thinking time.

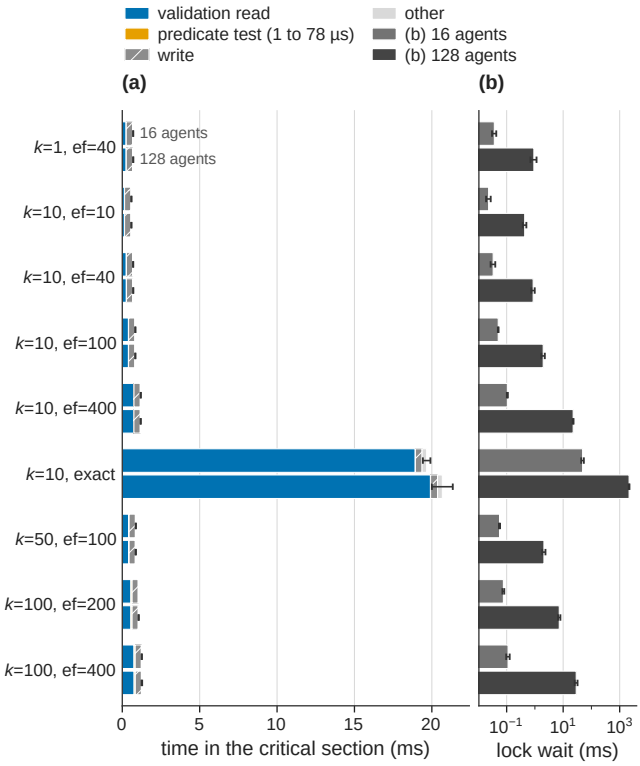


Figure A29: Time in the critical section by its parts, for retrieval depths k and search budgets of the validation read, and with an exact validation read (AG News, pgvector, $c = 1.1$). (a) Parts of the critical section, upper bar 16 and lower bar 128 agents; (b) lock wait, light 16 and dark 128 agents.

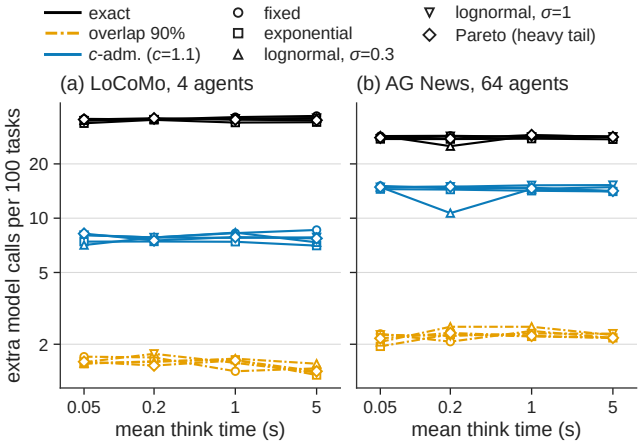


Figure A30: Thinking-time distributions.

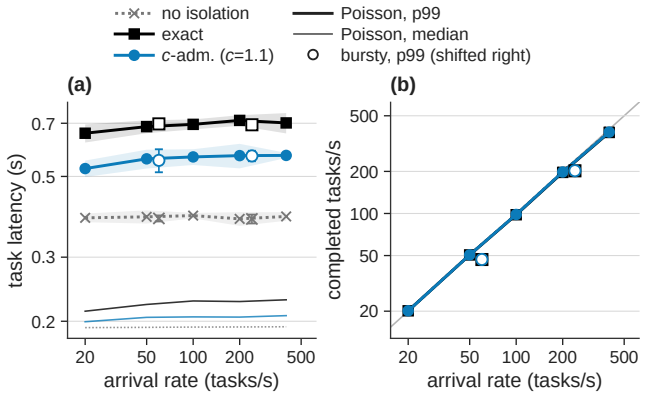


Figure A31: Open-loop arrivals. Bursty marks are drawn slightly to the right of their arrival rate so that they do not hide the Poisson marks. In (b) every protocol completes nearly the offered rate, so the marks coincide.

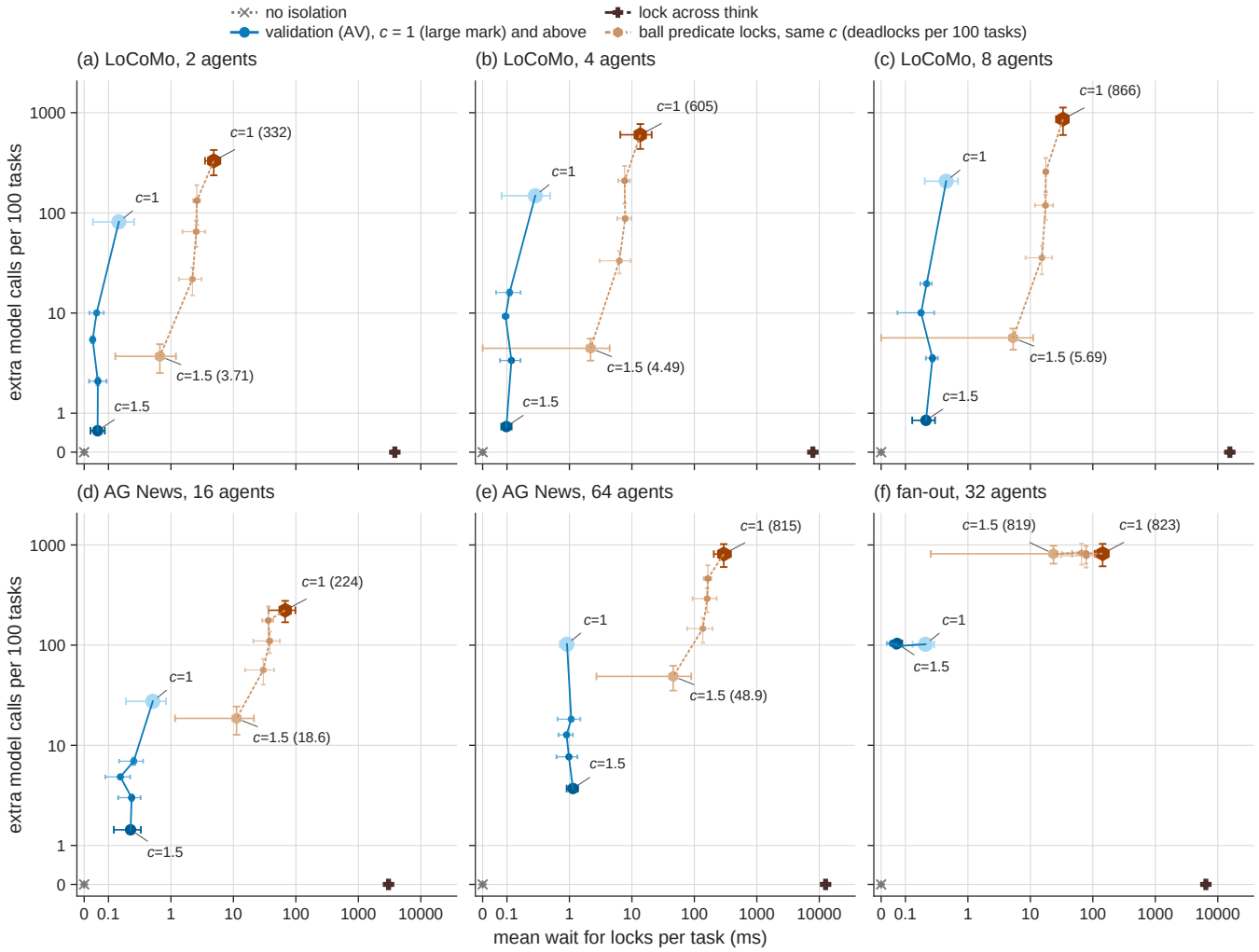


Figure A32: Validation against ball predicate locks, all six workloads. Each path is labeled at its ends ($c = 1$ and $c = 1.5$); numbers in parentheses are deadlocks per 100 tasks; horizontal intervals are clipped at 0.

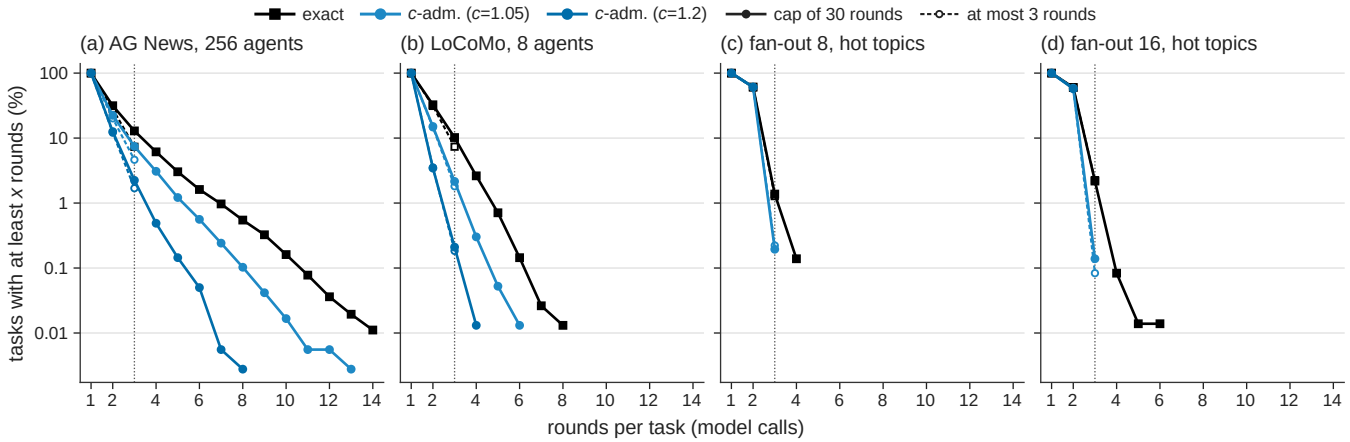


Figure A33: Rounds per task under worst-case workloads (CCDF), all with heavy-tailed thinking: AG News with 256 agents, LoCoMo with 8, and hot topics on the fan-out workload. Solid: a cap of 30 rounds; dashed, hollow marks: at most 3 rounds, then commit under the lock. On the hot-topic workloads, $c = 1.2$ finishes every task within 2 rounds and re-execution needs up to 6.

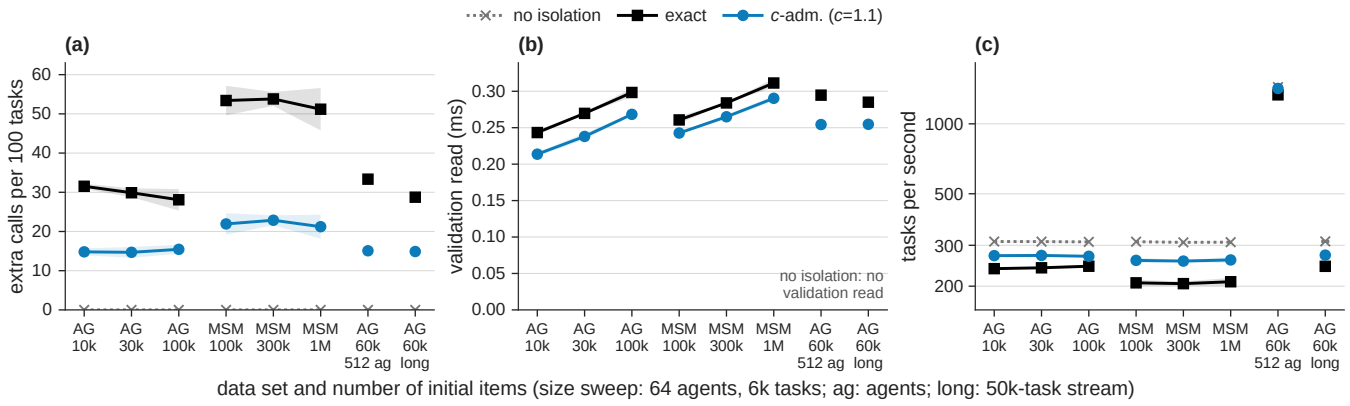


Figure A34: Size of the memory and of the run: (a) extra calls per 100 tasks, (b) mean time of the validation read (validating protocols only), (c) throughput (log scale). The size sweep uses 64 agents and 6,000 tasks; the two right-most cases start from 60k items, one with 512 agents (20,000 tasks), one with 50,000 tasks.

Table A6: Ablations (generated).

variant	extra calls	dup. writes	ρ_{99}	above c (%)	bound viol.	link differs (%)	crit. sect. (ms)	tasks / s
Validation predicate (AG News, 64 agents)								
exact	25.1 ± 0.7	0.057 ± 0.029	1.165 ± 0.030	1.57 ± 0.09	n/a	3.35 ± 0.12	0.77 ± 0.06	249 ± 2
set-exact	25.1 ± 0.5	0.043 ± 0.029	1.167 ± 0.036	1.61 ± 0.16	n/a	3.36 ± 0.14	0.76 ± 0.03	249 ± 1
top-1	9.84 ± 0.83	0.043 ± 0.029	1.295 ± 0.041	4.08 ± 0.21	n/a	3.28 ± 0.16	0.75 ± 0.05	283 ± 1
overlap 80%	1.16 ± 0.19	2.61 ± 0.13	2.927 ± 0.036	10.1 ± 0.3	n/a	11.6 ± 0.6	0.69 ± 0.04	307 ± 2
overlap 90%	4.51 ± 0.34	1.97 ± 0.36	2.384 ± 0.250	8.23 ± 0.31	n/a	9.71 ± 0.48	0.70 ± 0.02	297 ± 2
top-3	16.2 ± 0.1	0.063 ± 0.029	1.196 ± 0.004	2.06 ± 0.10	n/a	3.35	0.78 ± 0.11	268 ± 1
radius (c=1.1)	0.083 ± 0.038	2.86 ± 0.09	3.127 ± 0.011	10.8 ± 0.1	n/a	12.5 ± 0.4	0.69 ± 0.02	310 ± 2
rank-1 (c=1.1)	6.71 ± 0.20	0.057 ± 0.057	1.337 ± 0.057	4.94 ± 0.19	n/a	5.90 ± 0.38	0.70 ± 0.04	291 ± 1
sum-dist. (c=1.1)	0.91 ± 0.07	2.20 ± 0.26	2.526 ± 0.166	10.1 ± 0.5	n/a	11.6 ± 0.6	0.67 ± 0.08	308 ± 3
radius (c=1.2)	0	2.92 ± 0.12	3.152 ± 0.019	7.41 ± 0.41	n/a	12.6 ± 0.4	0.69 ± 0.03	310 ± 3
rank-1 (c=1.2)	5.43 ± 0.04	0.083 ± 0.072	1.333 ± 0.024	2.23 ± 0.22	n/a	7.17 ± 0.27	0.69 ± 0.03	295 ± 2
sum-dist. (c=1.2)	0.15 ± 0.12	2.77 ± 0.06	3.054 ± 0.044	7.19 ± 0.24	n/a	12.6	0.66 ± 0.02	310
c-adm. (c=1.1)	10.6 ± 0.3	0.057 ± 0.057	1.164 ± 0.062	1.61 ± 0.58	0	5.21 ± 0.60	0.73 ± 0.06	281 ± 2
c-adm. (c=1.2)	7.30 ± 0.34	0.043 ± 0.029	1.193 ± 0.004	0.75 ± 0.17	0	6.68 ± 0.20	0.67 ± 0.04	290 ± 1
certified (c=1.1)	12.6 ± 0.3	0	1.087 ± 0.001	0	0	3.76 ± 0.18	20.5 ± 1.1	47.4 ± 2.5
Scope of the commit lock (AG News, 64 agents, 8 teams)								
exact, global lock	24.1 ± 0.4	0.057 ± 0.017	1.137 ± 0.002	1.47 ± 0.05	n/a	3.17 ± 0.12	0.73 ± 0.01	251 ± 2
exact, lock per team	24.7 ± 0.5	0.11 ± 0.03	1.139 ± 0.003	1.51 ± 0.07	n/a	3.11 ± 0.24	0.74 ± 0.01	253 ± 1
exact, no lock	23.9 ± 0.3	0.13 ± 0.04	1.138 ± 0.003	1.48 ± 0.07	n/a	3.13 ± 0.24	0.73 ± 0.01	254 ± 2
c-adm. (c=1.1), global lock	12.4 ± 0.4	0.064 ± 0.016	1.164 ± 0.010	1.55 ± 0.07	0	5.02 ± 0.16	0.70 ± 0.01	278 ± 2
c-adm. (c=1.1), lock per team	12.1 ± 0.5	0.11 ± 0.02	1.168 ± 0.011	1.59 ± 0.09	0	4.88 ± 0.21	0.71 ± 0.01	280 ± 2
c-adm. (c=1.1), no lock	12.2 ± 0.6	0.14 ± 0.03	1.174 ± 0.011	1.65 ± 0.08	0	5.17 ± 0.16	0.71 ± 0.01	278 ± 2
After a rejected validation (AG News, 64 agents)								
exact, think again on F, at most 1 round	0	0.047 ± 0.029	1.193 ± 0.009	1.92 ± 0.17	n/a	3.98 ± 0.20	0.66 ± 0.05	310 ± 1
exact, think again on F, at most 2 rounds	17.3 ± 0.6	0.050	1.183 ± 0.008	1.73 ± 0.22	n/a	3.54 ± 0.23	0.73 ± 0.04	266 ± 2
exact, think again on F, at most 3 rounds	22.7 ± 0.2	0.043 ± 0.029	1.175 ± 0.046	1.71 ± 0.10	n/a	3.41 ± 0.03	0.77 ± 0.04	255 ± 3
exact, think again on F, at most 30 rounds	24.9 ± 1.0	0.050	1.174 ± 0.030	1.67	n/a	3.38 ± 0.04	0.78 ± 0.04	250 ± 1
exact, read again, at most 1 round	0	0.053 ± 0.029	1.199 ± 0.015	1.97 ± 0.01	n/a	4.04 ± 0.24	0.64 ± 0.02	311 ± 2
exact, read again, at most 2 rounds	17.5 ± 0.4	0.050	1.184 ± 0.028	1.71 ± 0.22	n/a	3.53 ± 0.41	0.76 ± 0.04	265 ± 2
exact, read again, at most 3 rounds	22.2 ± 1.6	0.050 ± 0.050	1.188 ± 0.018	1.74 ± 0.23	n/a	3.41 ± 0.13	0.75 ± 0.03	255 ± 4
exact, read again, at most 30 rounds	25.1 ± 0.2	0.057 ± 0.029	1.173 ± 0.032	1.63 ± 0.16	n/a	3.38 ± 0.01	0.79 ± 0.05	250 ± 2
exact, give up, at most 1 round	0	0.060 ± 0.050	1.195 ± 0.039	1.90 ± 0.28	n/a	3.95 ± 0.25	0.62 ± 0.02	311 ± 1
exact, give up, at most 2 rounds	0	0.047 ± 0.029	1.189 ± 0.020	1.82 ± 0.32	n/a	3.95 ± 0.19	0.66 ± 0.05	310
exact, give up, at most 3 rounds	0	0.053 ± 0.029	1.192 ± 0.008	1.86 ± 0.31	n/a	3.93 ± 0.32	0.68 ± 0.05	311 ± 2
exact, give up, at most 30 rounds	0	0.047 ± 0.029	1.192 ± 0.008	1.88 ± 0.28	n/a	3.96 ± 0.40	0.71 ± 0.02	310 ± 2
c-adm. (c=1.1), think again on F, at most 1 round	0	0.047 ± 0.014	1.178 ± 0.023	1.70 ± 0.20	0	5.69 ± 0.71	0.69 ± 0.06	311 ± 2
c-adm. (c=1.1), think again on F, at most 2 rounds	8.71 ± 0.53	0.037 ± 0.029	1.177 ± 0.022	1.65 ± 0.43	0	5.27 ± 0.28	0.73 ± 0.07	286 ± 3
c-adm. (c=1.1), think again on F, at most 3 rounds	10.3 ± 1.0	0.043 ± 0.029	1.178 ± 0.018	1.67	0	5.17 ± 0.14	0.76 ± 0.07	281 ± 2
c-adm. (c=1.1), think again on F, at most 30 rounds	10.8 ± 0.3	0.033 ± 0.038	1.164 ± 0.038	1.66 ± 0.31	0	5.25 ± 0.59	0.72 ± 0.09	280 ± 2
c-adm. (c=1.1), read again, at most 1 round	0	0.050	1.180 ± 0.019	1.66 ± 0.11	0	5.78 ± 0.37	0.68 ± 0.02	310 ± 2
c-adm. (c=1.1), read again, at most 2 rounds	8.96 ± 1.05	0.050	1.169 ± 0.037	1.67 ± 0.19	0	5.34 ± 0.17	0.71 ± 0.06	286 ± 2
c-adm. (c=1.1), read again, at most 3 rounds	10.4 ± 0.6	0.043 ± 0.029	1.165 ± 0.023	1.61 ± 0.04	0	5.04 ± 0.30	0.72 ± 0.06	281 ± 2
c-adm. (c=1.1), read again, at most 30 rounds	10.7 ± 0.5	0.050	1.170 ± 0.056	1.64 ± 0.39	0	5.18 ± 0.29	0.72 ± 0.11	281 ± 1
c-adm. (c=1.1), give up, at most 1 round	0	0.043 ± 0.063	1.182 ± 0.022	1.74 ± 0.17	0	5.73 ± 0.63	0.67 ± 0.01	310 ± 2
c-adm. (c=1.1), give up, at most 2 rounds	0	0.040 ± 0.043	1.177 ± 0.011	1.69 ± 0.16	0	5.70 ± 0.64	0.68 ± 0.03	310 ± 2
c-adm. (c=1.1), give up, at most 3 rounds	0	0.043 ± 0.014	1.176 ± 0.012	1.70 ± 0.13	0	5.71 ± 0.50	0.73 ± 0.02	310 ± 2
c-adm. (c=1.1), give up, at most 30 rounds	0	0.043 ± 0.014	1.175 ± 0.053	1.72 ± 0.30	0	5.76 ± 0.72	0.73 ± 0.07	310 ± 2
Search budget of the validation read (AG News, 64 agents)								
ef = 10	10.4 ± 0.5	0.043 ± 0.057	1.197 ± 0.007	1.93 ± 0.16	0	5.39 ± 0.27	0.67 ± 0.06	281 ± 3
ef = 40	10.8 ± 0.2	0.043 ± 0.029	1.176 ± 0.034	1.69 ± 0.13	0	5.28 ± 0.13	0.72 ± 0.05	280 ± 2
ef = 100	11.6 ± 0.3	0.020	1.095 ± 0.002	0.72 ± 0.06	0	4.28 ± 0.33	0.81 ± 0.03	278 ± 1
ef = 400	12.2 ± 0.3	0	1.088 ± 0.001	0.15 ± 0.06	0	3.81 ± 0.15	1.34 ± 0.10	276 ± 2
exact validation	12.6 ± 0.2	0	1.087 ± 0.001	0	0	3.82 ± 0.09	20.0 ± 0.9	48.6 ± 2.2
Isolation level of the phases, PostgreSQL (AG News, 64 agents)								
critical section RC, read phase RC	10.8 ± 0.9	0.050 ± 0.050	1.173 ± 0.007	1.41 ± 0.24	0	5.11 ± 0.20	0.71	271 ± 3
critical section RC, read phase RR	11.0 ± 1.4	0.007 ± 0.029	1.165 ± 0.043	1.26 ± 0.32	0	5.14 ± 1.04	0.71 ± 0.01	271 ± 4
critical section RR, read phase RC	10.8 ± 1.6	8.37 ± 1.35	4.402 ± 0.194	10.2 ± 1.4	0	14.1 ± 1.3	0.71 ± 0.01	270 ± 3
critical section RR, read phase RR	11.0 ± 1.9	8.53 ± 2.00	4.456 ± 0.080	10.5 ± 2.4	0	14.3 ± 1.9	0.71 ± 0.01	269 ± 5
Tolerance per operation (AG News, 64 agents, mixed ops; destr. = replace, delete)								
all: c=1.02	23.2 ± 3.1	0.20 ± 0.07	1.905 ± 0.342	15.5 ± 1.3	0	5.00 ± 0.60	0.74	253 ± 5
all: c=1.1	14.6 ± 0.6	0.22 ± 0.25	1.714 ± 0.460	3.09 ± 0.39	0	7.59 ± 0.84	0.72 ± 0.01	270 ± 4
all: c=1.5	8.36 ± 0.75	0.20 ± 0.06	1.779 ± 0.476	1.22 ± 0.33	0	13.0 ± 0.9	0.70 ± 0.01	286 ± 5
add: c=1.5; replace, delete: c=1.02	12.9 ± 1.9	0.24 ± 0.07	1.834 ± 0.573	1.31 ± 0.32	0	9.75 ± 0.31	0.71 ± 0.01	277 ± 6
add: c=1.5; replace, delete: exact	14.0 ± 0.8	0.25 ± 0.05	1.710 ± 0.098	1.22 ± 0.10	0	10.2 ± 0.7	0.71 ± 0.01	272 ± 1
add: re-decide; replace, delete: c=1.1	5.04 ± 0.62	0.18 ± 0.12	1.138 ± 0.026	1.51 ± 0.29	n/a	4.16 ± 0.25	0.69 ± 0.01	296 ± 3

D.5 Ablations and robustness (§6.7)

Table A6 is the generated ablation table with all blocks, including the strategy after a failed validation and the per-operation tolerance. The ablations are shown per figure: the components of the predicate (Figure A43), the scope of the commit lock (Figure A44), the strategy after a failed validation (Figure A35), the search budget of the validation read (Figure A36), the isolation levels of the two phases (Figure A37), and the per-operation tolerance (Figure A38). The robustness experiments cover several reads per transaction (Figure A45), the retrieval depth and the threshold (Figure A39), the share of read-only tasks (Figure A40), the stores (Figure A41), and client failures (Figure A42). Table A7 lists the stores.

Table A7: Portability across stores (generated).

	extra calls / 100 tasks		dup. writes / 100 tasks			safety	commits	crit. sect.	recall of
store	exact	c-adm. (c=1.1)	no isolation	exact	c-adm. (c=1.1)	violations	checked	(ms)	the index
pgvector HNSW	28.1 ± 1.5	14.2 ± 0.8	6.24 ± 1.19	0.46 ± 0.06	0.48 ± 0.21	0	9000	0.56	0.968 ± 0.003
pgvector IVFFlat	28.7 ± 1.9	15.3 ± 0.7	6.36 ± 0.20	0.45 ± 0.17	0.37	0	9000	0.56	0.969 ± 0.001
Qdrant	29.7 ± 3.2	15.0 ± 1.8	5.79 ± 1.41	0.35 ± 0.10	0.40 ± 0.07	0	9000	0.72 ± 0.01	0.968
Chroma	28.7 ± 2.7	14.4 ± 1.5	6.32 ± 0.91	0.42 ± 0.04	0.36 ± 0.20	0	9000	1.11 ± 0.03	0.968 ± 0.003
Milvus	28.9 ± 2.9	14.9 ± 3.4	6.31 ± 0.72	0.40 ± 0.43	0.38 ± 0.17	0	9000	0.94 ± 0.01	0.969 ± 0.002
LanceDB	29.4 ± 1.4	15.6 ± 1.2	6.82 ± 1.21	0.41 ± 0.09	0.42 ± 0.32	0	9000	0.84	0.968 ± 0.003
Weaviate	29.2 ± 2.8	15.0 ± 2.0	6.51 ± 1.41	0.42 ± 0.39	0.34 ± 0.06	0	9000	0.99 ± 0.01	0.968 ± 0.001
FAISS HNSW	28.5 ± 2.4	15.3 ± 1.1	6.11 ± 0.18	0.37 ± 0.09	0.40 ± 0.17	0	9000	0.18	0.968 ± 0.002
FAISS IVF	29.2 ± 2.7	14.6 ± 1.2	6.87 ± 1.94	0.41 ± 0.19	0.53 ± 0.21	0	9000	0.18	0.968 ± 0.002
FAISS flat	30.4 ± 1.8	15.9 ± 1.3	5.92 ± 1.99	0	0	0	9000	0.23	1.000

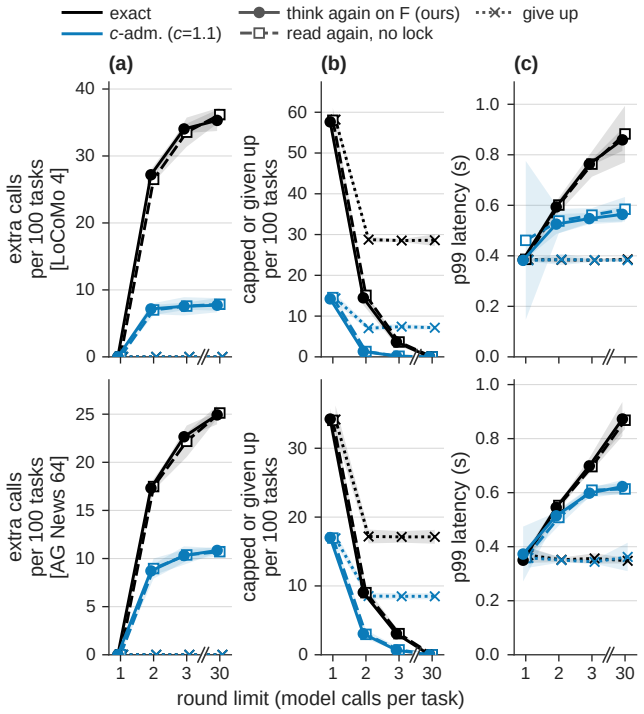


Figure A35: Strategy after a failed validation, by round limit (limits evenly spaced; the axis is broken before 30).

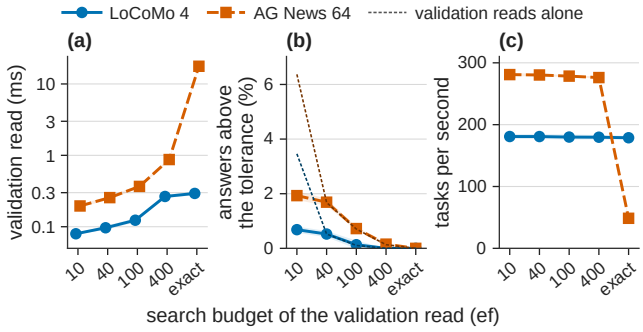


Figure A36: Search budget of the validation read: (a) read time (log scale), (b) answers above c, (c) throughput.

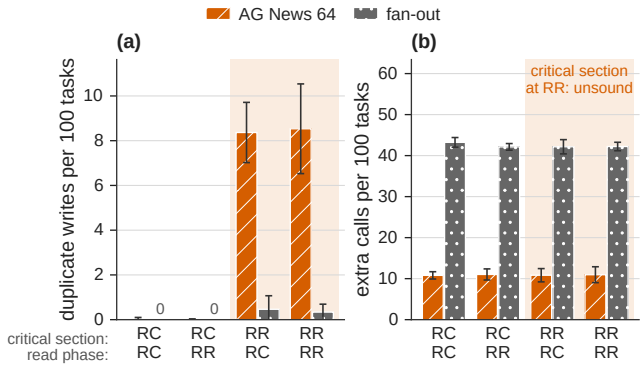


Figure A37: Isolation levels of the two phases on PostgreSQL.

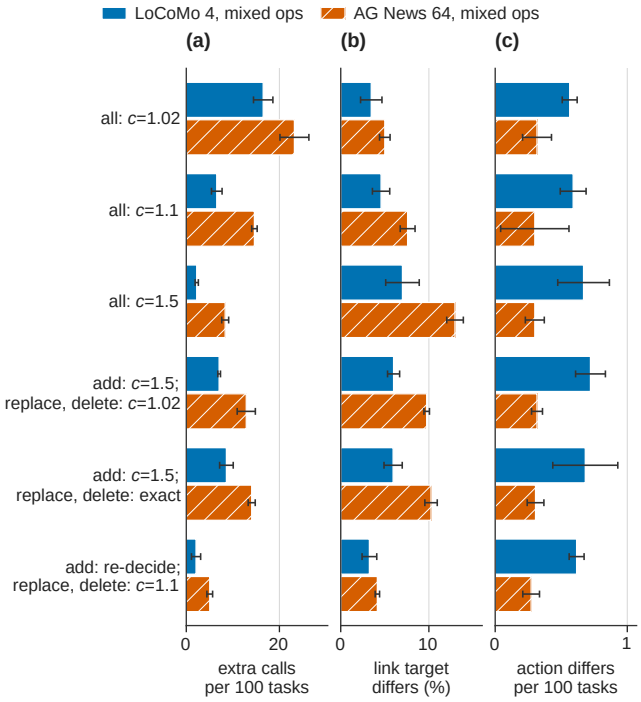


Figure A38: Per-operation tolerance.

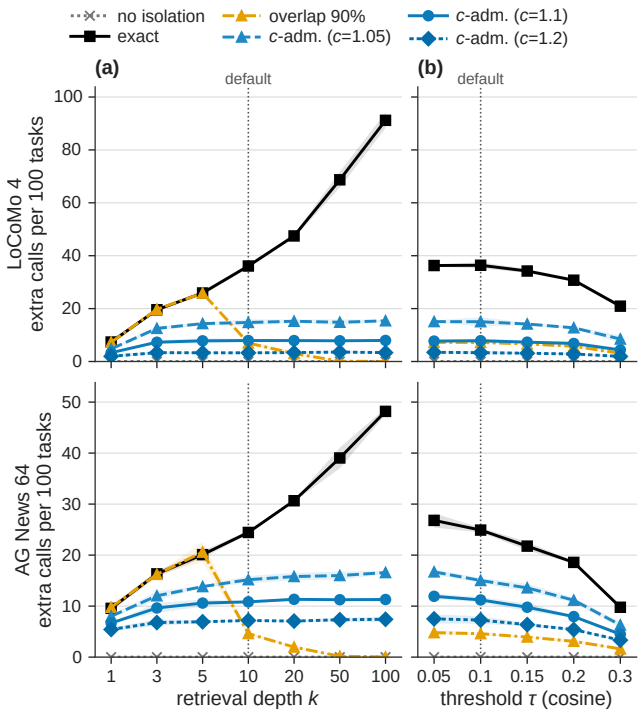


Figure A39: Extra calls against the retrieval depth and the threshold; each sweep varies one parameter with the other at its default (dotted line: $k = 10$, $\tau = 0.1$).

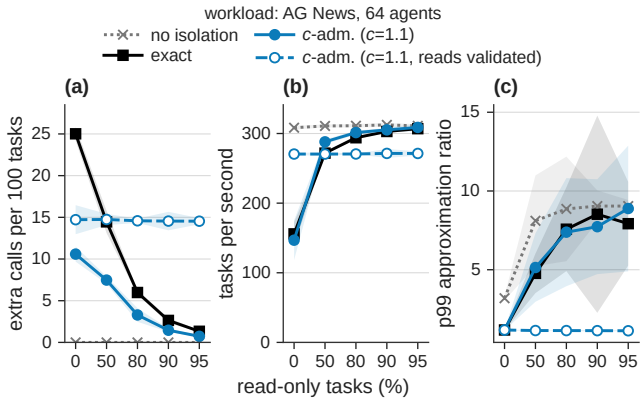


Figure A40: Share of read-only tasks (AG News, 64 agents).

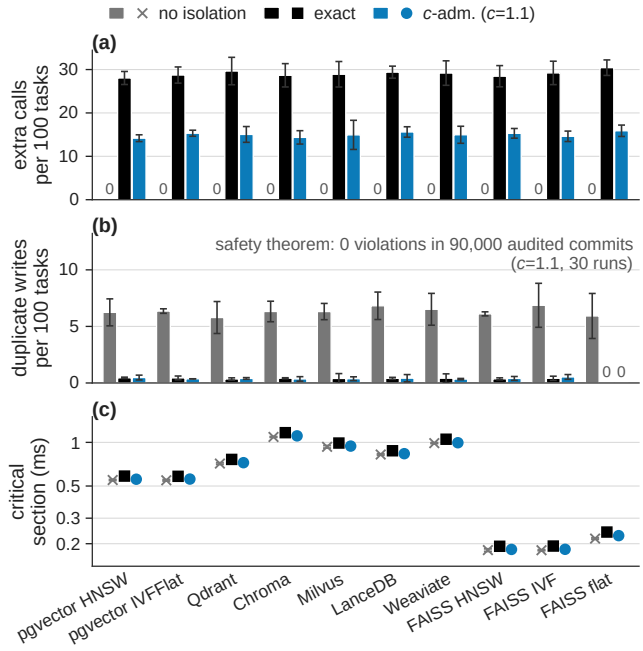


Figure A41: Stores: (a) extra calls, (b) duplicate writes, (c) mean time in the critical section (log scale; without isolation, the time of the write itself).

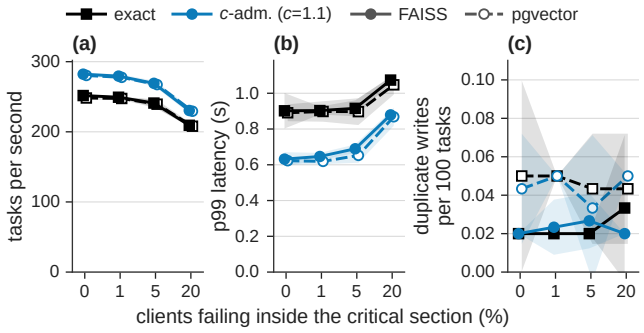


Figure A42: Client failures inside the critical section.

E Validating the Auditor

The auditor is the instrument behind every quality number, so we tested whether it detects protocols that are wrong.

Broken variants. We ran three variants that look correct: validation without mutual exclusion; one commit lock per team on a store shared by all teams; and, on PostgreSQL, a critical section at repeatable read, whose snapshot is taken before the advisory lock is granted. Figure A46 shows the outcome. Without a lock, co-arriving copies validate concurrently and are stored twice (0.64 duplicates per 100 tasks on the fan-out workload); the correct protocol writes only the duplicates that the index itself misses (0.09 per 100 tasks on the fan-out workload, its index floor). One lock per team is indistinguishable from the correct protocol on the fan-out workload, which has one team, and on LoCoMo, whose teams write about different conversations. On the news stream split into teams, which write about the same things, it stores 0.08 duplicates per 100 tasks against 0.06, a difference that three seeds do not resolve (the ten seeds of the lock-scope ablation do, Table A6): the variant is unsafe by construction, but this workload rarely exposes it. The repeatable-read variant is instructive: the inequality of Theorem 4.3 between the held answer and the fresh answer still holds, because the fresh answer is itself stale; the auditor exposes it through the ratio of the fresh answer measured in the commit state, which exceeds 1.1 for 2.4 to 4.1 points more of the tasks than at the index's own level, and through duplicate writes that the validation read should have prevented (Figure A46(a)).

Fault injection. Into the logs of correct runs we planted four kinds of faults and audited again: a commit that used the answer of its first read although validation had rejected it (3543 of 3552 detected as violations of the safety bound; each of the others still satisfies the bound, because the fresh answer at commit is itself approximate), a commit position moved back in the log (482 of 1332; a move that does not cross a relevant write is not an error), an answer that contains an item absent from the commit state (1451 of 1451), and distances misreported by the store (1452 of 1452; the auditor recomputes all distances from the vectors). The unmodified logs of the correct runs have no finding.

F Reproducibility

All experiments, audits, analyses, figures, and tables are produced by one framework from one command per experiment, with some seed counts and the MS MARCO subset set on the command line (`experiments/vldb/run_perf.sh`); two steps are manual (building the patched pgvector, and the dense counterfactual replay of Figure 4 of the paper); the repository is linked in the artifact block of the paper. Every experiment is a configuration file that states its question, its grid, and the audits that run on its logs; every measured number in the paper and in this appendix is read from the result files by a script (only software versions, the model names, the machines, and a few counts (data sets, baselines, models, each checked against the runs) are entered by hand), and the submission build fails if a number, a figure, or a table is missing. Each run logs, per task, the answers it reasoned about, the fresh answers, the commit-log positions, and the decision, and a commit log of all writes; the auditor reconstructs the exact state at any log position from the workload and the log, so audits need neither the store nor the index. The PostgreSQL tests reproduce the findings of §3 on a stock installation, including the debugger-based reproduction, and the patch is included. The reported runs use PostgreSQL 16.15 with pgvector 0.8.7 and the embedding model WordLlama (256 dimensions). Every experiment whose results include a time or a rate (throughput, latency, critical section, lock wait and occupancy, insertion time) and both language-model experiments ran on two dedicated 64-core servers (Intel Xeon Platinum 8375C, SMT off, 256 GB of memory): one runs the timing experiments, store and clients together, and the other the model and count experiments, whose open model is served by vLLM on one NVIDIA L40S (48 GB). Every time and every rate in the paper and in this report therefore comes from these servers. The models that decide are GPT-4o mini (gpt-4o-mini-2024-07-18) and Qwen2.5-7B-Instruct; GPT-4o mini answers and judges the downstream questions on the final memory of every seed. The remaining experiments only count events (the races of §3.1, the safety and separation audits, the conflict-law runs, the counterfactual replays, the index and tolerance sweeps, and the robustness runs over embedding models, seeds, retrieval depths, and metrics) and ran on the second of these servers; a third, a 192-core AMD EPYC 9R14 server, repeats the cost experiment on every workload but MS MARCO. On the cells both ran, the extra calls agree within 0.4% on average (both ends of the 90% interval of the mean offset, a random-effects estimate over 156 cells) and within 7% on every cell with five or more extra calls per 100 tasks. Numbers are means over seeds; an interval is the 95% Student t interval of the mean over seeds unless noted. In the generated tables, "$a \pm h$" gives the mean a and the half-width h of that interval (half the range with two seeds), and a mean printed without ± has an interval that rounds to zero. Every configuration is run with three seeds, except the model experiments, the conflict-law runs on real streams, the planted-law simulations, and the exchangeable controls (five seeds), the seed-variance experiment and the lock-scope ablation (ten), the anomaly taxonomy (forty), and the long run and the Euclidean-distance experiment (one seed each); the cost experiment has eight seeds on L2, L4, and A64 and three elsewhere.

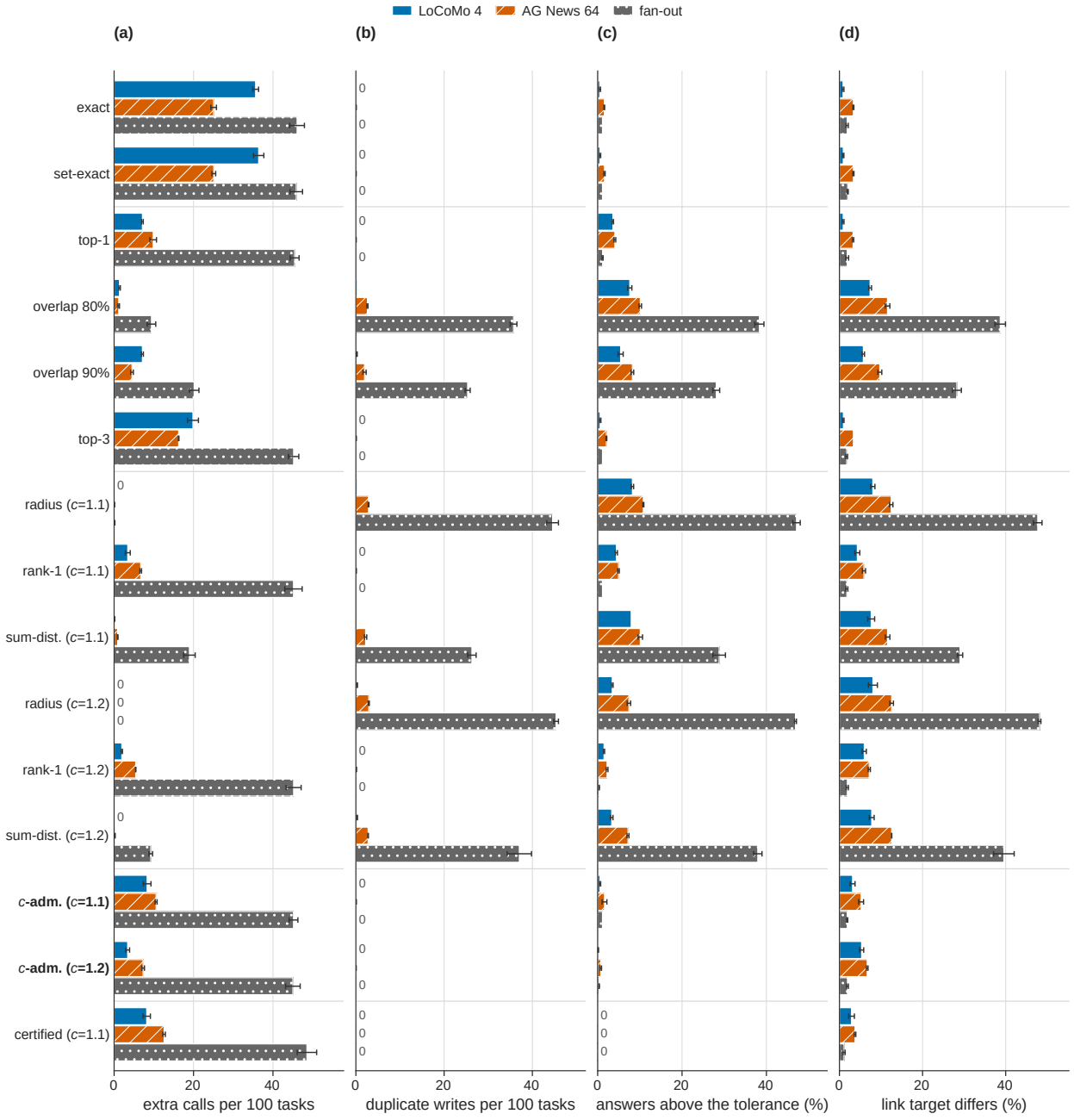


Figure A43: Predicate components.

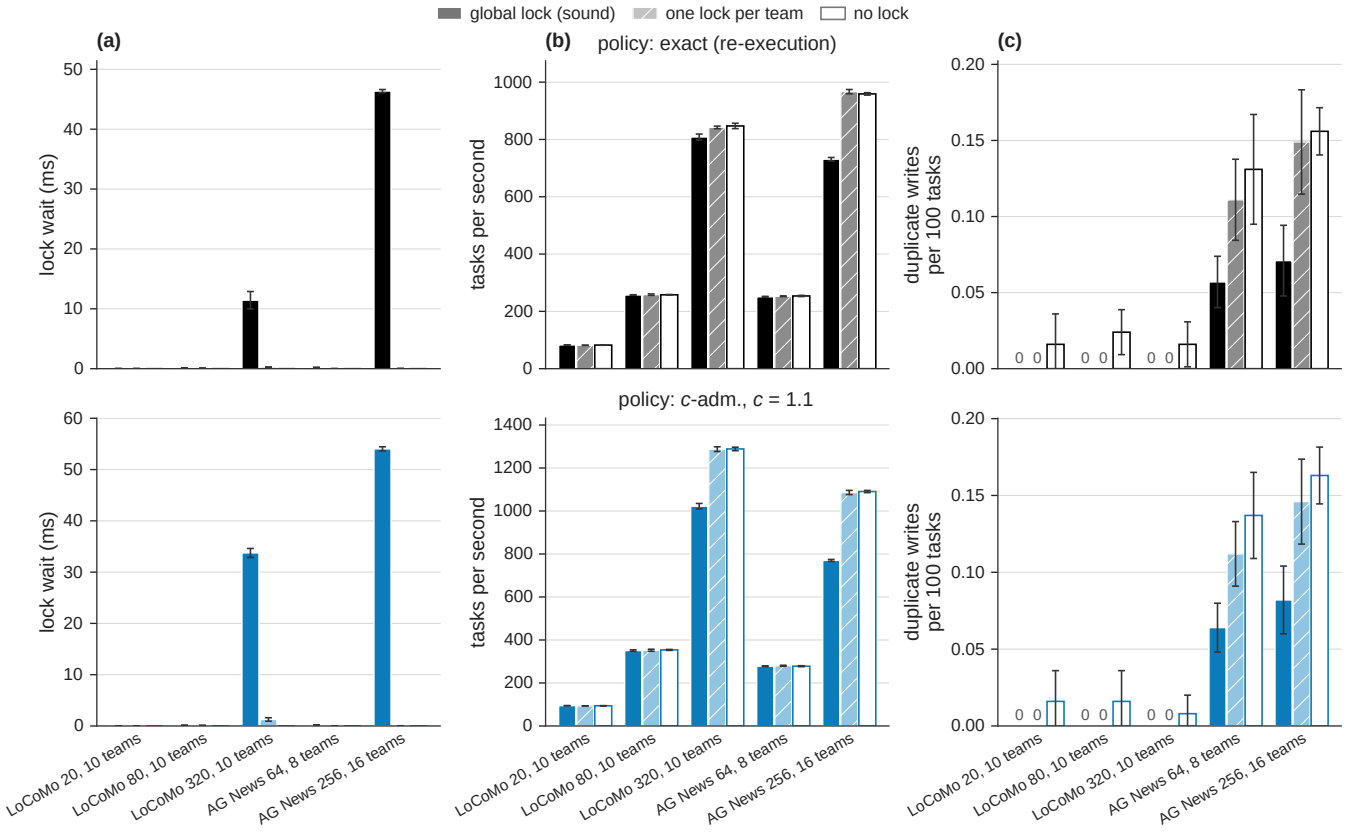


Figure A44: Scope of the commit lock (top: re-execution validation; bottom: $c = 1.1$; fill and hatch: lock scope). LoCoMo has one team per conversation: 10 teams of 2, 8 or 32 agents.

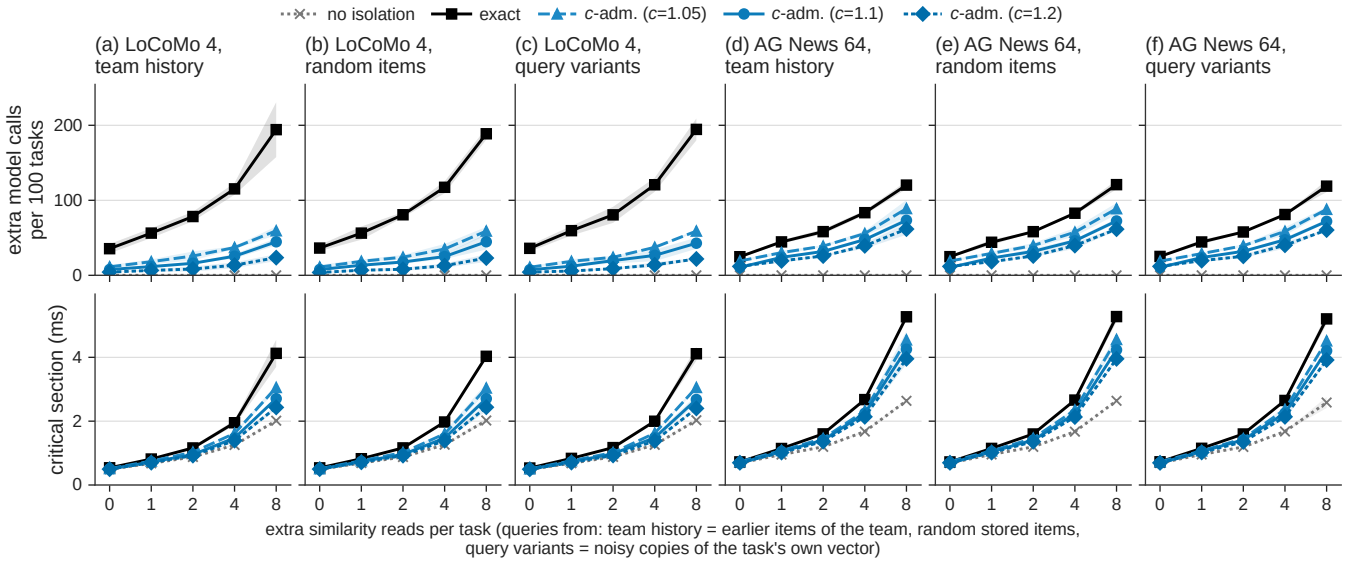


Figure A45: Transactions with several reads.

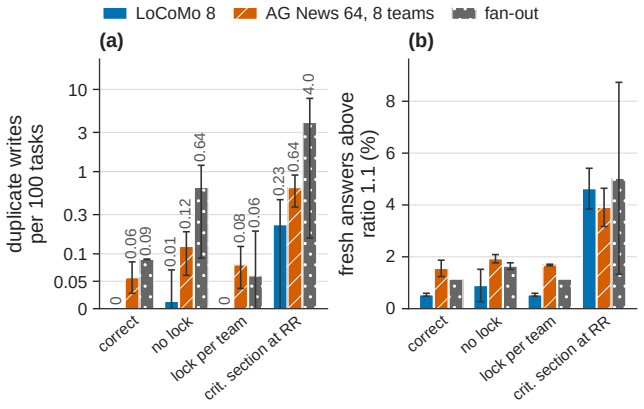


Figure A46: Deliberately broken variants of the protocol, as seen by the auditor: (a) duplicate writes per 100 tasks (symmetric log scale; means on the bars), (b) fresh answers above ratio 1.1 (%). Duplicates counted from the validation reads equal those in (a) in every run.

G Related Work

Isolation and concurrency control. Our definition builds on predicate- and state-based accounts of isolation [1, 2, 13, 14, 21, 35, 40, 48, 114] and on the analysis of snapshot isolation [18, 42, 117]. Phantom protection for ordered indexes [47, 77, 86, 99, 100] and main-memory engines [36, 66, 75, 106, 143], evaluations of concurrency control [160, 172], weaker models [8, 9], and linearizability [54] assume exact reads. History checkers [16, 70, 141] share the intent of our auditor, which must first decide which answers are legal for an approximate read. Epsilon serializability [119, 122], conits [10, 170], relaxed specifications of concurrent objects [3, 4, 53], and intermediate value linearizability [126, 127, 146] bound numeric error; admissibility bounds a ranked set, by a ratio at every rank, in a serial order of multi-operation transactions.

Multidimensional and nearest-neighbor queries. R-trees and GiST [51, 52, 72] protect region predicates against phantoms through predicate and node locks [73], granular locks [22, 23], and grid cells [76, 134], for predicates known before the search over low-dimensional data; linearizable nearest-neighbor search exists for exact, low-dimensional trees [24, 94, 96]. Deciding which open reads an insertion affects is a reverse nearest-neighbor problem [5, 19, 71, 132, 142, 161], akin to continuous query monitoring [17, 88, 103, 164, 173]. Nearest-neighbor and top-k processing [15, 41, 55, 60, 61, 128], duplicate detection [26, 39], and metric integrity constraints [67] do not treat isolation.

Approximate indexes and vector databases. Graph and quantization indexes [7, 29, 38, 43–45, 63, 65, 80, 92, 93, 136, 149], their analyses [25, 37, 57, 62, 78, 154], filtered, parallel, and GPU variants [46, 95, 110, 116], and work that keeps an index fresh under updates [49, 101, 102, 133, 162, 165, 167, 171, 180] or versioned [33, 152] reason about the index, not about what a read in a transaction may return. Vector database systems [27, 50, 58, 82, 83, 85, 107, 109, 118, 130, 138, 144, 148, 157, 168, 176, 178], surveys [32, 64, 90, 112, 113, 140, 147], and store documentation [31, 74, 98, 108, 120, 121, 155, 156] discuss replica consistency and snapshot visibility; reliability studies [150, 163] call for testing, public safety analyses cover no vector store yet [69], and a practitioner note suspects a repeatable-read problem in pgvector [135].

Agent memory and transactional agents. Agent memory architectures [30, 59, 111, 115, 123, 125, 137, 151, 153, 166, 175, 179, 181], multi-agent frameworks [56, 79, 159], and semantic caches [12, 129] produce our workload. Analyses of multi-agent failures and data systems for agents [20, 84, 97, 169, 174] motivate transactional agents, and recent proposals bring concurrency control to tool calls, objects, and workflow resources [68, 81, 89, 104, 139]; none, including the remark that agent searches resemble predicate reads [105], defines what a similarity read may return. We use standard, publicly available embedding models and benchmarks [6, 11, 28, 34, 91, 124, 131, 145, 158, 177], among them LoCoMo and AG News.

H Notation and Terms

This appendix collects every symbol and term of the paper and of this report, with the place where it is defined. Symbols used only inside one proof are defined there and not repeated here.

Stores, reads, and answers (§2).

S, S_0, N	state (the set of stored items), initial state, number of stored items.
q, k, $\textsc{Read}(q,k)$	query vector, number of neighbors, similarity read returning k distinct items in non-decreasing distance from q.
d, $\delta_i(S,q)$	metric on the vectors (chord distance in the evaluation, $\sqrt{2}$ times the square root of the cosine distance); i-th smallest distance from q to S.
M, ef	HNSW degree; search budget, the number of candidates a search keeps.
c, $\mathrm{Adm}_c(S,q,k)$	tolerance ($c \geq 1$); the set of c-admissible answers, lists whose i-th item is at most c times as far from q as the true i-th neighbor, for every i (Definition 2.1).
$\rho(A \mid S,q)$	approximation ratio of the list A in S, the least c for which A is c-admissible.
ρ	a bound on the approximation ratios of the answers a store returns (§4.2); $\rho = 1$ for an exact search. ρ_{99} is the 99th percentile of a ratio over transactions (Table 4).
c-store	a store whose read may return any element of Adm_c (§A.2).

Transactions and isolation (§§2 and 4).

T, write set	transaction; its insertions and deletions, buffered and applied atomically at commit. A replacement is a deletion and an insertion.
read-think-write	a transaction whose computation between its reads and its writes is a model call.
τ, τ-deduplicating insertion	deduplication threshold (stated in cosine distance); a transaction that issues one read and inserts q only if its nearest item is farther than τ.
s-separated	a state in which any two items are farther apart than s (Proposition 4.2).
held answer A	the answer a transaction acted on.
execution, π, S_j^π	an initial state and a set of committed transactions; a total order of them; the state after the write sets of the first j transactions in π.
c-serializable	there is an order π in which every held answer is c-admissible in the state just before its transaction (Definition 4.1).

Protocol and system (§§4.2 and 4.3).

fresh answer F, validation read	the answer of the read repeated at commit, under the commit lock.
$\sqsubseteq_c$	validation test: $\|A\| = \|F\|$, every item of A is still stored, and $d(q, a_i) \le c\, d(q, f_i)$ for every i.
admissibility validation	Algorithm 1 with the test $\sqsubseteq_c$; with an exact validation read it is *certified*.
re-execution validation	validation that commits only if $A = F$ (the predicate exact).
round	one pass through the loop of Algorithm 1: think, validate, and commit or retry; a transaction aborts after 30 failed rounds.
commit lock, critical section	the lock under which validation reads and writes run; the time spent holding it ("c.s." in the tables).
auditor	the tool that replays the commit log, rebuilds the exact state at every commit, and reports the approximation ratio of every held and fresh answer.

Conflicts (§5).

$r_0, \dots, r_k$	$r_0 = 0$ and the distances of the exact answer, $r_1 < \cdots < r_k$; the read's ball has radius r_k.
in-ball insertion, t, p	an insertion at distance $t < r_k$ from q; its rank p in the new list.
conflict, $R_c(q)$	an insertion that makes the held answer inadmissible; the conflict radius, inside which an insertion conflicts (Corollary 5.2).
y	position $\ln(r_k/t)$ of an in-ball insertion.
overlap	a write overlaps a read if it commits between the read and its validation; n is the number of background writes that overlap a read.
D, intrinsic dimension	the exponent of the growth of the number of points with distance around q; λs^D points lie within distance s (§A.6).
mass coordinate ω, μ_i, B_i	$\omega = \lambda s^D$; masses of the stored neighbors; ratios μ_i/μ_{i+1} of consecutive masses (§A.6).
background write	a write unrelated to the read, exchangeable with the stored items.
sibling write, D_s	a write about the same thing as the read (a rendering of the same fact, the next fact of the same conversation); its radial density is proportional to t^{D_s-1} with sibling dimension $D_s \le D$.
α, x	dimension ratio D_s/D; $x = c^{-D}$.
$\theta_\alpha(p)$, $\varphi_\alpha(p)$, $P_\alpha(c)$	probability that an in-ball insertion has rank p; that it has rank p and leaves the answer admissible; conflict probability of one in-ball insertion (Theorem 5.3).
σ, two scales	share of sibling writes among in-ball insertions; a mixture of two sibling dimensions with their shares σ_j, fitted when one dimension does not describe a workload.
exchangeable model, closed form, semi-parametric form	the law with every write treated as background; Theorem 5.3 with the pooled intrinsic dimension; the sibling law evaluated on the read's actual distances (§5.2).

The lag kernel (§5.3 and §A.7).

lag ℓ, lag bucket b	distance in the stream between the reading item and the source of a write, in the same stream (the same conversation or feed); a range of lags.
h_ℓ, D_ℓ; $\hat{h}_b$, $\hat{D}_b$	probability that the item at lag ℓ falls into the read's ball, and its sibling dimension; their estimates per lag bucket.
o_ℓ, o_b, ν	expected number of overlapping writes at lag ℓ (or in bucket b); expected in-ball insertions per read.
pooled parameters	sibling dimensions and shares fitted over all lags of one run.
static stream	the estimate of the kernel from the stream alone, without any execution.
transfer	a prediction for a run from parameters fitted on another run (another concurrency level, k, time, collection, or the static stream).
w_{bg}, f_{bg}	background share of in-ball insertions, fixed from first principles; background density of y (§A.7).

Evaluation (§6).

fan-out workload	600 facts, each written by one, two, or four agents as perturbed copies (§3.1).
L2, L4, L8	LoCoMo with teams of 2, 4, or 8 agents per conversation; g is the team size.
A16, A64, A64k50, A256, A64-long	AG News with 16, 64, or 256 agents; A64k50 reads $k = 50$; A256 writes 12,000 items, A64-long 30,000.
LME4, QQP64, MSM64	LongMemEval with teams of four agents; Quora questions with 64 agents; MS MARCO passages with 64 agents.
extra calls	model calls beyond one per task, per 100 tasks.
decision proxies	dup (is the nearest item within τ), tag (majority label of the k items), link (identity of the nearest item), top3 (identity of the three nearest), set (identity of all k).
predicates	none (no isolation), exact (re-execution), set, overlap (at least 90% common identifiers), objects (the items read still exist), the k-th distance (the radius), the first distance, and the sum of distances only.
index floor	the share of transactions whose decision on the fresh answer differs from that on the exact answer at commit.

References

[1] Atul Adya. 1999. *Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions.* Ph.D. dissertation. MIT. https://pmg.csail.mit.edu/pubs/adya99__weak_consis-abstract.html

[2] Atul Adya, Barbara Liskov, and Patrick E. O'Neil. 2000. Generalized Isolation Level Definitions. In *Proc. ICDE'00.* IEEE Computer Society, 67–78.

[3] Yehuda Afek, Guy Korland, and Eitan Yanovsky. 2010. Quasi-Linearizability: Relaxed Consistency for Improved Concurrency. In *Proc. OPODIS'10 (Lecture Notes in Computer Science)*, Vol. 6490. Springer, 395–410.

[4] Dan Alistarh, Trevor Brown, Justin Kopinsky, Jerry Z. Li, and Giorgi Nadiradze. 2018. Distributionally Linearizable Data Structures. *CoRR* abs/1804.01018 (2018).

[5] David Arthur and Steve Y. Oudot. 2010. Reverse Nearest Neighbors Search in High Dimensions using Locality-Sensitive Hashing. *CoRR* abs/1011.4955 (2010).

[6] Martin Aumüller, Erik Bernhardsson, and Alexander Faithfull. 2020. ANN-Benchmarks: A benchmarking tool for approximate nearest neighbor algorithms. *Inf. Syst.* 87 (2020), 101374.

[7] Artem Babenko and Victor Lempitsky. 2012. The Inverted Multi-Index. In *Proc. CVPR'12.* IEEE Computer Society, 3069–3076.

[8] Peter Bailis, Aaron Davidson, Alan Fekete, Ali Ghodsi, Joseph M. Hellerstein, and Ion Stoica. 2013. Highly Available Transactions: Virtues and Limitations. *Proc. VLDB Endow.* 7, 3 (2013), 181–192.

[9] Peter Bailis, Alan Fekete, Michael J. Franklin, Ali Ghodsi, Joseph M. Hellerstein, and Ion Stoica. 2014. Coordination Avoidance in Database Systems. *Proc. VLDB Endow.* 8, 3 (2014), 185–196.

[10] Peter Bailis, Shivaram Venkataraman, Michael J. Franklin, Joseph M. Hellerstein, and Ion Stoica. 2012. Probabilistically Bounded Staleness for Practical Partial Quorums. *Proc. VLDB Endow.* 5, 8 (2012), 776–787.

[11] Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, Mir Rosenberg, Xia Song, Alina Stoica, Saurabh Tiwary, and Tong Wang. 2016. MS MARCO: A Human Generated MAchine Reading COmprehension Dataset. *CoRR* abs/1611.09268 (2016).

[12] Fu Bang. 2023. GPTCache: An Open-Source Semantic Cache for LLM Applications Enabling Faster Answers and Cost Savings. In *Proc. NLP-OSS'23.* Association for Computational Linguistics, 212–218.

[13] Hal Berenson, Philip A. Bernstein, Jim Gray, Jim Melton, Elizabeth J. O'Neil, and Patrick E. O'Neil. 1995. A Critique of ANSI SQL Isolation Levels. In *Proc. SIGMOD'95.* ACM, 1–10.

[14] Philip A. Bernstein, Vassos Hadzilacos, and Nathan Goodman. 1987. *Concurrency Control and Recovery in Database Systems.* Addison-Wesley.

[15] Kevin S. Beyer, Jonathan Goldstein, Raghu Ramakrishnan, and Uri Shaft. 1999. When Is "Nearest Neighbor" Meaningful?. In *Proc. ICDT'99 (Lecture Notes in Computer Science)*, Vol. 1540. Springer, 217–235.

[16] Ranadeep Biswas and Constantin Enea. 2019. On the Complexity of Checking Transactional Consistency. *Proc. ACM Program. Lang.* 3, OOPSLA (2019), 165:1–165:28.

[17] Christian Böhm, Beng Chin Ooi, Claudia Plant, and Ying Yan. 2007. Efficiently Processing Continuous k-NN Queries on Data Streams. In *Proc. ICDE'07.* IEEE Computer Society, 156–165.

[18] Michael J. Cahill, Uwe Röhm, and Alan D. Fekete. 2008. Serializable Isolation for Snapshot Databases. In *Proc. SIGMOD'08.* ACM, 729–738.

[19] Guillaume Casanova, Elias Englmeier, Michael E. Houle, Peer Kröger, Michael Nett, Erich Schubert, and Arthur Zimek. 2017. Dimensional Testing for Reverse k-Nearest Neighbor Search. *Proc. VLDB Endow.* 10, 7 (2017), 769–780.

[20] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. 2025. Why Do Multi-Agent LLM Systems Fail?. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025).* 135676–135729.

[21] Andrea Cerone, Giovanni Bernardi, and Alexey Gotsman. 2015. A Framework for Transactional Consistency Models with Atomic Visibility. In *Proc. CONCUR'15 (LIPIcs)*, Vol. 42. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 58–71.

[22] Kaushik Chakrabarti and Sharad Mehrotra. 1998. Dynamic Granular Locking Approach to Phantom Protection in R-Trees. In *Proc. ICDE'98.* IEEE Computer Society, 446–454.

[23] Kaushik Chakrabarti and Sharad Mehrotra. 1999. Efficient Concurrency Control in Multidimensional Access Methods. In *Proc. SIGMOD'99.* ACM, 25–36.

[24] Bapi Chatterjee, Ivan Walulya, and Philippas Tsigas. 2015. *Concurrent Linearizable Nearest Neighbour Search in LockFree-kD-tree.* Technical Report 2015-08. Department of Computer Science and Engineering, Chalmers University of Technology and Göteborg University. https://research.chalmers.se/en/publication/232185

[25] Manos Chatzakis, Yannis Papakonstantinou, and Themis Palpanas. 2025. DARTH: Declarative Recall Through Early Termination for Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data* 3, 4 (2025), 242:1–242:26.

[26] Surajit Chaudhuri, Venkatesh Ganti, and Raghav Kaushik. 2006. A Primitive Operator for Similarity Joins in Data Cleaning. In *Proc. ICDE'06.* IEEE Computer Society, 5.

[27] Cheng Chen, Chenzhe Jin, Yunan Zhang, Sasha Podolsky, Chun Wu, Szu-Po Wang, Eric Hanson, Zhou Sun, Robert Walzer, and Jianguo Wang. 2024. SingleStore-V: An Integrated Vector Database System in SingleStore. *Proc. VLDB Endow.* 17, 12 (2024), 3772–3785.

[28] Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In *Proc. ACL Findings'24.* Association for Computational Linguistics, 2318–2335.

[29] Qi Chen, Bing Zhao, Haidong Wang, Mingqin Li, Chuanjie Liu, Zengzhong Li, Mao Yang, and Jingdong Wang. 2021. SPANN: Highly-efficient Billion-scale Approximate Nearest Neighborhood Search. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021).* 5199–5212.

[30] Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory. In *Proc. ECAI'25 (Frontiers in Artificial Intelligence and Applications)*, Vol. 413. IOS Press.

[31] Chroma. 2026. Chroma: Open-source Search and Retrieval Database for AI Applications. https://github.com/chroma-core/chroma. Version 1.5.9.

[32] Yannis Chronis, Helena Caminal, Yannis Papakonstantinou, Fatma Özcan, and Anastasia Ailamaki. 2025. Filtered Vector Search: State-of-the-art and Research Opportunities. *Proc. VLDB Endow.* 18, 12 (2025), 5488–5492.

[33] Jun Woo Chung and Weijie Zhao. 2026. A Versioned Unified Graph Index for Dynamic Timestamp-Aware Nearest Neighbor Search. *CoRR* abs/2608.27663 (2026).

[34] Brian F. Cooper, Adam Silberstein, Erwin Tam, Raghu Ramakrishnan, and Russell Sears. 2010. Benchmarking Cloud Serving Systems with YCSB. In *Proc. SoCC'10.* ACM, 143–154.

[35] Natacha Crooks, Youer Pu, Lorenzo Alvisi, and Allen Clement. 2017. Seeing is Believing: A Client-Centric Specification of Database Isolation. In *Proc. PODC'17.* ACM, 73–82.

[36] Cristian Diaconu, Craig Freedman, Erik Ismert, Per-Åke Larson, Pravin Mittal, Ryan Stonecipher, Nitin Verma, and Mike Zwilling. 2013. Hekaton: SQL Server's Memory-Optimized OLTP Engine. In *Proc. SIGMOD'13.* ACM, 1243–1254.

[37] Haya Diwan, Jinrui Gou, Cameron Musco, Christopher Musco, and Torsten Suel. 2024. Navigable Graphs for High-Dimensional Nearest Neighbor Search: Constructions and Limits. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024).* 59513–59531.

[38] Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. 2026. The Faiss Library. *IEEE Trans. Big Data* 12, 2 (2026), 346–361.

[39] Ahmed K. Elmagarmid, Panagiotis G. Ipeirotis, and Vassilios S. Verykios. 2007. Duplicate Record Detection: A Survey. *IEEE Trans. Knowl. Data Eng.* 19, 1 (2007), 1–16.

[40] Kapali P. Eswaran, Jim N. Gray, Raymond A. Lorie, and Irving L. Traiger. 1976. The Notions of Consistency and Predicate Locks in a Database System. *Commun. ACM* 19, 11 (1976), 624–633.

[41] Ronald Fagin, Amnon Lotem, and Moni Naor. 2001. Optimal Aggregation Algorithms for Middleware. In *Proc. PODS'01*. ACM, 102–113.

[42] Alan Fekete, Dimitrios Liarokapis, Elizabeth O'Neil, Patrick O'Neil, and Dennis Shasha. 2005. Making Snapshot Isolation Serializable. *ACM Trans. Database Syst.* 30, 2 (2005), 492–528.

[43] Cong Fu, Chao Xiang, Changxu Wang, and Deng Cai. 2019. Fast Approximate Nearest Neighbor Search With The Navigating Spreading-out Graph. *Proc. VLDB Endow.* 12, 5 (2019), 461–474.

[44] Jianyang Gao, Yutong Gou, Yuexuan Xu, Yongyi Yang, Cheng Long, and Raymond Chi-Wing Wong. 2025. Practical and Asymptotically Optimal Quantization of High-Dimensional Vectors in Euclidean Space for Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data* 3, 3 (2025), 202:1–202:26.

[45] Jianyang Gao and Cheng Long. 2024. RaBitQ: Quantizing High-Dimensional Vectors with a Theoretical Error Bound for Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data* 2, 3 (2024), 167:1–167:27.

[46] Siddharth Gollapudi, Neel Karia, Varun Sivashankar, Ravishankar Krishnaswamy, Nikit Begwani, Swapnil Raz, Yiyong Lin, Yin Zhang, Neelam Mahapatro, Premkumar Srinivasan, Amit Singh, and Harsha Vardhan Simhadri. 2023. Filtered-DiskANN: Graph Algorithms for Approximate Nearest Neighbor Search with Filters. In *Proc. WWW'23*. ACM, 3406–3416.

[47] Goetz Graefe. 2010. A Survey of B-Tree Locking Techniques. *ACM Trans. Database Syst.* 35, 3 (2010), 16:1–16:26.

[48] Jim Gray, Raymond A. Lorie, Gianfranco R. Putzolu, and Irving L. Traiger. 1976. Granularity of Locks and Degrees of Consistency in a Shared Data Base. In *Proc. IFIP Working Conference on Modelling in Data Base Management Systems*. North-Holland.

[49] Hao Guo and Youyou Lu. 2026. OdinANN: Direct Insert for Consistently Stable Performance in Billion-Scale Graph-Based Vector Search. In *Proc. FAST'26*. USENIX Association, 133–147.

[50] Rentong Guo, Xiaofan Luan, Long Xiang, Xiao Yan, Xiaomeng Yi, Jigao Luo, Qianya Cheng, Weizhi Xu, Jiarui Luo, Frank Liu, Zhenshan Cao, Yanliang Qiao, Ting Wang, Bo Tang, and Charles Xie. 2022. Manu: A Cloud Native Vector Database Management System. *Proc. VLDB Endow.* 15, 12 (2022), 3548–3561.

[51] Antonin Guttman. 1984. R-Trees: A Dynamic Index Structure for Spatial Searching. In *Proc. SIGMOD'84*. ACM, 47–57.

[52] Joseph M. Hellerstein, Jeffrey F. Naughton, and Avi Pfeffer. 1995. Generalized Search Trees for Database Systems. In *Proc. VLDB'95*. Morgan Kaufmann, 562–573.

[53] Thomas A. Henzinger, Christoph M. Kirsch, Hannes Payer, Ali Sezgin, and Ana Sokolova. 2013. Quantitative Relaxation of Concurrent Data Structures. In *Proc. POPL'13*. ACM, 317–328.

[54] Maurice P. Herlihy and Jeannette M. Wing. 1990. Linearizability: A Correctness Condition for Concurrent Objects. *ACM Trans. Program. Lang. Syst.* 12, 3 (1990), 463–492.

[55] Gísli R. Hjaltason and Hanan Samet. 1999. Distance Browsing in Spatial Databases. *ACM Trans. Database Syst.* 24, 2 (1999), 265–318.

[56] Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiawu Zheng, Yuheng Cheng, Jinlin Wang, Ceyao Zhang, Zili Wang, Steven Yau, Zijuan Lin, Liyang Zhou, Chenyu Ran, Lingfeng Xiao, Chenglin Wu, and Jürgen Schmidhuber. 2024. MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. In *Proc. ICLR'24*. OpenReview.net, 23247–23275.

[57] Sonia Horchidan, Fabian Zeiher, Henrik Boström, and Paris Carbone. 2025. ConANN: Conformal Approximate Nearest Neighbor Search. *Proc. VLDB Endow.* 19, 1 (2025), 29–42.

[58] Guoyu Hu, Shaofeng Cai, Tien Tuan Anh Dinh, Zhongle Xie, Cong Yue, Gang Chen, and Beng Chin Ooi. 2025. HAKES: Scalable Vector Database for Embedding Search Service. *CoRR* abs/2505.12524 (2025).

[59] Yuyang Hu, Shichun Liu, Yanwei Yue, Guibin Zhang, Boyang Liu, Fangyi Zhu, Jiahang Lin, Honglin Guo, Shihan Dou, Zhiheng Xi, Senjie Jin, Jiejun Tan, Yanbin Yin, Jiongnan Liu, Zeyu Zhang, Zhongxiang Sun, Yutao Zhu, Hao Sun, Boci Peng, Zhenrong Cheng, Xuanbo Fan, Jiaxin Guo, Xinlei Yu, Zhenhong Zhou, Zewen Hu, Jiahao Huo, Junhao Wang, Yuwei Niu, Yu Wang, Zhenfei Yin, Xiaobin Hu, Yue Liao, Qiankun Li, Kun Wang, Wangchunshu Zhou, Yixin Liu, Dawei Cheng, Qi Zhang, Tao Gui, Shirui Pan, Yan Zhang, Philip Torr, Zhicheng Dou, Ji-Rong Wen, Xuanjing Huang, Yu-Gang Jiang, and Shuicheng Yan. 2025. Memory in the Age of AI Agents. *CoRR* abs/2512.13564 (2025).

[60] Ihab F. Ilyas, George Beskales, and Mohamed A. Soliman. 2008. A Survey of Top-k Query Processing Techniques in Relational Database Systems. *ACM Comput. Surv.* 40, 4 (2008), 1–58.

[61] Piotr Indyk and Rajeev Motwani. 1998. Approximate Nearest Neighbors: Towards Removing the Curse of Dimensionality. In *Proc. STOC'98*. ACM, 604–613.

[62] Piotr Indyk and Haike Xu. 2023. Worst-case Performance of Popular Approximate Nearest Neighbor Search Implementations: Guarantees and Limitations. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. 66239–66256.

[63] Hervé Jégou, Matthijs Douze, and Cordelia Schmid. 2011. Product Quantization for Nearest Neighbor Search. *IEEE Trans. Pattern Anal. Mach. Intell.* 33, 1 (2011), 117–128.

[64] Zhi Jing, Yongye Su, Yikun Han, Bo Yuan, Haiyun Xu, Chunjiang Liu, Kehai Chen, and Min Zhang. 2024. When Large Language Models Meet Vector Databases: A Survey. *CoRR* abs/2402.01763 (2024).

[65] Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2021. Billion-Scale Similarity Search with GPUs. *IEEE Trans. Big Data* 7, 3 (2021), 535–547.

[66] J. R. Jordan, J. Banerjee, and R. B. Batman. 1981. Precision Locks. In *Proc. SIGMOD'81*. ACM, 143–147.

[67] Youri Kaminsky, Benny Kimelfeld, Ester Livshits, Felix Naumann, and David Wajc. 2025. Repairing Databases over Metric Spaces with Coincidence Constraints. In *Proc. ICDT'25 (LIPIcs)*, Vol. 328. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 14:1–14:18.

[68] Sajjad Khan. 2026. S-Bus: Automatic Read-Set Reconstruction for Multi-Agent LLM State Coordination. *CoRR* abs/2605.17076 (2026).

[69] Kyle Kingsbury. 2026. Jepsen Analyses. https://jepsen.io/analyses.

[70] Kyle Kingsbury and Peter Alvaro. 2020. Elle: Inferring Isolation Anomalies from Experimental Observations. *Proc. VLDB Endow.* 14, 3 (2020), 268–280.

[71] Flip Korn and S. Muthukrishnan. 2000. Influence Sets Based on Reverse Nearest Neighbor Queries. In *Proc. SIGMOD'00*. ACM, 201–212.

[72] Marcel Kornacker and Douglas Banks. 1995. High-Concurrency Locking in R-Trees. In *Proc. VLDB'95*. Morgan Kaufmann, 134–145.

[73] Marcel Kornacker, C. Mohan, and Joseph M. Hellerstein. 1997. Concurrency and Recovery in Generalized Search Trees. In *Proc. SIGMOD'97*. ACM, 62–72.

[74] LanceDB. 2026. LanceDB: Embedded Retrieval Engine for Multimodal AI. https://github.com/lancedb/lancedb.

[75] Per-Åke Larson, Spyros Blanas, Cristian Diaconu, Craig Freedman, Jignesh M. Patel, and Mike Zwilling. 2011. High-Performance Concurrency Control Mechanisms for Main-Memory Databases. *Proc. VLDB Endow.* 5, 4 (2011), 298–309.

[76] Seok Jae Lee, Seok Il Song, and Jae Soo Yoo. 2007. Phantom Protection Method for Multi-dimensional Index Structures. *International Journal of Contents* 3, 2 (2007), 6–17.

[77] Philip L. Lehman and S. Bing Yao. 1981. Efficient Locking for Concurrent Operations on B-Trees. *ACM Trans. Database Syst.* 6, 4 (1981), 650–670.

[78] Conglong Li, Minjia Zhang, David G. Andersen, and Yuxiong He. 2020. Improving Approximate Nearest Neighbor Search through Learned Adaptive Early Termination. In *Proc. SIGMOD'20*. ACM, 2539–2554.

[79] Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. 2023. CAMEL: Communicative Agents for "Mind" Exploration of Large Language Model Society. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. 51991–52008.

[80] Wen Li, Ying Zhang, Yifang Sun, Wei Wang, Mingjie Li, Wenjie Zhang, and Xuemin Lin. 2020. Approximate Nearest Neighbor Search on High Dimensional Data — Experiments, Analyses, and Improvement. *IEEE Trans. Knowl. Data Eng.* 32, 8 (2020), 1475–1488.

[81] Xiaoyang Li, Yiqi Wang, Haohui Lu, Zhi Chen, Mo Li, Pingan Song, Mingkai Zheng, and Taotao Cai. 2026. MemTX: Transactional Belief Commit for Stateful Agent Memory. *CoRR* abs/2607.23929 (2026).

[82] Jiayi Liu, Te Guo, and Jianguo Wang. 2026. Building An Integrated Vector Database System in PostgreSQL. *CoRR* abs/2608.15994 (2026).

[83] Jiayi Liu, Yunan Zhang, Chenzhe Jin, Aditya Gupta, Shige Liu, and Jianguo Wang. 2026. Fast Vector Search in PostgreSQL: A Decoupled Approach. In *Proc. CIDR'26*. www.cidrdb.org. https://vldb.org/cidrdb/2026/fast-vector-search-in-postgresql-a-decoupled-approach.html

[84] Shu Liu, Soujanya Ponnapalli, Shreya Shankar, Sepanta Zeighami, Alan Zhu, Shubham Agarwal, Ruiqi Chen, Samion Suwito, Shuo Yuan, Ion Stoica, Matei Zaharia, Alvin Cheung, Natacha Crooks, Joseph E. Gonzalez, and Aditya G. Parameswaran. 2026. Supporting Our AI Overlords: Redesigning Data Systems to be Agent-First. In *Proc. CIDR'26*. www.cidrdb.org. https://vldb.org/cidrdb/2026/supporting-our-ai-overlords-redesigning-data-systems-to-be-agent-first.html

[85] Shige Liu, Zhifang Zeng, Li Chen, Adil Ainihaer, Arun Ramasami, Songting Chen, Yu Xu, Mingxi Wu, and Jianguo Wang. 2025. TigerVector: Supporting Vector Search in Graph Databases for Advanced RAGs. In *Proc. SIGMOD Companion'25*. ACM, 553–565.

[86] David B. Lomet. 1993. Key Range Locking Strategies for Improved Concurrency. In *Proc. VLDB'93*. Morgan Kaufmann, 655–664.

[87] Chang Lu and Yizheng Zhao. 2026. HNSW under SERIALIZABLE: Two Concurrent Check-then-Insert Transactions Can Both Commit. pgvector issue #1055, https://github.com/pgvector/pgvector/issues/1055.

[88] Duo Lu, Siming Feng, Jonathan Zhou, Franco Solleza, Malte Schwarzkopf, and Uğur Çetintemel. 2025. VectraFlow: Integrating Vectors into Stream Processing. In *Proc. CIDR'25*. www.cidrdb.org. https://vldb.org/cidrdb/2025/vectraflow-integrating-vectors-into-stream-processing.html

[89] Hongtao Lyu, Dingyan Zhang, Mingyu Wu, Xingda Wei, and Haibo Chen. 2026. CoAgent: Concurrency Control for Multi-Agent Systems. *CoRR* abs/2606.15376 (2026).

[90] Le Ma, Ran Zhang, Yikun Han, Shirui Yu, Zaitian Wang, Zhiyuan Ning, Jinghan Zhang, Ping Xu, Pengjiang Li, Ziyue Qiao, Wei Ju, Chong Chen, Dongjie

Wang, Kunpeng Liu, Pengyang Wang, Pengfei Wang, Yanjie Fu, Chunjiang Liu, Yuanchun Zhou, and Chang-Tien Lu. 2023. A Comprehensive Survey on Vector Database: Storage and Retrieval Technique, Challenge. *CoRR* abs/2310.11703 (2023).

[91] Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating Very Long-Term Conversational Memory of LLM Agents. In *Proc. ACL'24*. Association for Computational Linguistics, 13851–13870.

[92] Yury Malkov, Alexander Ponomarenko, Andrey Logvinov, and Vladimir Krylov. 2014. Approximate nearest neighbor algorithm based on navigable small world graphs. *Inf. Syst.* 45 (2014), 61–68.

[93] Yu. A. Malkov and D. A. Yashunin. 2020. Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs. *IEEE Trans. Pattern Anal. Mach. Intell.* 42, 4 (2020), 824–836.

[94] Magdalen Dobson Manohar. 2025. *New Techniques for Parallelism and Concurrency in Nearest Neighbor Search*. Ph.D. Dissertation. Carnegie Mellon University.

[95] Magdalen Dobson Manohar, Zheqi Shen, Guy Blelloch, Laxman Dhulipala, Yan Gu, Harsha Vardhan Simhadri, and Yihan Sun. 2024. ParlayANN: Scalable and Deterministic Parallel Graph-Based Approximate Nearest Neighbor Search Algorithms. In *Proc. PPoPP'24*. ACM, 270–285.

[96] Magdalen Dobson Manohar, Yuanhao Wei, and Guy E. Blelloch. 2025. CLEANN: Lock-Free Augmented Trees for Low-Dimensional k-Nearest Neighbor Search. In *Proc. SPAA'25*. ACM, 131–143.

[97] Yanki Margalit, Nurit Cohen-Inger, Erni Avram, Ran Taig, and Oded Margalit. 2026. Governed Shared Memory for Multi-Agent LLM Systems. *CoRR* abs/2606.24535 (2026).

[98] Milvus. 2026. Milvus Documentation: Consistency Level. https://milvus.io/docs/consistency.md.

[99] C. Mohan. 1990. ARIES/KVL: A Key-Value Locking Method for Concurrency Control of Multiaction Transactions Operating on B-Tree Indexes. In *Proc. VLDB'90*. Morgan Kaufmann, 392–405.

[100] C. Mohan and Frank Levine. 1992. ARIES/IM: An Efficient and High Concurrency Index Management Method Using Write-Ahead Logging. In *Proc. SIGMOD'92*. ACM, 371–380.

[101] Jason Mohoney, Anil Pacaci, Shihabur Rahman Chowdhury, Umar Farooq Minhas, Jeffery Pound, Cedric Renggli, Nima Reyhani, Ihab F. Ilyas, Theodoros Rekatsinas, and Shivaram Venkataraman. 2024. Incremental IVF Index Maintenance for Streaming Vector Search. *CoRR* abs/2411.00970 (2024).

[102] Jason Mohoney, Devesh Sarda, Mengze Tang, Shihabur Rahman Chowdhury, Anil Pacaci, Ihab F. Ilyas, Theodoros Rekatsinas, and Shivaram Venkataraman. 2025. Quake: Adaptive Indexing for Vector Search. In *Proc. OSDI'25*. USENIX Association, 153–169.

[103] Kyriakos Mouratidis, Marios Hadjieleftheriou, and Dimitris Papadias. 2005. Conceptual Partitioning: An Efficient Method for Continuous Nearest Neighbor Monitoring. In *Proc. SIGMOD'05*. ACM, 634–645.

[104] Barzan Mozafari. 2026. BEGIN AI TRANSACTION: Semantic Isolation for Durable AI Workflows. *CoRR* abs/2608.05412 (2026).

[105] Oto Mraz, Rares Şerban, Kyriakos Psarakis, Burcu Kulahcioglu Ozkan, and Asterios Katsifodimos. 2026. Tracking State Footprints: How Agents Can Transact. *CoRR* abs/2610.03140 (2026).

[106] Thomas Neumann, Tobias Mühlbauer, and Alfons Kemper. 2015. Fast Serializable Multi-Version Concurrency Control for Main-Memory Database Systems. In *Proc. SIGMOD'15*. ACM, 677–689.

[107] Zhaojie Niu, Xinhui Tian, Xindong Peng, and Xing Chen. 2025. BlendHouse: A Cloud-Native Vector Database System in ByteHouse. In *Proc. ICDE'25*. IEEE, 4332–4345.

[108] nmslib. 2026. hnswlib: Header-only C++/Python library for fast approximate nearest neighbors. https://github.com/nmslib/hnswlib.

[109] Sukjoon Oh, Minki Kang, Dohyun Kim, Baotong Lu, Jing Liu, Qianxi Zhang, Qi Chen, and Youjip Won. 2026. Aker: Density-Aware Approximate Caching for Vector Search. *Proc. VLDB Endow.* 19, 10 (2026), 2727–2740.

[110] Hiroyuki Ootomo, Akira Naruse, Corey Nolet, Ray Wang, Tamas Feher, and Yong Wang. 2024. CAGRA: Highly Parallel Graph Construction and Approximate Nearest Neighbor Search for GPUs. In *Proc. ICDE'24*. IEEE, 4236–4247.

[111] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as Operating Systems. *CoRR* abs/2310.08560 (2023).

[112] James Jie Pan, Jianguo Wang, and Guoliang Li. 2024. Survey of Vector Database Management Systems. *VLDB J.* 33, 5 (2024), 1591–1615.

[113] James Jie Pan, Jianguo Wang, and Guoliang Li. 2024. Vector Database Management Techniques and Systems. In *Proc. SIGMOD Companion'24*. ACM, 597–604.

[114] Christos H. Papadimitriou. 1979. The Serializability of Concurrent Database Updates. *J. ACM* 26, 4 (1979), 631–653.

[115] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proc. UIST'23*. ACM, 1–22.

[116] Liana Patel, Peter Kraft, Carlos Guestrin, and Matei Zaharia. 2024. ACORN: Performant and Predicate-Agnostic Search Over Vector Embeddings and Structured Data. *Proc. ACM Manag. Data* 2, 3 (2024), 120:1–120:27.

[117] Dan R. K. Ports and Kevin Grittner. 2012. Serializable Snapshot Isolation in PostgreSQL. *Proc. VLDB Endow.* 5, 12 (2012), 1850–1861.

[118] Jeffrey Pound, Floris Chabert, Arjun Bhushan, Ankur Goswami, Anil Pacaci, and Shihabur Rahman Chowdhury. 2025. MicroNN: An On-device Disk-resident Updatable Vector Database. In *Proc. SIGMOD Companion'25*. ACM, 608–621.

[119] Calton Pu and Avraham Leff. 1991. Replica Control in Distributed Systems: An Asynchronous Approach. In *Proc. SIGMOD'91*. ACM, 377–386.

[120] Qdrant. 2026. Qdrant Documentation: Consistency Guarantees. https://qdrant.tech/documentation/scaling/consistency-guarantees/.

[121] Qdrant. 2026. Qdrant: Vector Database and Vector Search Engine. https://github.com/qdrant/qdrant. Version 1.15.4.

[122] Krithi Ramamritham and Calton Pu. 1995. A Formal Characterization of Epsilon Serializability. *IEEE Trans. Knowl. Data Eng.* 7, 6 (1995), 997–1007.

[123] Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. 2025. Zep: A Temporal Knowledge Graph Architecture for Agent Memory. *CoRR* abs/2501.13956 (2025).

[124] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proc. EMNLP-IJCNLP'19*. Association for Computational Linguistics, 3980–3990.

[125] Alireza Rezazadeh, Zichao Li, Ange Lou, Yuying Zhao, Wei Wei, and Yujia Bao. 2025. Collaborative Memory: Multi-User Memory Sharing in LLM Agents with Dynamic Access Control. *CoRR* abs/2505.18279 (2025).

[126] Arik Rinberg and Idit Keidar. 2020. Intermediate Value Linearizability: A Quantitative Correctness Criterion. In *Proc. DISC'20 (LIPIcs)*, Vol. 179. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2:1–2:17.

[127] Arik Rinberg, Alexander Spiegelman, Edward Bortnikov, Eshcar Hillel, Idit Keidar, Lee Rhodes, and Hadar Serviansky. 2020. Fast Concurrent Data Sketches. In *Proc. PPoPP'20*. ACM, 117–129.

[128] Nick Roussopoulos, Stephen Kelley, and Frédéric Vincent. 1995. Nearest Neighbor Queries. In *Proc. SIGMOD'95*. ACM, 71–79.

[129] Luis Gaspar Schroeder, Aditya Desai, Alejandro Cuadron, Kyle Chu, Shu Liu, Mark Zhao, Stephan Krusche, Alfons Kemper, Matei Zaharia, and Joseph E. Gonzalez. 2026. vCache: Verified Semantic Prompt Caching. In *Proc. ICLR'26*. OpenReview.net. https://openreview.net/forum?id=zF0A0xw3HZ

[130] Gaurav Sehgal and Semih Salihoğlu. 2025. NaviX: A Native Vector Index Design for Graph DBMSs With Robust Predicate-Agnostic Search Performance. *Proc. VLDB Endow.* 18, 11 (2025), 4438–4450.

[131] Harsha Vardhan Simhadri, Martin Aumüller, Matthijs Douze, Dmitry Baranchuk, Amir Ingber, Edo Liberty, George Williams, Ben Landrum, Magdalen Manohar, Mazin Karjikar, Laxman Dhulipala, Meng Chen, Yue Chen, Rui Ma, Kai Zhang, Yuzheng Cai, Jiayang Shi, Weiguo Zheng, Yizhuo Chen, Jie Yin, and Ben Huang. 2025. Results of the Big ANN: NeurIPS'23 competition. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. https://papers.nips.cc/paper_files/paper/2025/hash/63092d79154adebd7305dfd498cbff70-Abstract-Datasets_and_Benchmarks_Track.html

[132] Amit Singh, Hakan Ferhatosmanoglu, and Ali Şaman Tosun. 2003. High Dimensional Reverse Nearest Neighbor Queries. In *Proc. CIKM'03*. ACM, 91–98.

[133] Aditi Singh, Suhas Jayaram Subramanya, Ravishankar Krishnaswamy, and Harsha Vardhan Simhadri. 2021. FreshDiskANN: A Fast and Accurate Graph-Based ANN Index for Streaming Similarity Search. *CoRR* abs/2105.09613 (2021).

[134] Seok Il Song, Seok Jae Lee, Tae Ho Kang, and Jae Soo Yoo. 2005. An Efficient Phantom Protection Method for Multi-dimensional Index Structures. In *Proc. DASFAA'05 (Lecture Notes in Computer Science)*, Vol. 3453. Springer, 875–887.

[135] Zhao Song. 2025. Is pgvector breaking PostgreSQL's Repeatable Read isolation? https://kernelmaker.github.io/pgvector_rr.

[136] Suhas Jayaram Subramanya, Fnu Devvrit, Harsha Vardhan Simhadri, Ravishankar Krishnaswamy, and Rohan Kadekodi. 2019. DiskANN: Fast Accurate Billion-point Nearest Neighbor Search on a Single Node. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*.

[137] Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. 2024. Cognitive Architectures for Language Agents. *Trans. Mach. Learn. Res.* (2024). https://openreview.net/forum?id=1i6ZCvflQJ

[138] Ji Sun, Guoliang Li, James Pan, Jiang Wang, Yongqing Xie, Ruicheng Liu, and Wen Nie. 2025. GaussDB-Vector: A Large-Scale Persistent Real-Time Vector Database for LLM Applications. *Proc. VLDB Endow.* 18, 12 (2025), 4951–4963.

[139] Zhaoyan Sun, Xiaoxiao Wang, and Guoliang Li. 2026. Agentic Transaction: Towards ACID-Compliant Agent Systems. *CoRR* abs/2608.13900 (2026).

[140] Toni Taipalus. 2024. Vector database management systems: Fundamental concepts, use-cases, and current challenges. *Cogn. Syst. Res.* 85 (2024), 101216.

[141] Cheng Tan, Changgeng Zhao, Shuai Mu, and Michael Walfish. 2020. Cobra: Making Transactional Key-Value Stores Verifiably Serializable. In *Proc. OSDI'20*. USENIX Association, 63–80.

[142] Yufei Tao, Dimitris Papadias, and Xiang Lian. 2004. Reverse kNN Search in Arbitrary Dimensionality. In *Proc. VLDB'04*. Morgan Kaufmann, 744–755.

[143] Stephen Tu, Wenting Zheng, Eddie Kohler, Barbara Liskov, and Samuel Madden. 2013. Speedy Transactions in Multicore In-Memory Databases. In *Proc. SOSP'13*. ACM, 18–32.

[144] Nitish Upreti, Harsha Vardhan Simhadri, Hari Sudan Sundar, Krishnan Sundaram, Samer Boshra, Balachandar Perumalswamy, Shivam Atri, Martin Chisholm, Revti Raman Singh, Greg Yang, Tamara Hass, Nitesh Dudhey, Subramanyam Pattipaka, Mark Hildebrand, Magdalen Manohar, Jack Moffitt, Haiyang Xu, Naren Datha, Suryansh Gupta, Ravishankar Krishnaswamy, Prashant Gupta, Abhishek Sahu, Hemeswari Varada, Sudhanshu Barthwal, Ritika Mor, James Codella, Shaun Cooper, Kevin Pilch, Simon Moreno, Aayush Kataria, Santosh Kulkarni, Neil Deshpande, Amar Sagare, Dinesh Billa, Zishan Fu, and Vipul Vishal. 2025. Cost-Effective, Low Latency Vector Search with Azure Cosmos DB. *Proc. VLDB Endow.* 18, 12 (2025), 5166–5183.

[145] Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2019. GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding. In *Proc. ICLR'19*. OpenReview.net.

[146] Congying Wang, Nithin Sastry Tellapuri, Sphoorthi Keshannagari, Dylan Zinsley, Zhuoyue Zhao, and Dong Xie. 2023. Approximate Queries over Concurrent Updates. *Proc. VLDB Endow.* 16, 12 (2023), 3986–3989.

[147] Jianguo Wang, Eric Hanson, Guoliang Li, Yannis Papakonstantinou, Harsha Simhadri, and Charles Xie. 2024. Vector Databases: What's Really New and What's Next? (VLDB 2024 Panel). *Proc. VLDB Endow.* 17, 12 (2024), 4505–4506.

[148] Jianguo Wang, Xiaomeng Yi, Rentong Guo, Hai Jin, Peng Xu, Shengjun Li, Xiangyu Wang, Xiangzhou Guo, Chengming Li, Xiaohai Xu, Kun Yu, Yuxing Yuan, Yinghao Zou, Jiquan Long, Yudong Cai, Zhenxiang Li, Zhifeng Zhang, Yihua Mo, Jun Gu, Ruiyi Jiang, Yi Wei, and Charles Xie. 2021. Milvus: A Purpose-Built Vector Data Management System. In *Proc. SIGMOD'21*. ACM, 2614–2627.

[149] Mengzhao Wang, Xiaoliang Xu, Qiang Yue, and Yuxiang Wang. 2021. A Comprehensive Survey and Experimental Comparison of Graph-Based Approximate Nearest Neighbor Search. *Proc. VLDB Endow.* 14, 11 (2021), 1964–1978.

[150] Shenao Wang, Yanjie Zhao, Yinglin Xie, Zhao Liu, Xinyi Hou, Quanchen Zou, and Haoyu Wang. 2025. Towards Reliable Vector Database Management Systems: A Software Testing Roadmap for 2030. *CoRR* abs/2502.20812 (2025).

[151] Yu Wang and Xi Chen. 2025. MIRIX: Multi-Agent Memory System for LLM-Based Agents. *CoRR* abs/2507.07957 (2025).

[152] Yuxiang Wang, Ziyuan He, Yongxin Tong, Zimu Zhou, and Yiman Zhong. 2025. Timestamp Approximate Nearest Neighbor Search over High-Dimensional Vector Data. In *Proc. ICDE'25*. IEEE, 3043–3055.

[153] Ziming Wang. 2026. TOKI: A Bitemporal Operator Algebra for Contradiction Resolution in LLM-Agent Persistent Memory. *CoRR* abs/2606.06240 (2026).

[154] Zeyu Wang, Qitong Wang, Xiaoxing Cheng, Peng Wang, Themis Palpanas, and Wei Wang. 2024. Steiner-Hardness: A Query Hardness Measure for Graph-Based ANN Indexes. *Proc. VLDB Endow.* 17, 13 (2024), 4668–4682.

[155] Weaviate. 2026. Weaviate: An Open-source Vector Database. https://github.com/weaviate/weaviate.

[156] Weaviate. 2026. Weaviate Documentation: Consistency. https://docs.weaviate.io/weaviate/concepts/replication-architecture/consistency.

[157] Chuangxian Wei, Bin Wu, Sheng Wang, Renjie Lou, Chaoqun Zhan, Feifei Li, and Yuanzhe Cai. 2020. AnalyticDB-V: A Hybrid Analytical Engine Towards Query Fusion for Structured and Unstructured Data. *Proc. VLDB Endow.* 13, 12 (2020), 3152–3165.

[158] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2025. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. In *Proc. ICLR'25*. OpenReview.net.

[159] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. 2024. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. In *Proc. COLM'24*. https://arxiv.org/abs/2308.08155

[160] Yingjun Wu, Joy Arulraj, Jiexi Lin, Ran Xian, and Andrew Pavlo. 2017. An Empirical Evaluation of In-Memory Multi-Version Concurrency Control. *Proc. VLDB Endow.* 10, 7 (2017), 781–792.

[161] Wenxuan Xia, Wentao Li, Mingyu Yang, and Wei Wang. 2026. HRNN: A Hybrid Graph Index for Approximate Reverse k-Nearest Neighbor Search on High-Dimensional Vectors. *CoRR* abs/2606.03225 (2026).

[162] Wentao Xiao, Yueyang Zhan, Rui Xi, Mengshu Hou, and Jianming Liao. 2024. Enhancing HNSW Index for Real-Time Updates: Addressing Unreachable Points and Performance Degradation. *CoRR* abs/2407.07871 (2024).

[163] Yinglin Xie, Xinyi Hou, Yanjie Zhao, Shenao Wang, Kai Chen, and Haoyu Wang. 2025. Toward Understanding Bugs in Vector Database Management Systems. *CoRR* abs/2506.02617 (2025).

[164] Xiaopeng Xiong, Mohamed F. Mokbel, and Walid G. Aref. 2005. SEA-CNN: Scalable Processing of Continuous K-Nearest Neighbor Queries in Spatio-temporal Databases. In *Proc. ICDE'05*. IEEE Computer Society, 643–654.

[165] Haike Xu, Magdalen Dobson Manohar, Philip A. Bernstein, Badrish Chandramouli, Richard Wen, and Harsha Vardhan Simhadri. 2025. In-Place Updates of a Graph Index for Streaming Approximate Nearest Neighbor Search. *CoRR* abs/2502.13826 (2025).

[166] Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025. A-MEM: Agentic Memory for LLM Agents. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. 17577–17604.

[167] Yuming Xu, Hengyu Liang, Jin Li, Shuotao Xu, Qi Chen, Qianxi Zhang, Cheng Li, Ziyue Yang, Fan Yang, Yuqing Yang, Peng Cheng, and Mao Yang. 2023. SPFresh: Incremental In-Place Update for Billion-Scale Vector Search. In *Proc. SOSP'23*. ACM, 545–561.

[168] Wen Yang, Tao Li, Gai Fang, and Hong Wei. 2020. PASE: PostgreSQL Ultra-High-Dimensional Approximate Nearest Neighbor Search Extension. In *Proc. SIGMOD'20*. ACM, 2241–2253.

[169] Xin Yang, Letian Li, Zimo Ji, Terry Jingchen Zhang, and Wenyuan Jiang. 2026. Position: Multi-Agent Systems Should Prioritize Concurrency Control. In *Proc. ICML'26 (Proceedings of Machine Learning Research)*, Vol. 306. PMLR.

[170] Haifeng Yu and Amin Vahdat. 2002. Design and Evaluation of a Conit-Based Continuous Consistency Model for Replicated Services. *ACM Trans. Comput. Syst.* 20, 3 (2002), 239–282.

[171] Song Yu, Shengyuan Lin, Shufeng Gong, Yongqing Xie, Ruicheng Liu, Yijie Zhou, Ji Sun, Yanfeng Zhang, Guoliang Li, and Ge Yu. 2025. A Topology-Aware Localized Update Strategy for Graph-Based ANN Index. *Proc. VLDB Endow.* 19, 3 (2025), 495–508.

[172] Xiangyao Yu, George Bezerra, Andrew Pavlo, Srinivas Devadas, and Michael Stonebraker. 2014. Staring into the Abyss: An Evaluation of Concurrency Control with One Thousand Cores. *Proc. VLDB Endow.* 8, 3 (2014), 209–220.

[173] Xiaohui Yu, Ken Q. Pu, and Nick Koudas. 2005. Monitoring k-Nearest Neighbor Queries over Moving Objects. In *Proc. ICDE'05*. IEEE Computer Society, 631–642.

[174] Zhongming Yu, Naicheng Yu, Hejia Zhang, Wentao Ni, Mingrui Yin, Jiaying Yang, Yujie Zhao, and Jishen Zhao. 2026. Multi-Agent Memory from a Computer Architecture Perspective: Visions and Challenges Ahead. *CoRR* abs/2603.10062 (2026).

[175] Guibin Zhang, Muxin Fu, Guancheng Wan, Miao Yu, Kun Wang, and Shuicheng Yan. 2025. G-Memory: Tracing Hierarchical Memory for Multi-Agent Systems. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. 14587–14617.

[176] Qianxi Zhang, Shuotao Xu, Qi Chen, Guoxin Sui, Jiadong Xie, Zhizhen Cai, Yaoqi Chen, Yinxuan He, Yuqing Yang, Fan Yang, Mao Yang, and Lidong Zhou. 2023. VBASE: Unifying Online Vector Similarity Search and Relational Queries via Relaxed Monotonicity. In *Proc. OSDI'23*. USENIX Association, 377–395.

[177] Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level Convolutional Networks for Text Classification. In *Advances in Neural Information Processing Systems 28 (NIPS 2015)*. 649–657.

[178] Yunan Zhang, Shige Liu, and Jianguo Wang. 2024. Are There Fundamental Limitations in Supporting Vector Data Management in Relational Databases? A Case Study of PostgreSQL. In *Proc. ICDE'24*. IEEE, 3640–3653.

[179] Zeyu Zhang, Quanyu Dai, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. 2025. A Survey on the Memory Mechanism of Large Language Model-based Agents. *ACM Trans. Inf. Syst.* 43, 6 (2025).

[180] Ziyu Zhang, Yuanhao Wei, Joshua Engels, and Julian Shun. 2025. CleANN: Efficient Full Dynamism in Graph-based Approximate Nearest Neighbor Search. *CoRR* abs/2507.19802 (2025).

[181] Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. MemoryBank: Enhancing Large Language Models with Long-Term Memory. In *Proc. AAAI'24*. AAAI Press, 19724–19731.